

Depression Connect Forum

Master Analysis Report

Dataset variant: Old Data Only

EDA · Time series · CDS prevalence · LIWC · Longitudinal trends · LIWC-22 validation

1. EDA - All Users

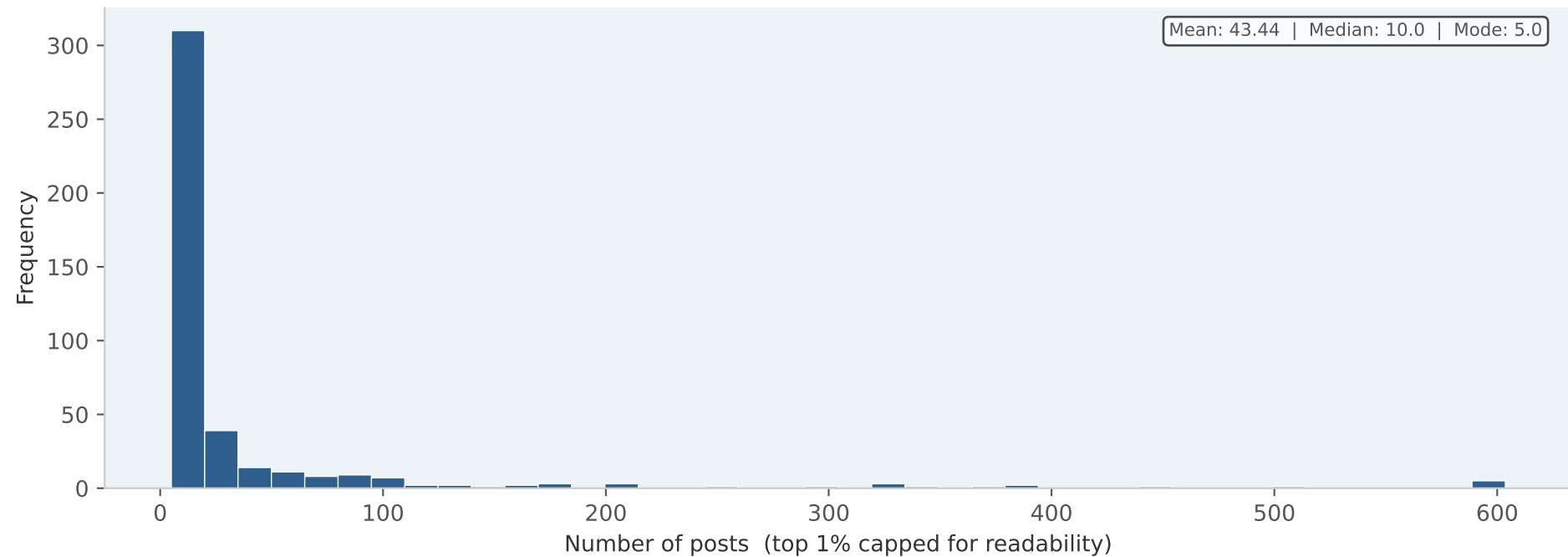
Depression Connect Forum

Exploratory Data Analysis Report — All Users

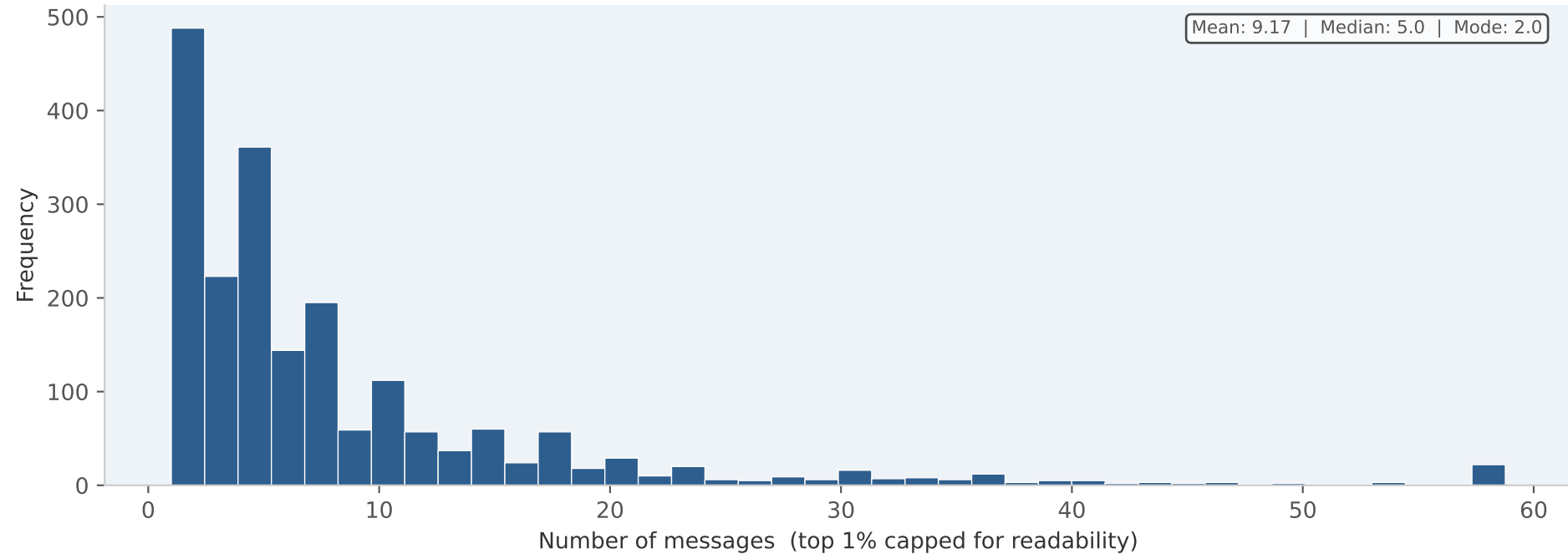
Summary Statistics

Metric	Mean	Median	Mode	Min	Max
Posts per user	43.44	10.0	5.0	5	1646
Messages per thread (total)	9.17	5.0	2.0	1	562
Replies per thread	9.26	5.0	1.0	1	561
Activity span (days)	187.55	55.0	2.0	0	1183
Active days per user	19.17	6.0	4.0	1	512
Words per post	112.46	69.0	6.0	0	2724
Words per user (total)	4885.65	1135.0	584.0	114	316713
'ik' count per user	205.63	60.0	18.0	4	10049
'ik' % per user	5.23	5.36	2.52	0.35	11.79
'mijn' count per user	40.97	13.0	6.0	0	2438
'mijn' % per user	1.14	1.06	0.0	0.0	4.62

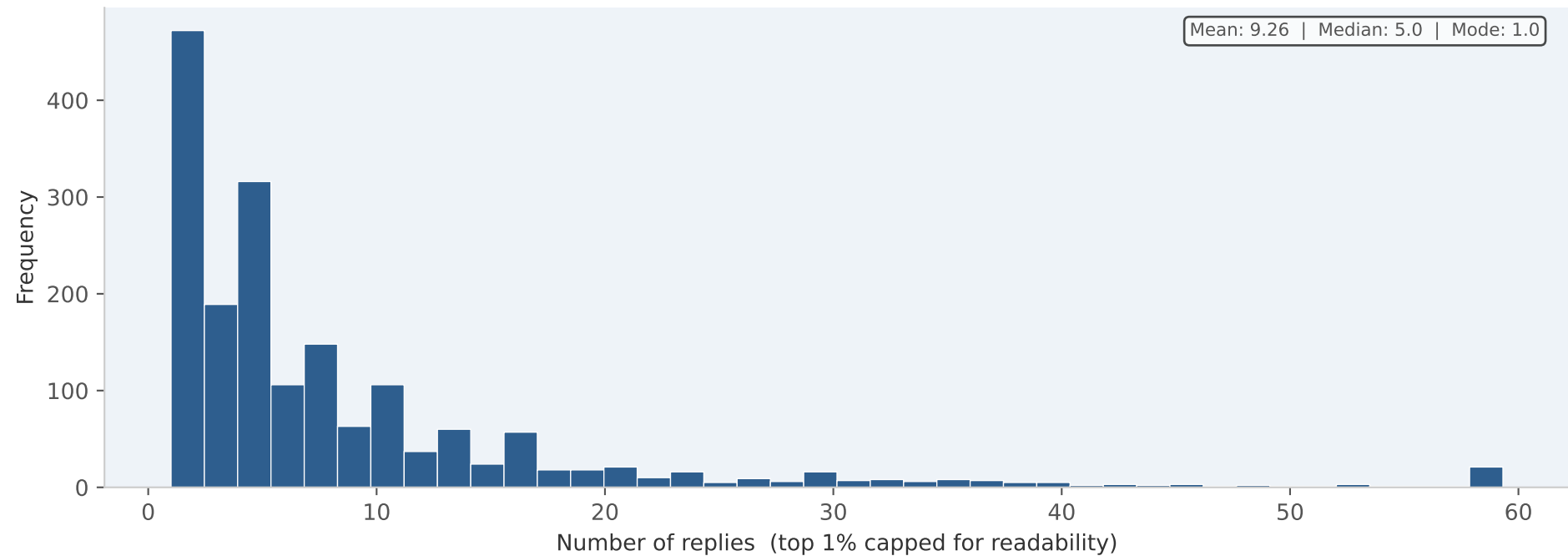
Distribution: Posts per User



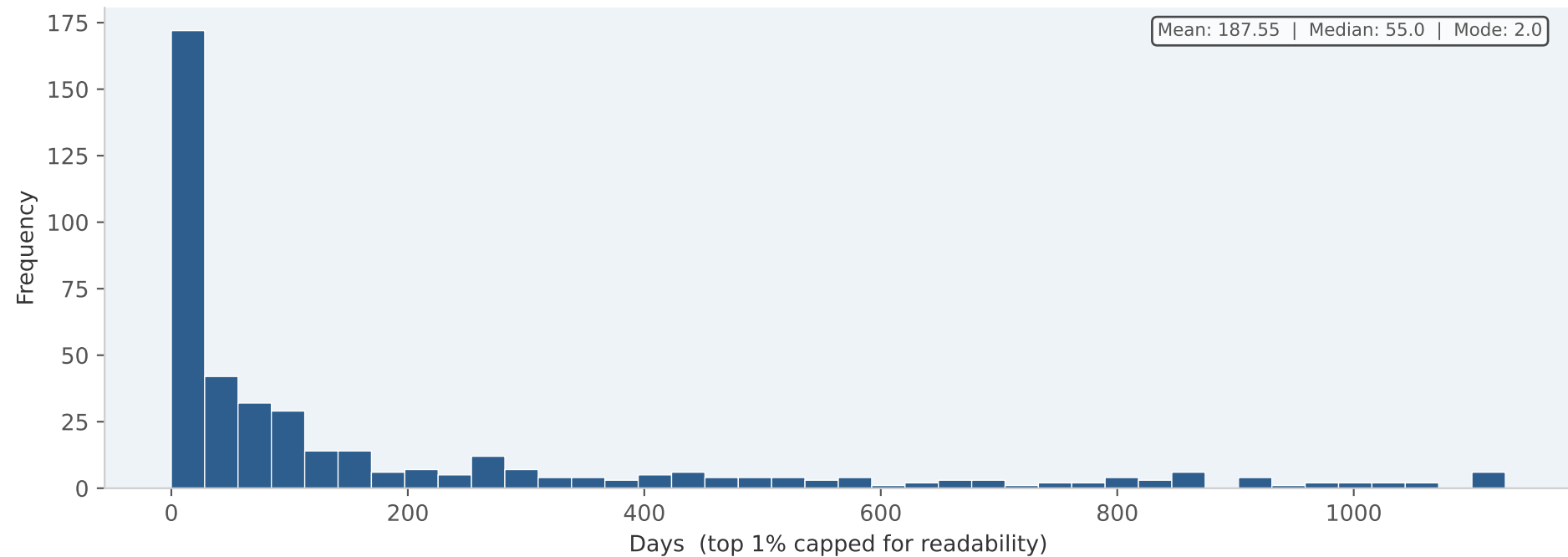
Distribution: Total Messages per Thread



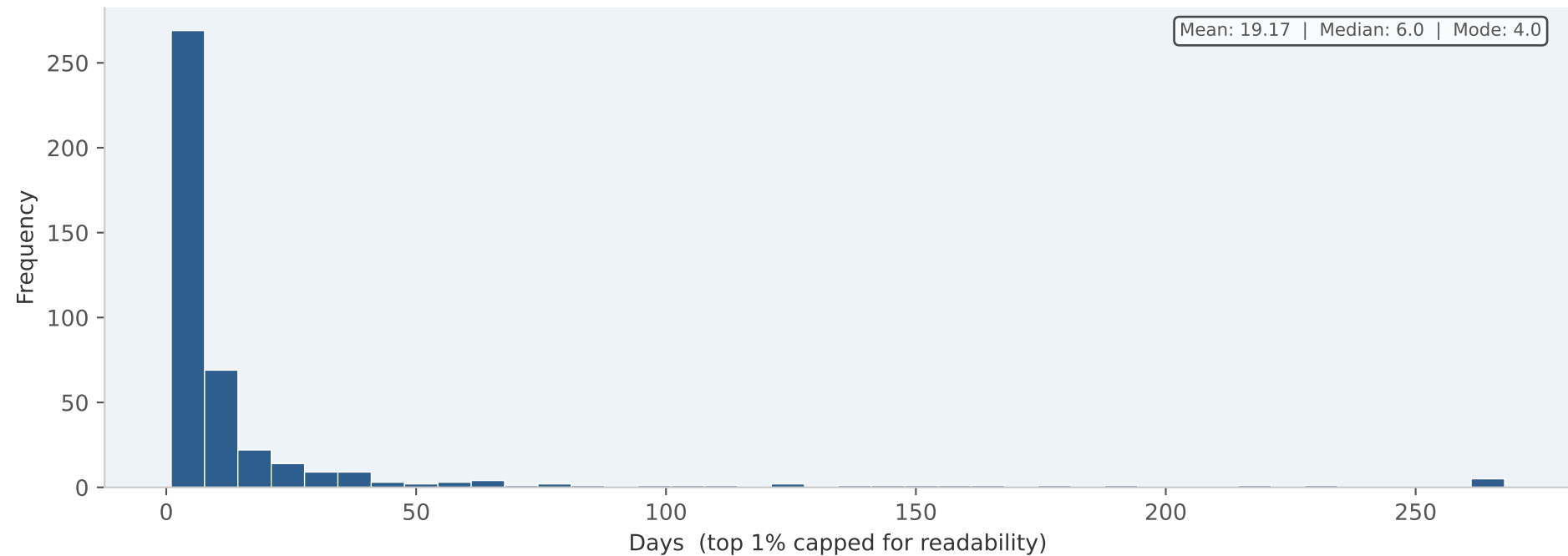
Distribution: Replies per Thread (excluding opening post)



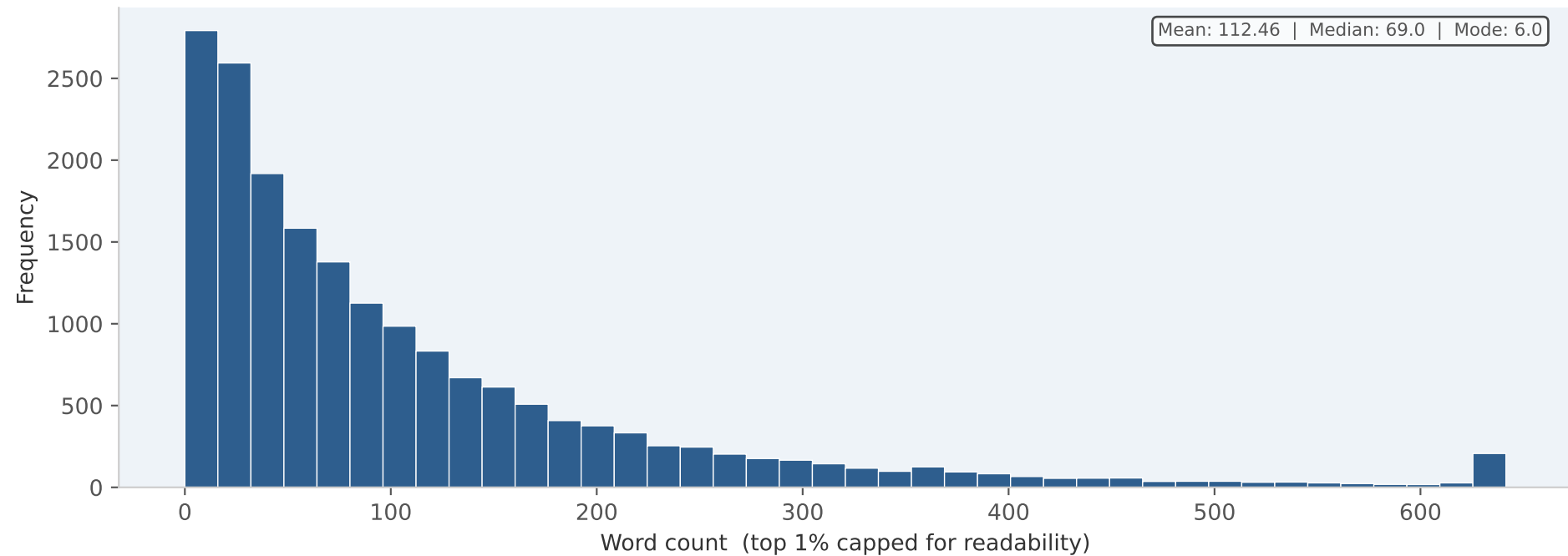
Distribution: User Activity Span (days between first and last post)



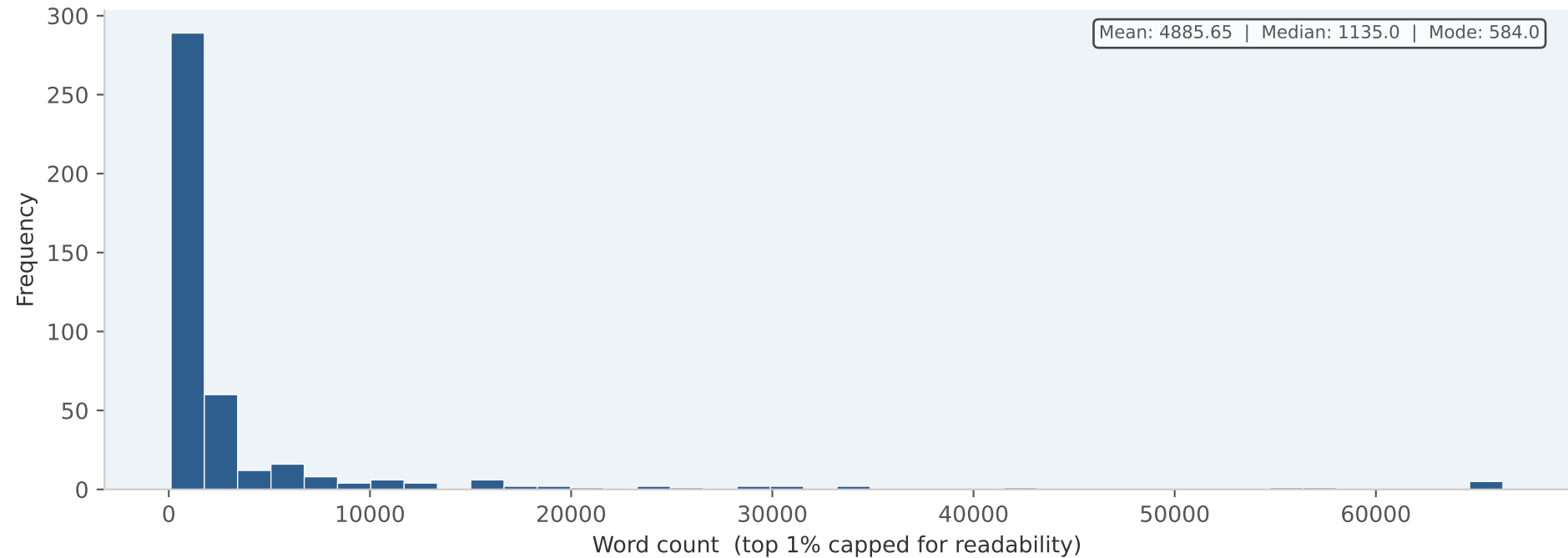
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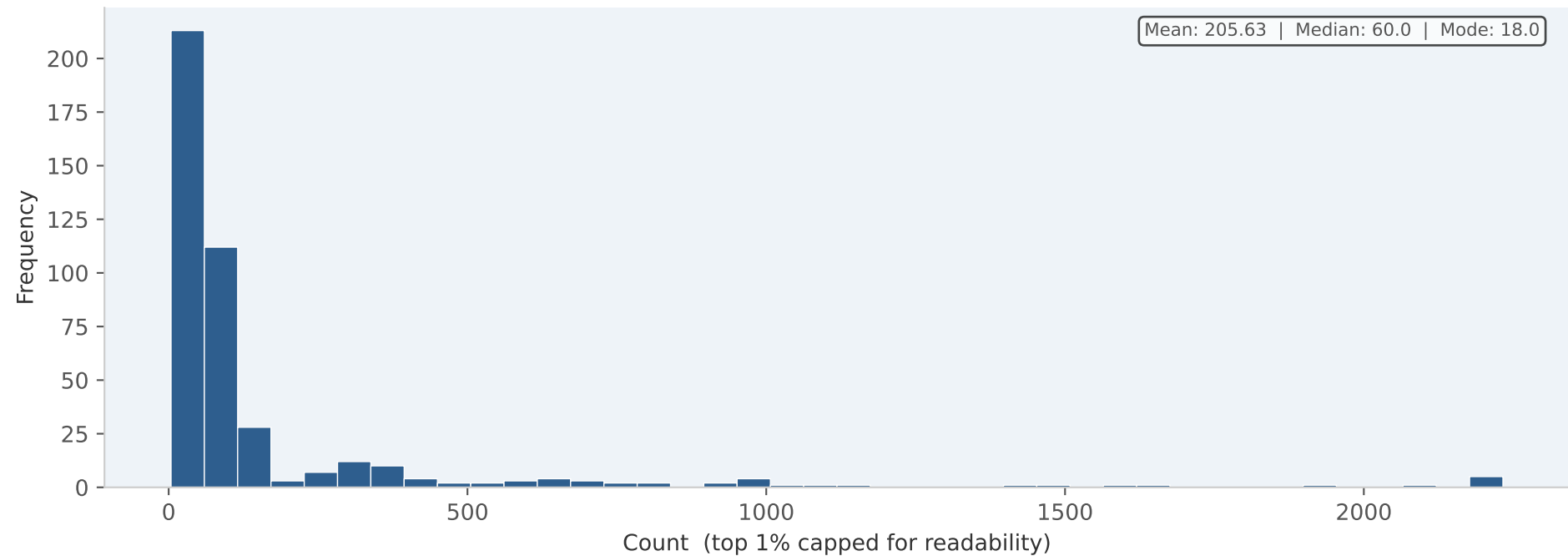
Distribution: Words per Post



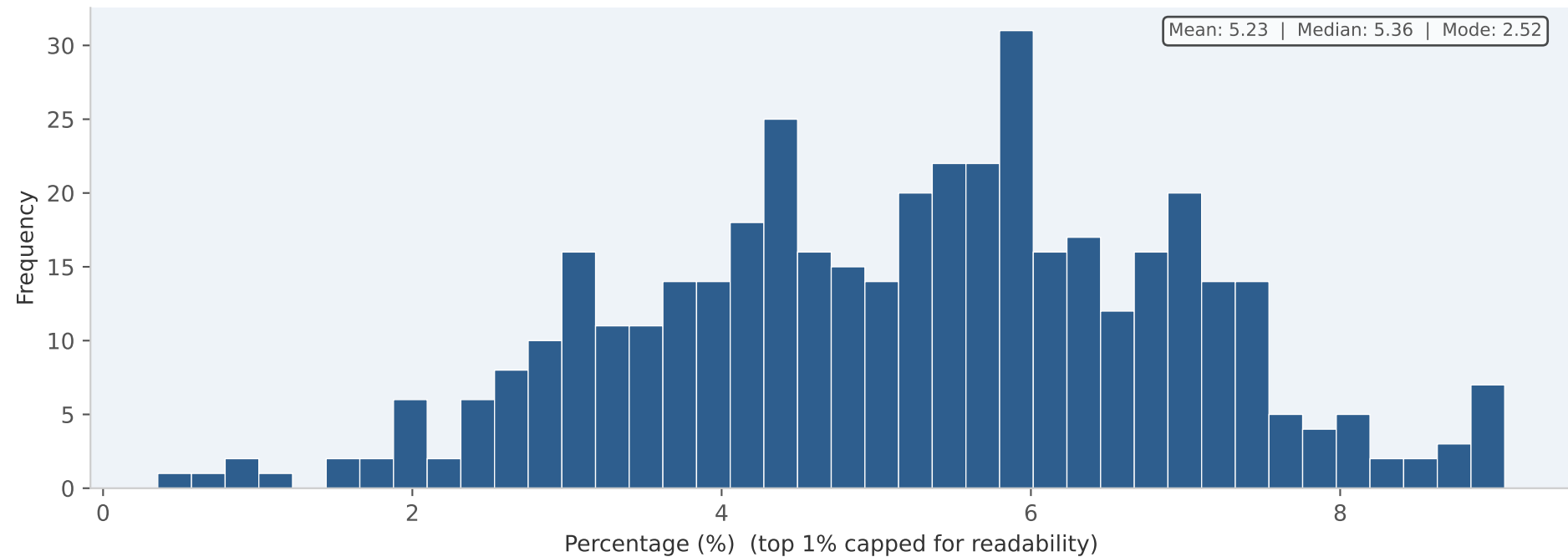
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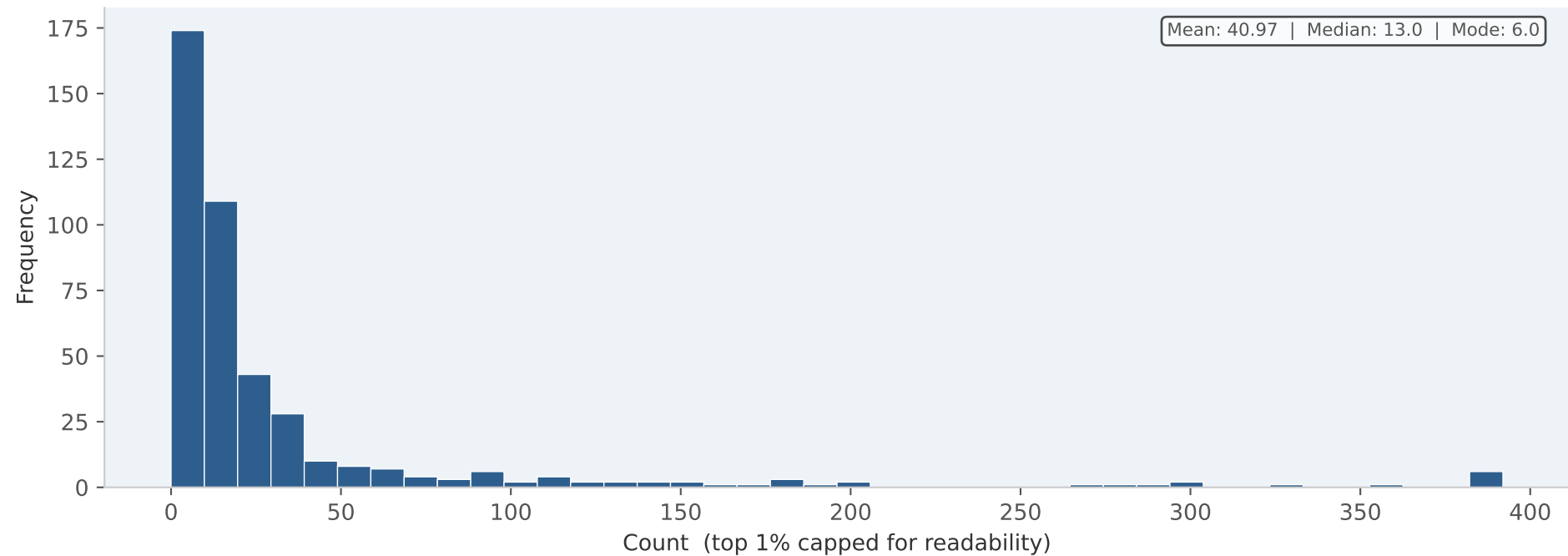
Distribution: 'ik' Count per User



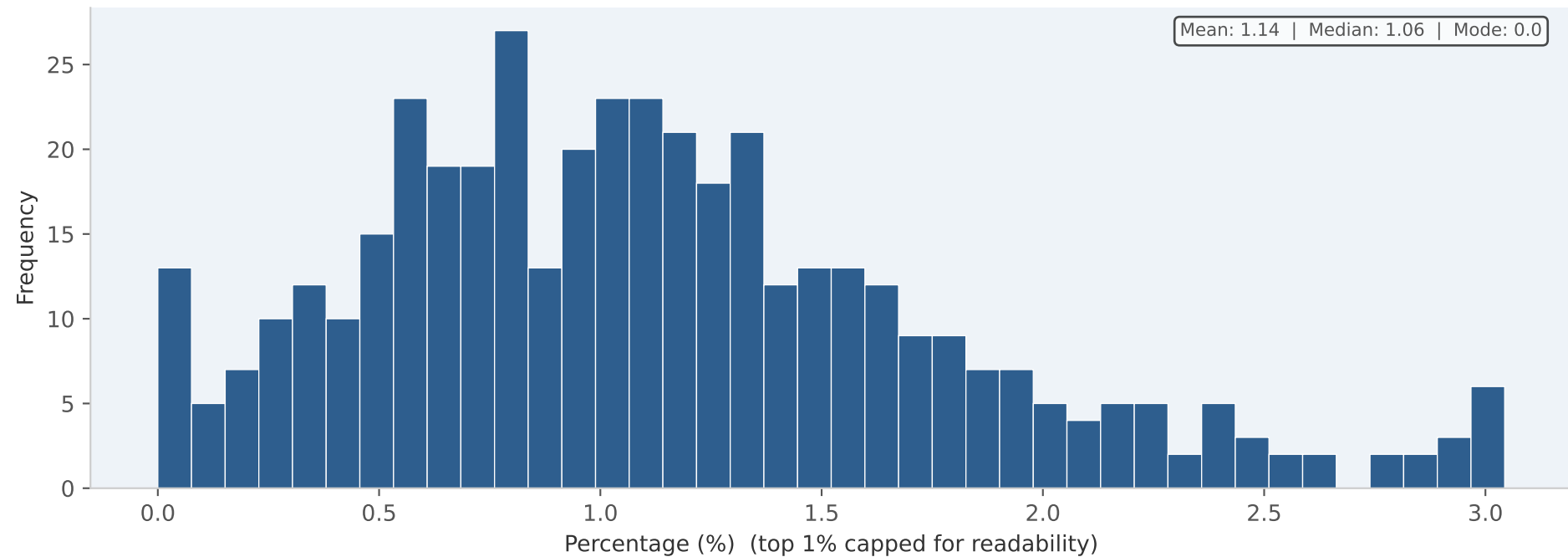
Distribution: 'ik' as % of Total Words per User



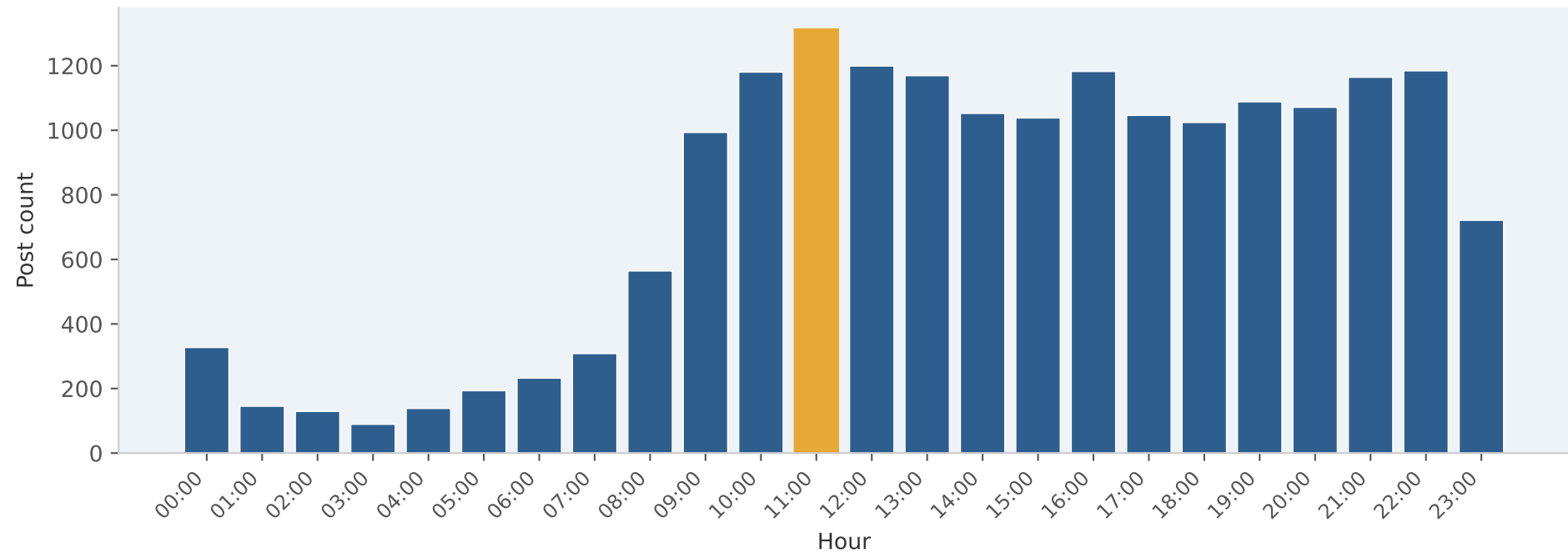
Distribution: 'mijn' Count per User



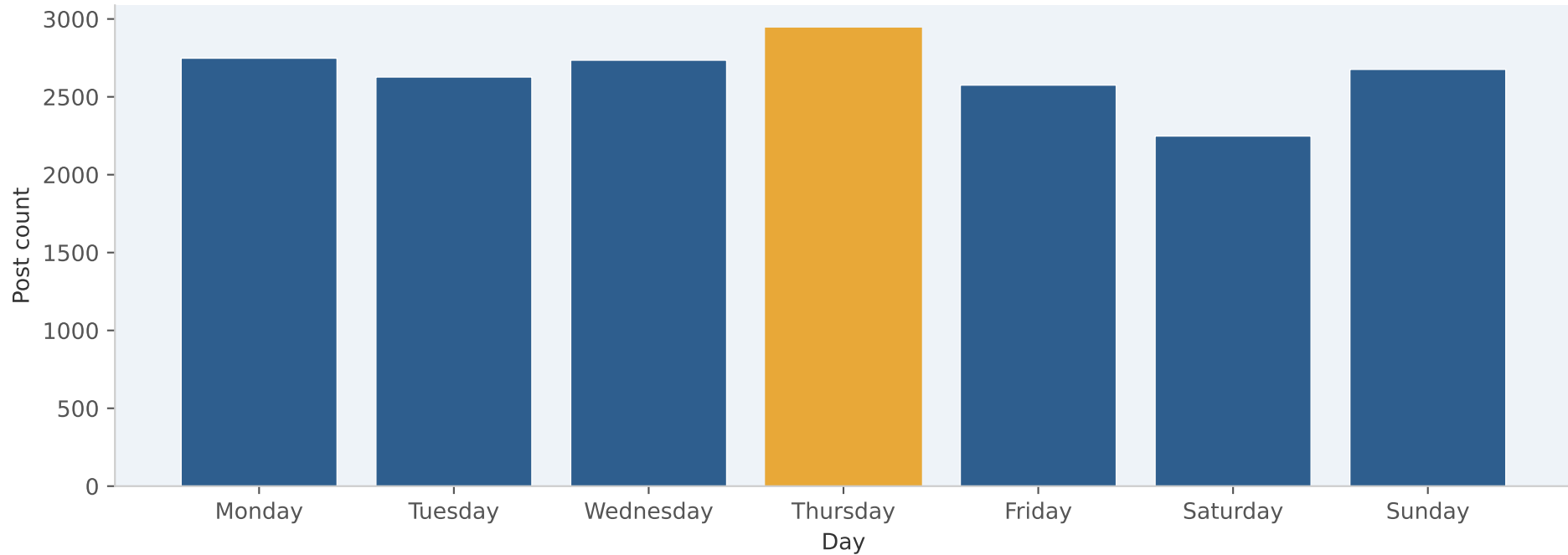
Distribution: 'mijn' as % of Total Words per User



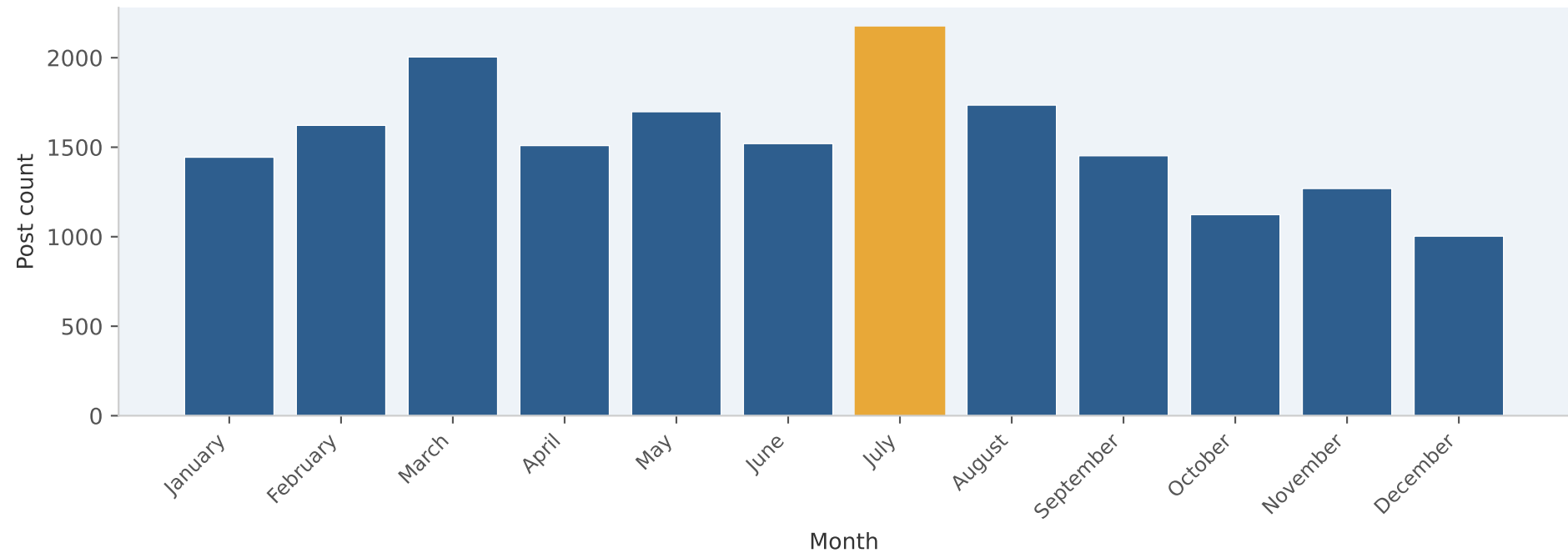
Popular Hours of Day



Popular Days of Week

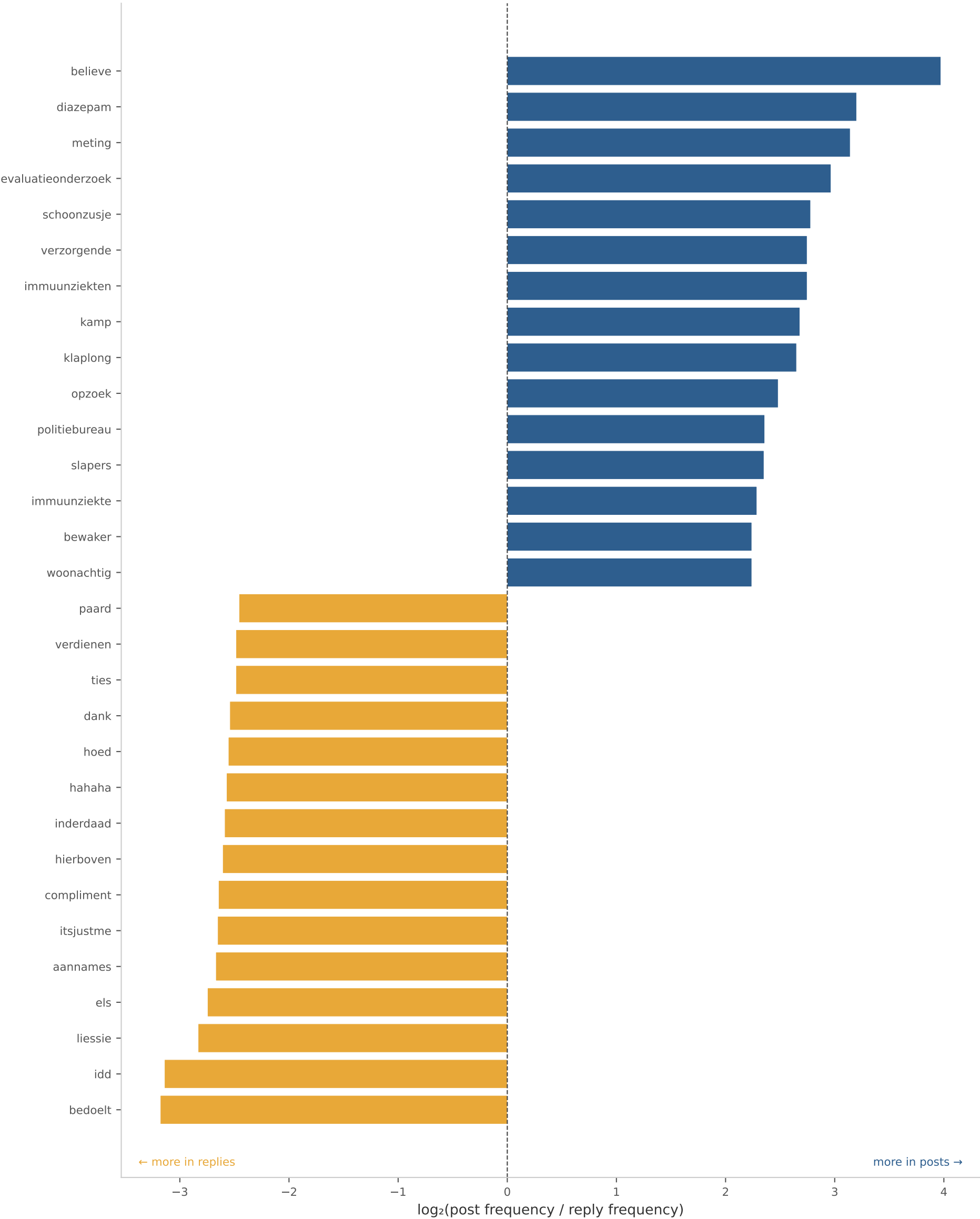


Popular Months



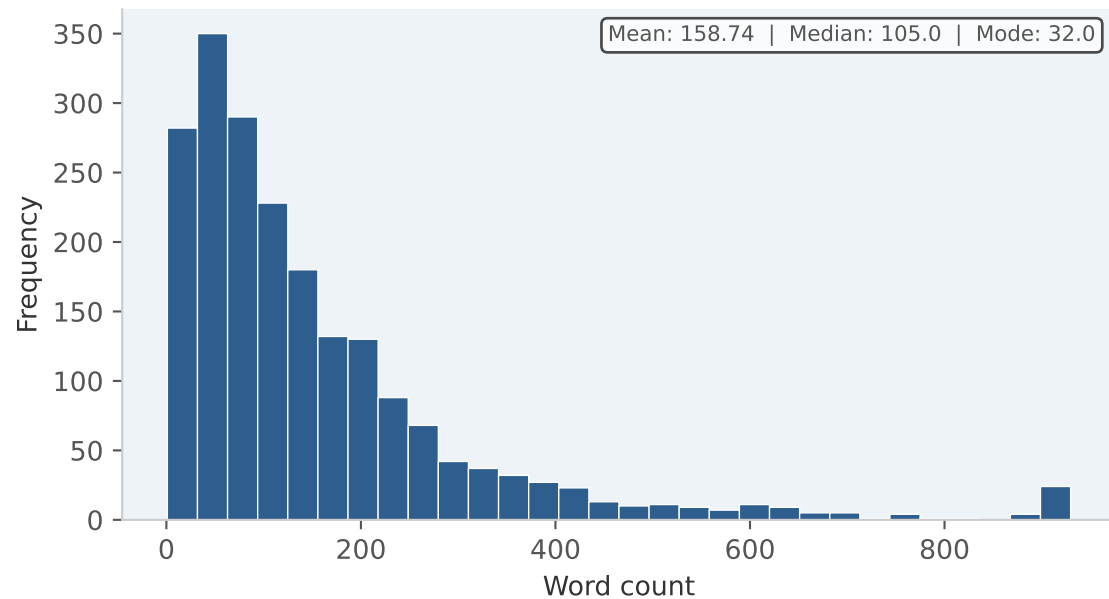
Role-Based Analysis — Posts vs Replies

Word Choice: Initial Posts vs Replies

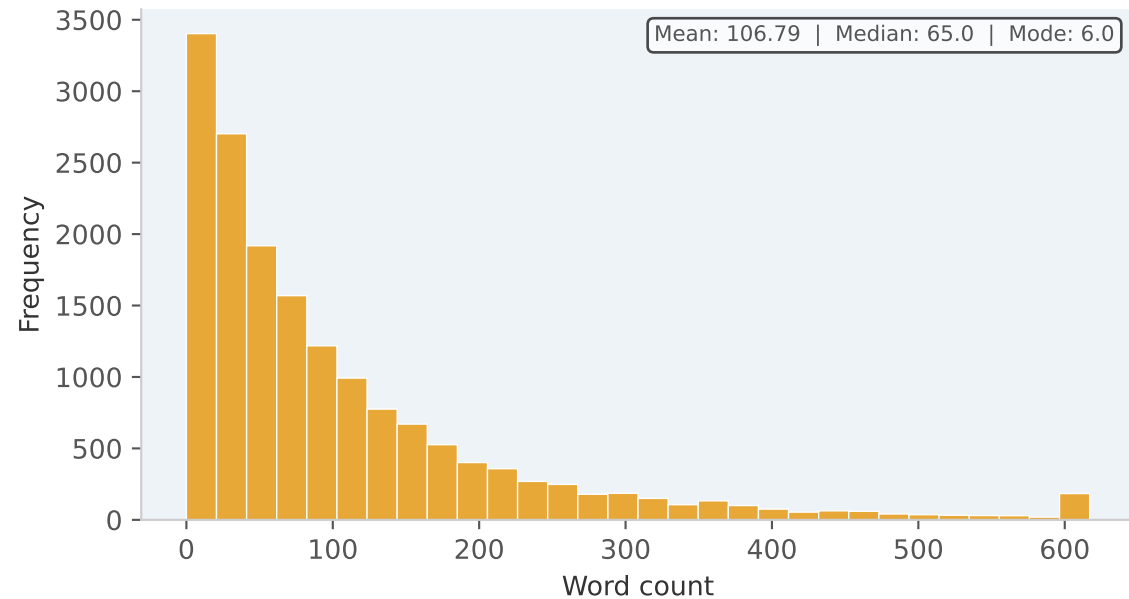


Words per Message: Initial Posts vs Replies

Initial Posts

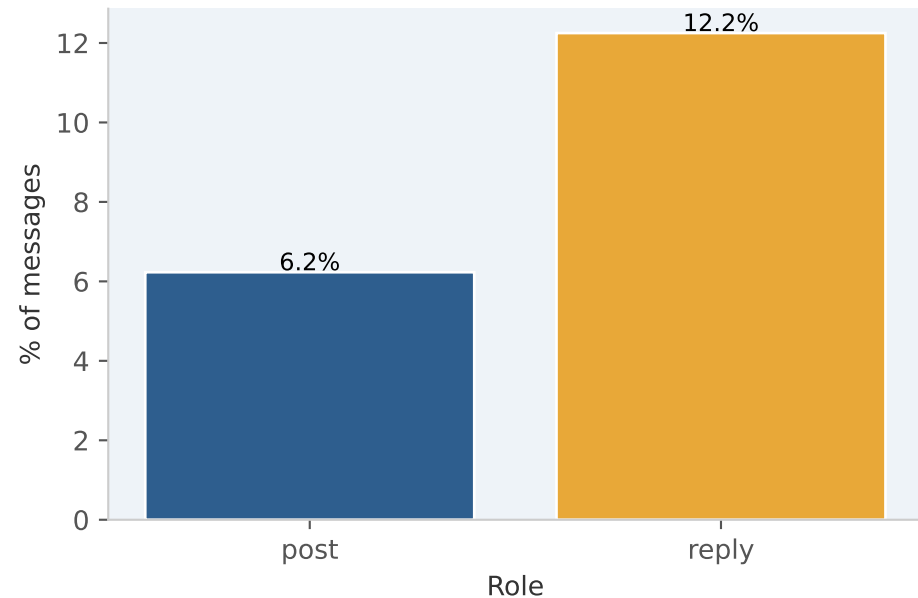


Replies

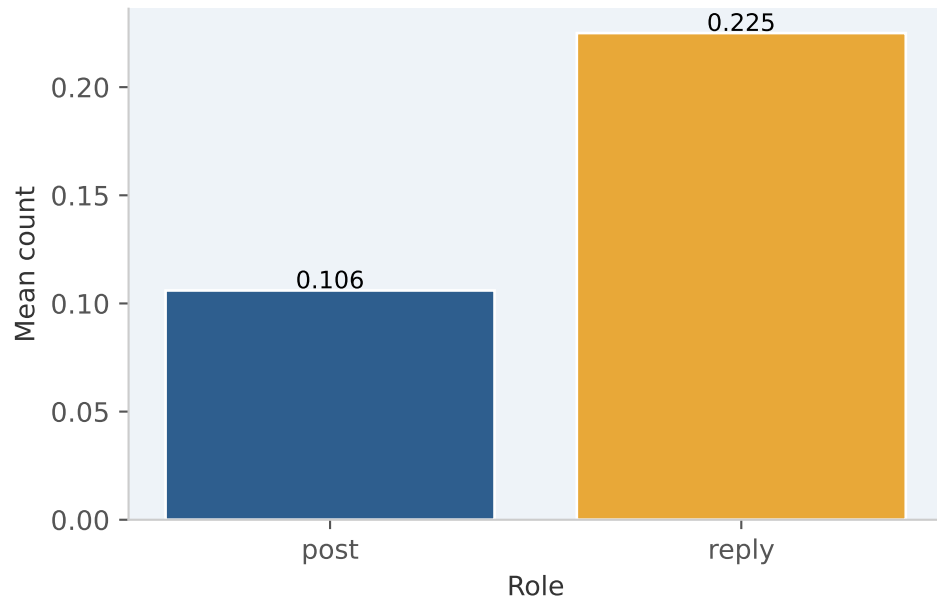


Emoji Use by Message Role

% Messages Containing Emoji



Mean Emoji Count per Message



Top 20 Most Frequent Emojis — Posts vs Replies

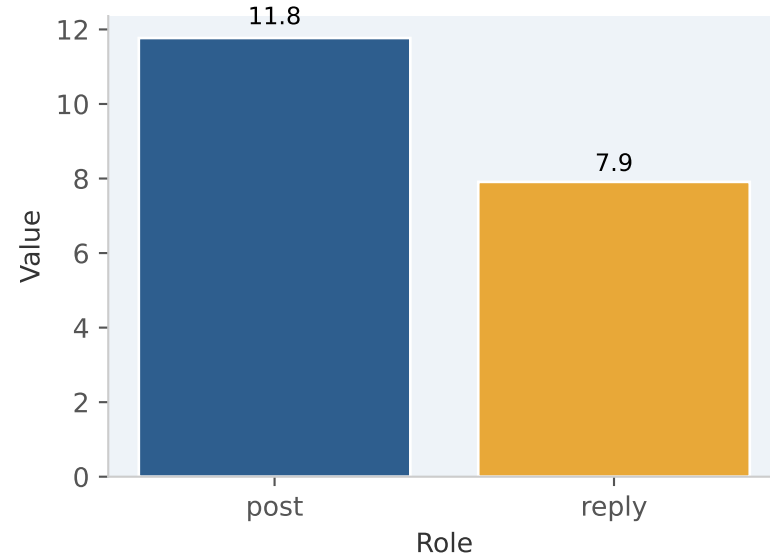
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1	four leaf clover	28	13.08	four leaf clover	452	12.13
2	winking face	12	5.61	winking face	266	7.14
3	puzzle piece	12	5.61	smiling face with open hands	244	6.55
4	loudly crying face	11	5.14	upside-down face	231	6.2
5	grinning face with sweat drops	9	4.21	smiling face with smiling eyes	180	4.83
6	upside-down face	7	3.27	grinning face with sweat drops	158	4.24
7	thinking face	6	2.8	thumbs up	106	2.85
8	smiling face with smiling eyes	6	2.8	slightly smiling face	92	2.47
9	smiling face with open hands	6	2.8	beaming face with smiling eyes	90	2.42
10	see-no-evil monkey	5	2.34	smiling face	76	2.04
11	slightly smiling face	5	2.34	face with rolling eyes	65	1.74
12	zany face	4	1.87	thinking face	62	1.66
13	pensive face	4	1.87	crying face	60	1.61
14	ZZZ	4	1.87	rolling on the floor laughing	60	1.61
15	crying face	4	1.87	face blowing a kiss	55	1.48
16	sleepy face	4	1.87	folded hands	54	1.45
17	raising hands	4	1.87	pensive face	49	1.32
18	copyright	3	1.4	flexed biceps	45	1.21
19	beaming face with smiling eyes	3	1.4	face with tears of joy	44	1.18
20	sad but relieved face	3	1.4	sunflower	42	1.13

Top 20 Most Frequent Words — Posts vs Replies

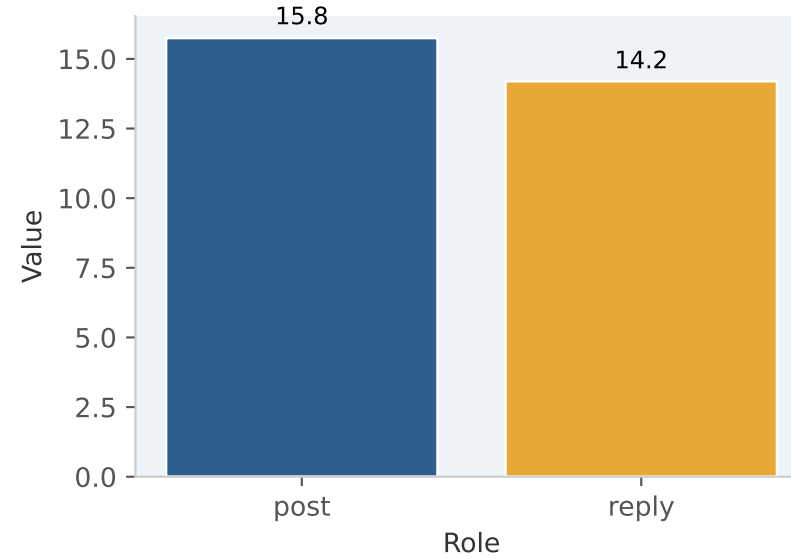
#	Post word	Count	%	Reply word	Count	%
1	mijn	3979	2.333	mijn	13514	1.481
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4	depressie	1310	0.768	weer	7538	0.826
5	veel	1245	0.73	goed	7537	0.826
6	naar	1233	0.723	veel	6844	0.75
7	hier	1084	0.636	naar	6093	0.668
8	goed	1080	0.633	dus	5616	0.615
9	heel	1076	0.631	over	5562	0.61
10	over	1018	0.597	daar	5084	0.557
11	hoe	956	0.561	hoe	5052	0.554
12	geen	952	0.558	gaat	4809	0.527
13	gaan	893	0.524	iets	4597	0.504
14	wil	884	0.518	geen	4586	0.503
15	moet	819	0.48	echt	4536	0.497
16	dus	791	0.464	toch	4529	0.496
17	mensen	776	0.455	doen	4489	0.492
18	doen	773	0.453	depressie	4461	0.489
19	leven	759	0.445	hier	4455	0.488
20	voel	729	0.427	gaan	4397	0.482

Sentence Structure — Initial Posts vs Replies

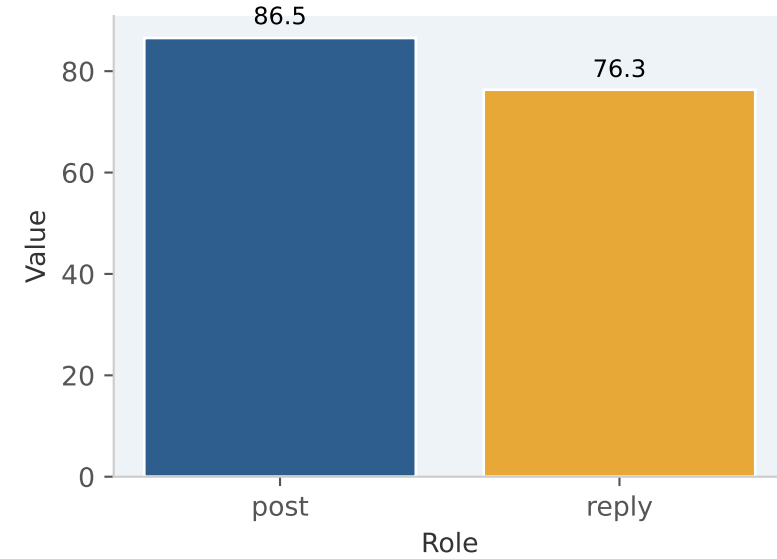
Avg Sentences per Message



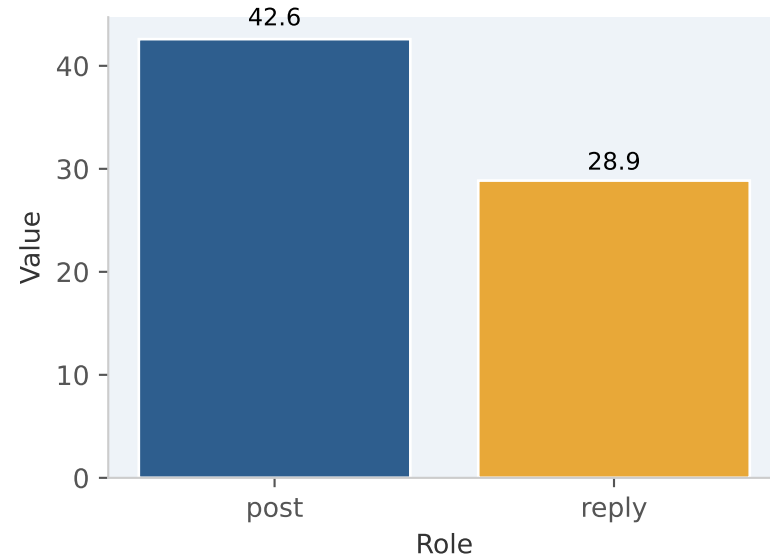
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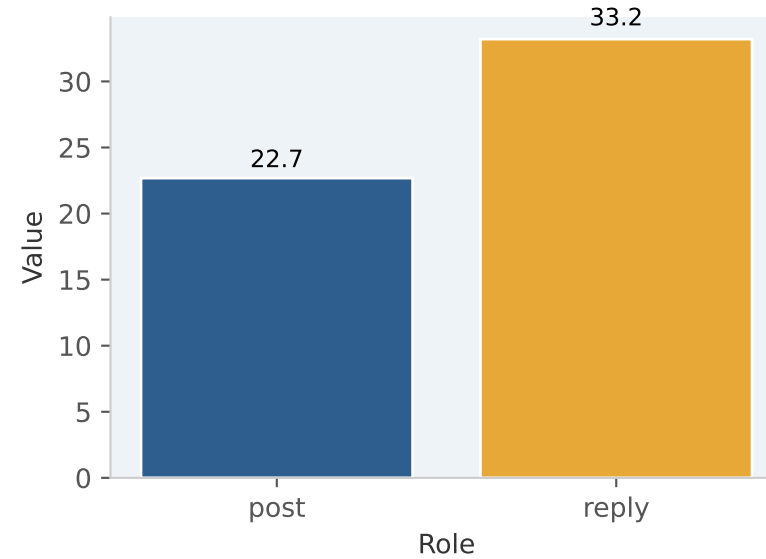
Avg Chars per Sentence



% Messages with '?'



% Messages with '!'



% Messages with '...'



2. EDA - Multi-Posters Only

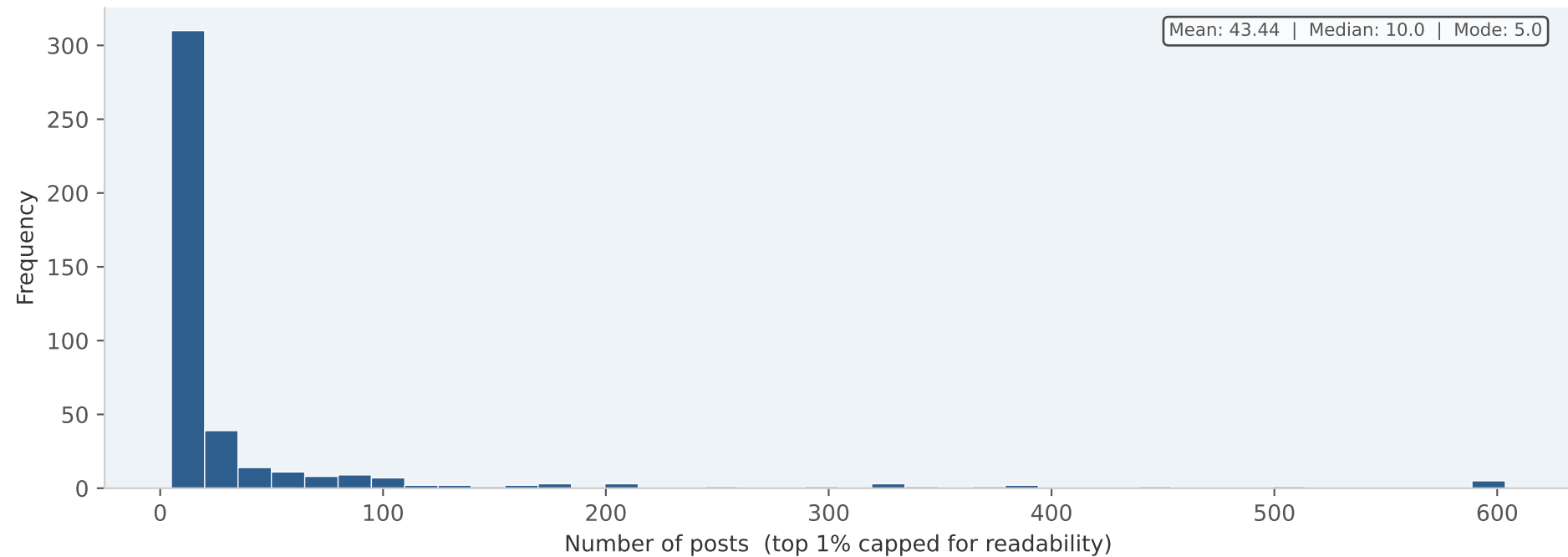
Depression Connect Forum

Exploratory Data Analysis Report — Multi-Posters Only

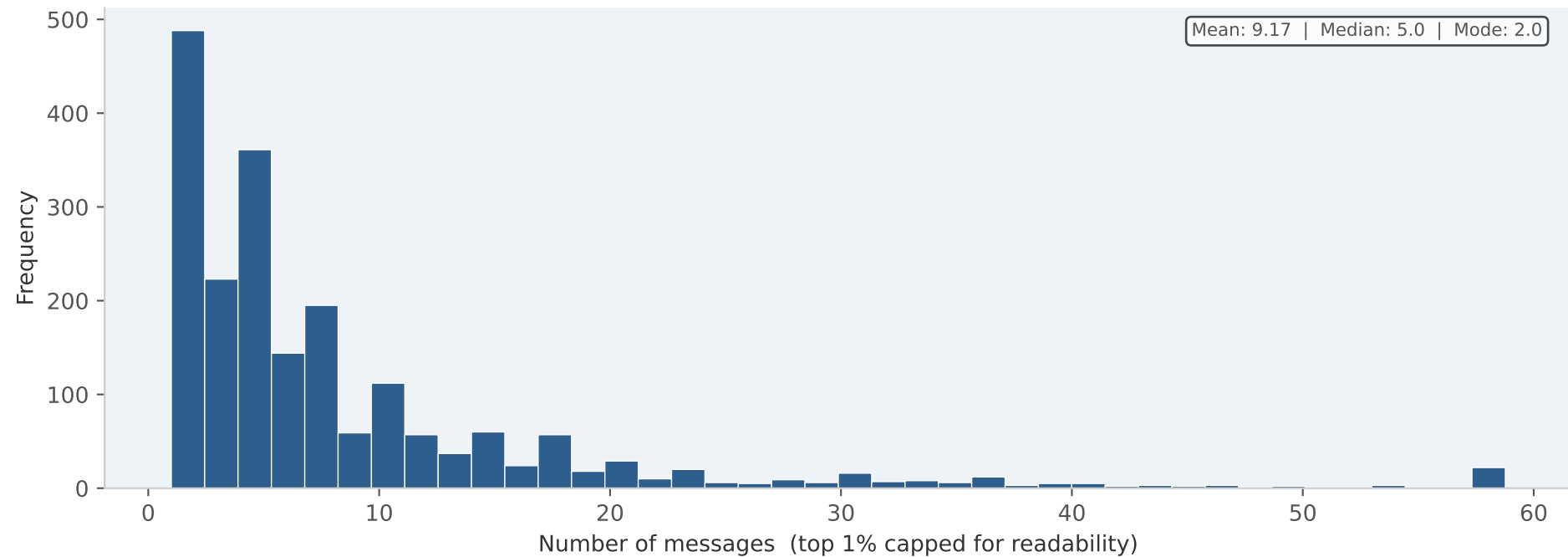
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Distribution: Posts per User

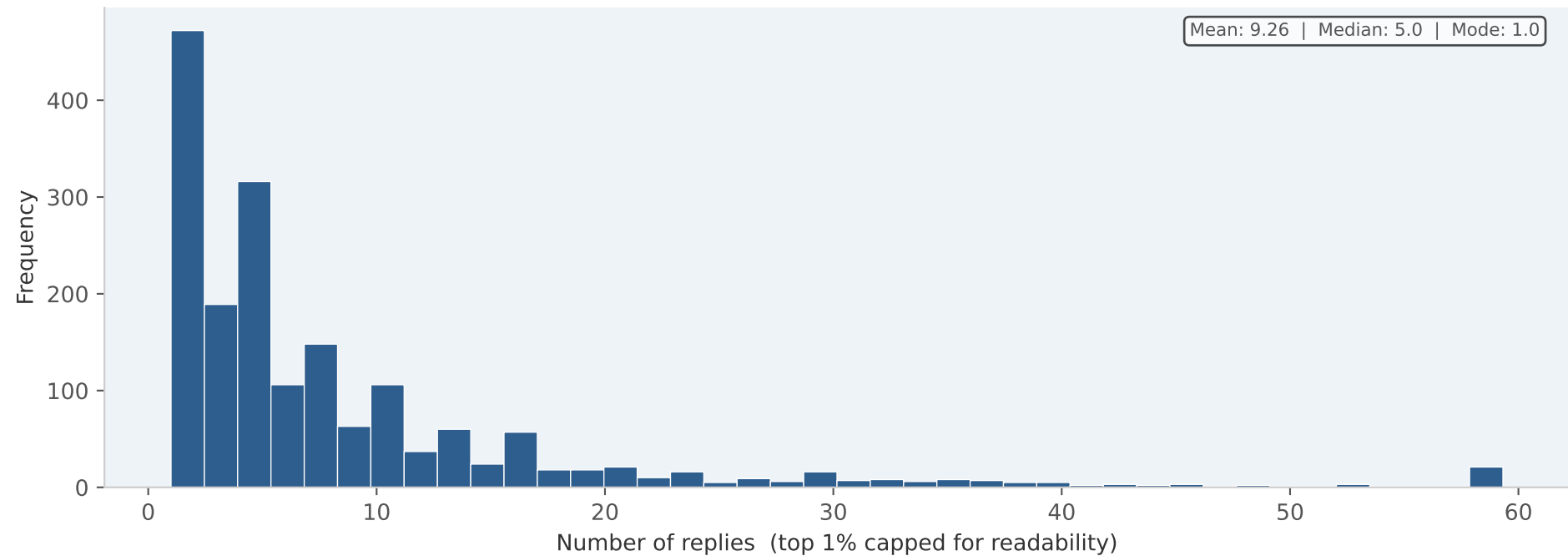


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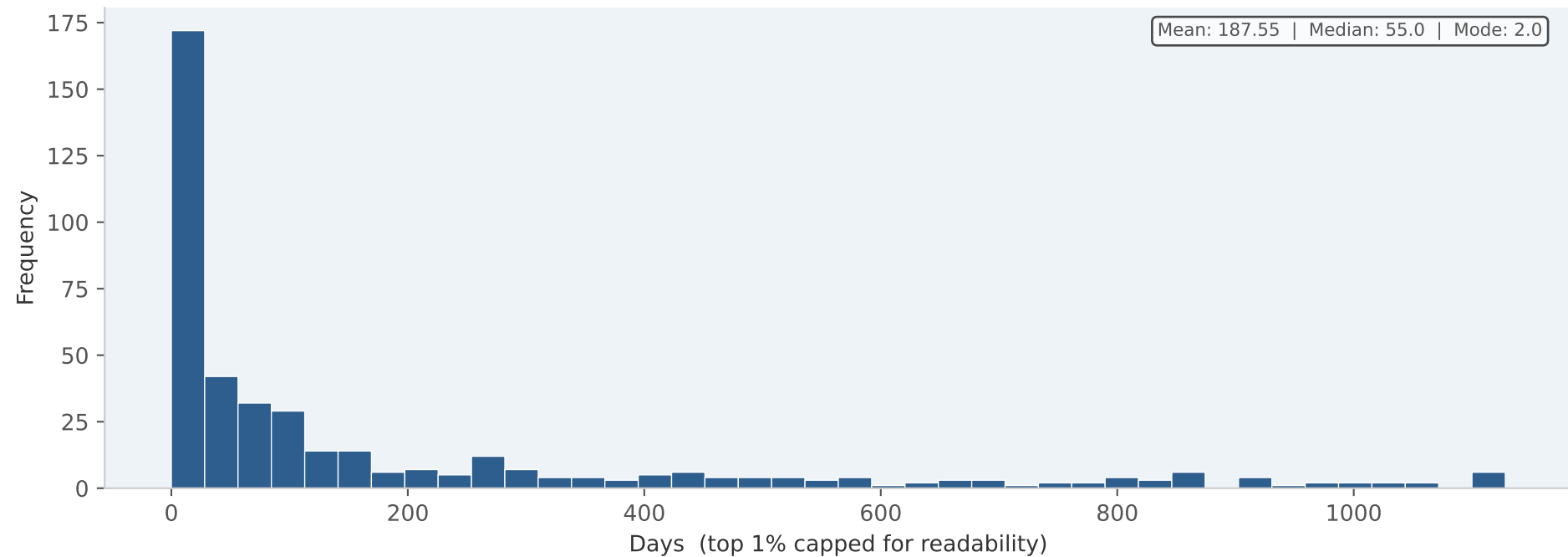


Distribution: Replies per Thread (excluding opening post)

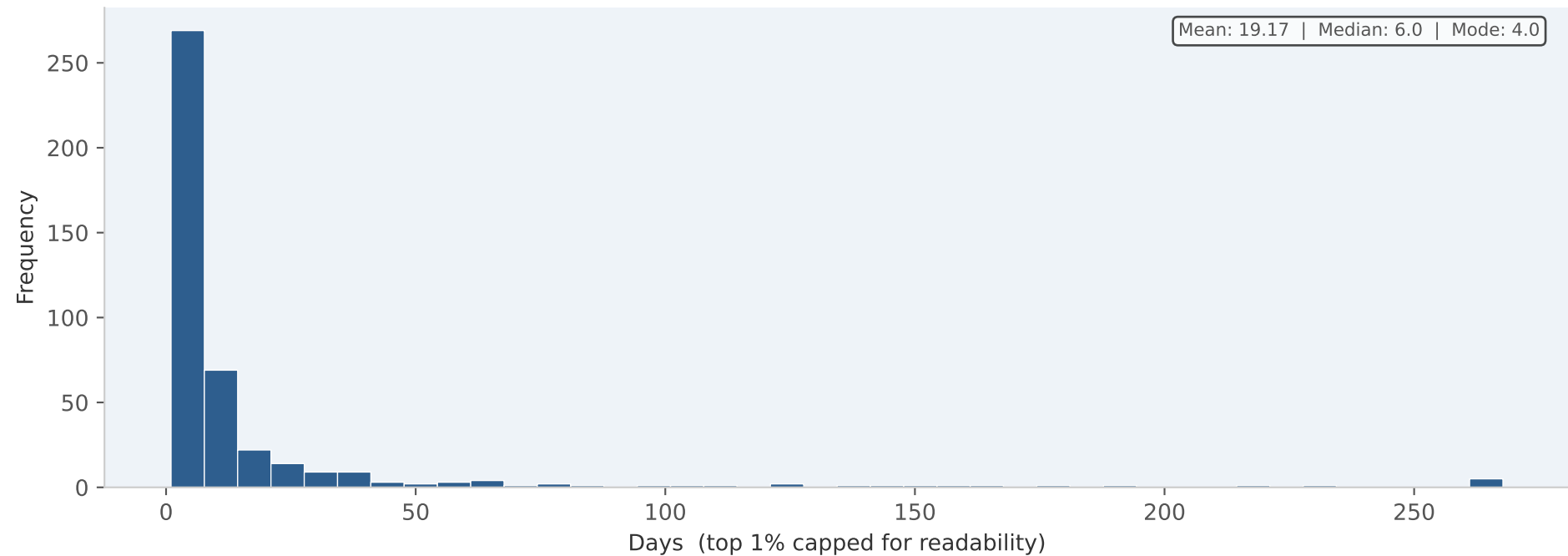
Mean: 9.26 | Median: 5.0 | Mode: 1.0



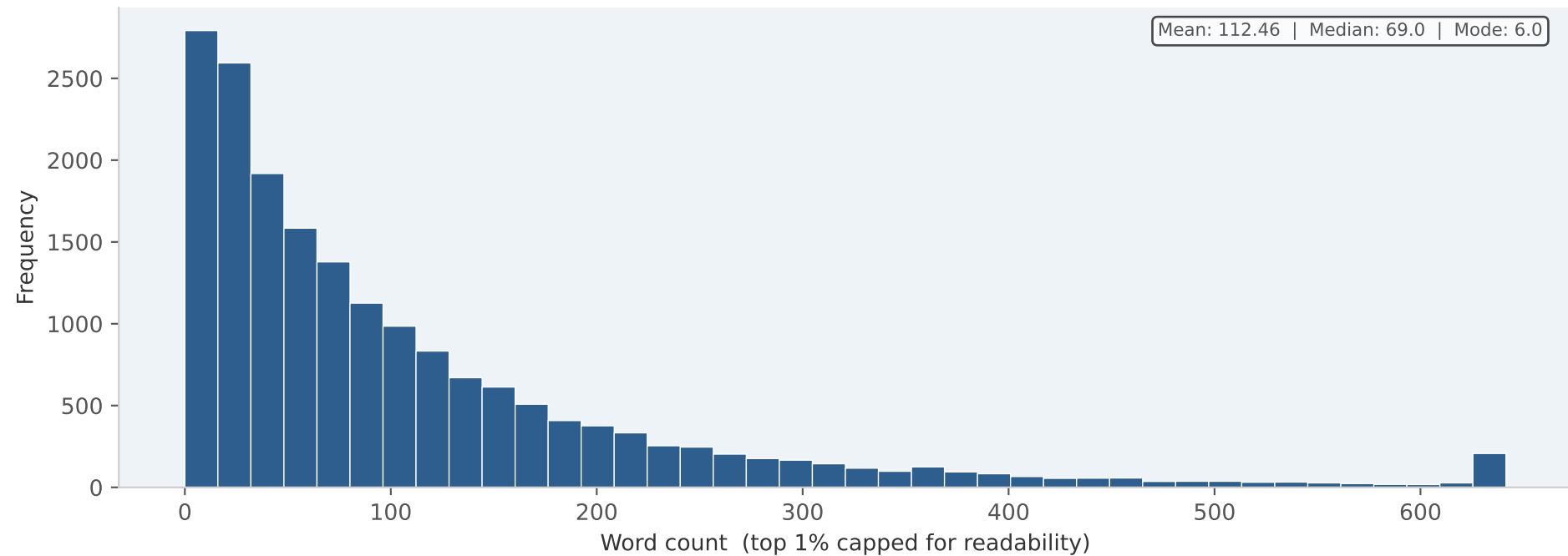
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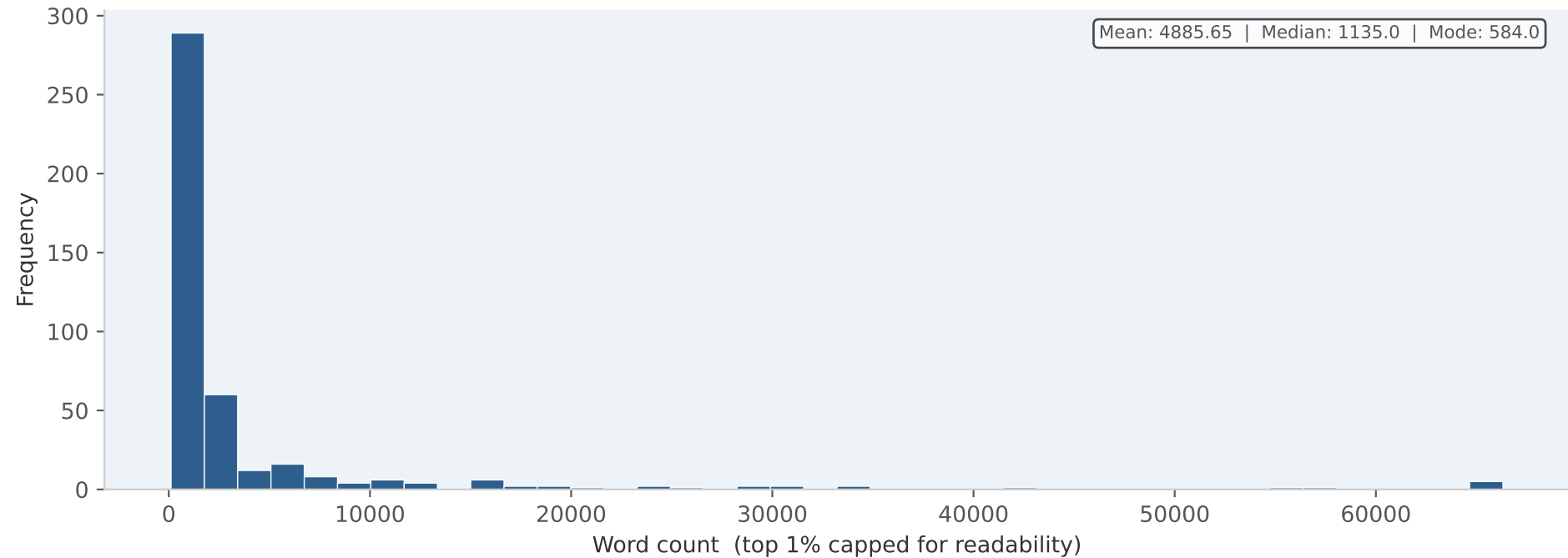
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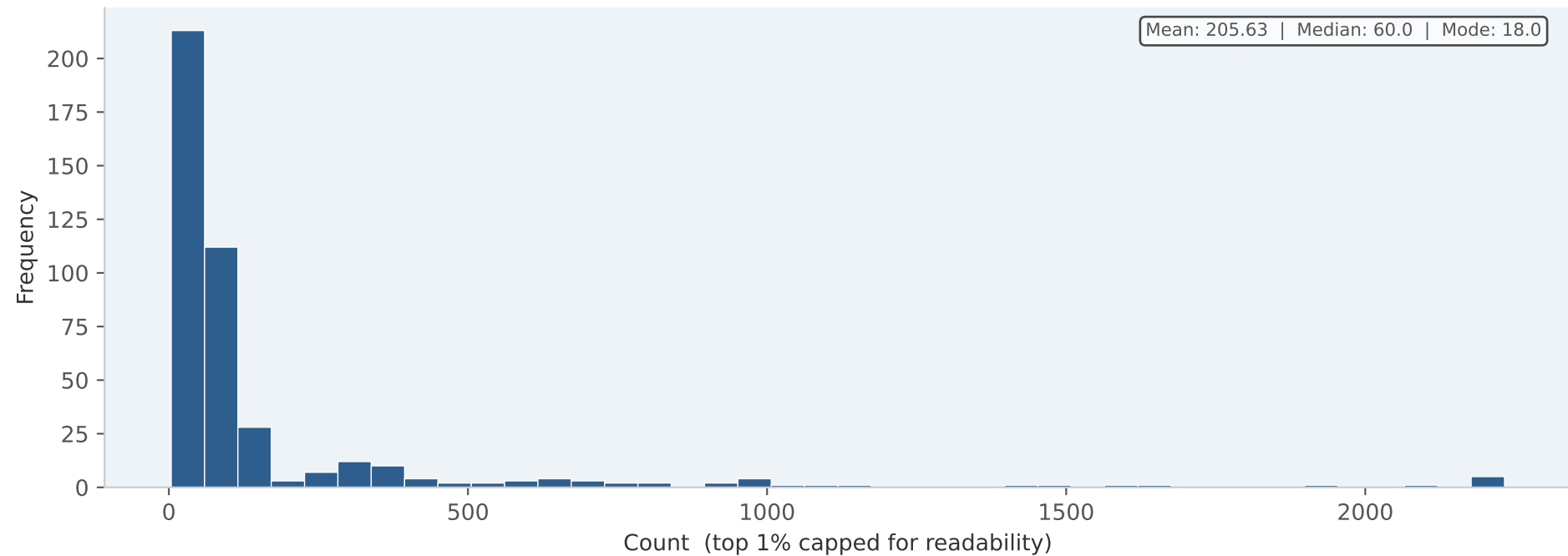
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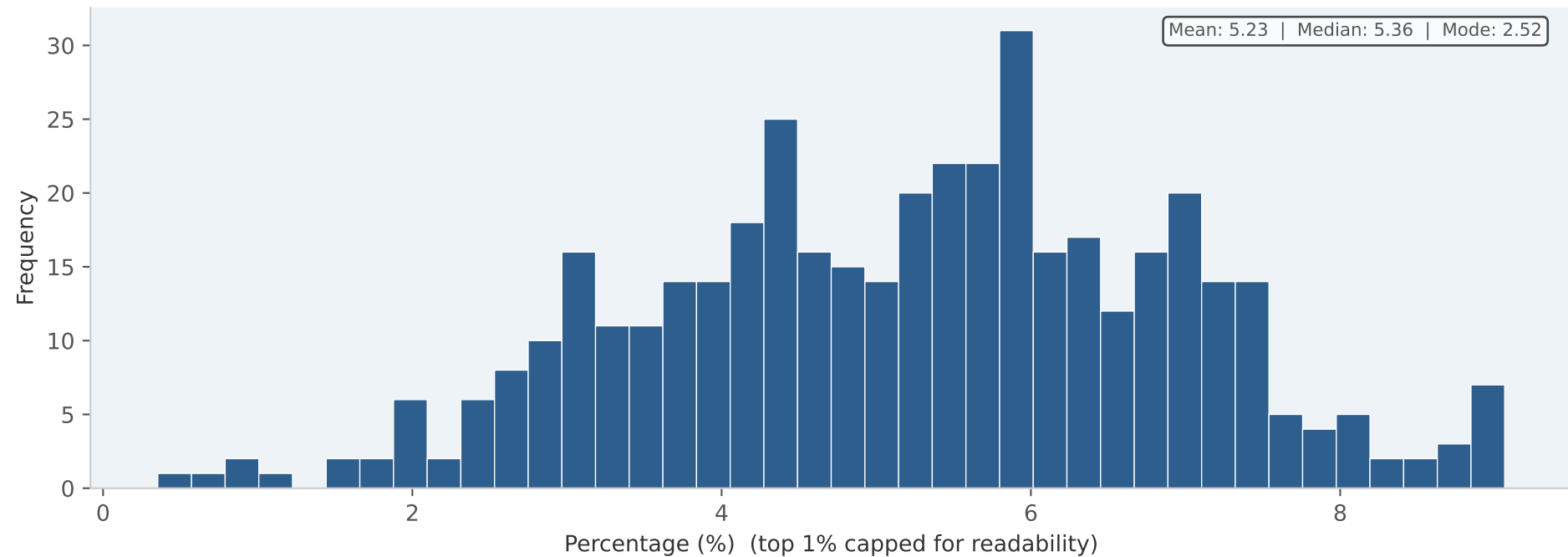
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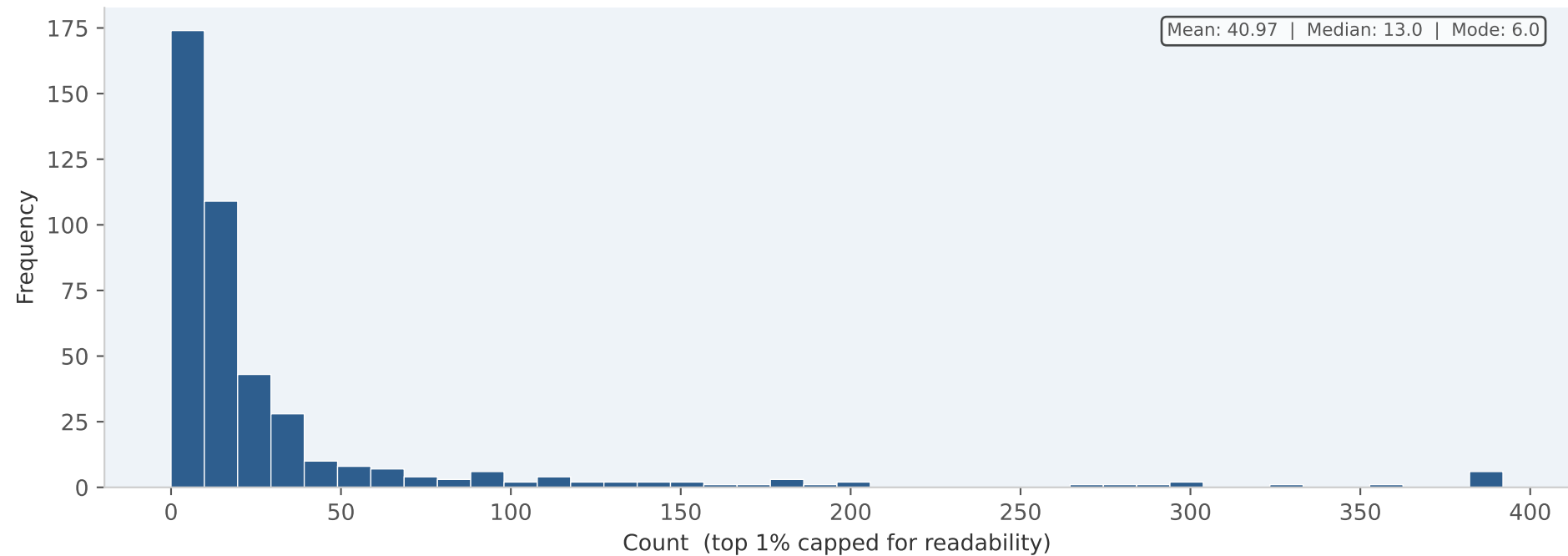
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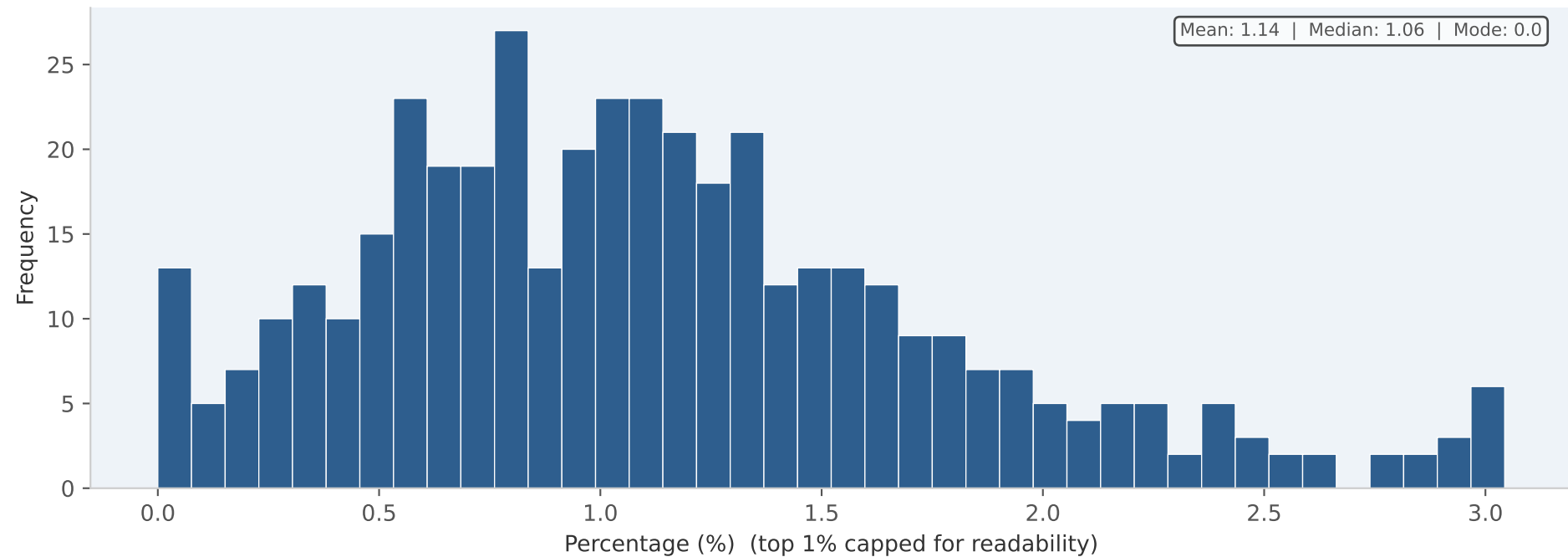
Distribution: 'ik' as % of Total Words per User



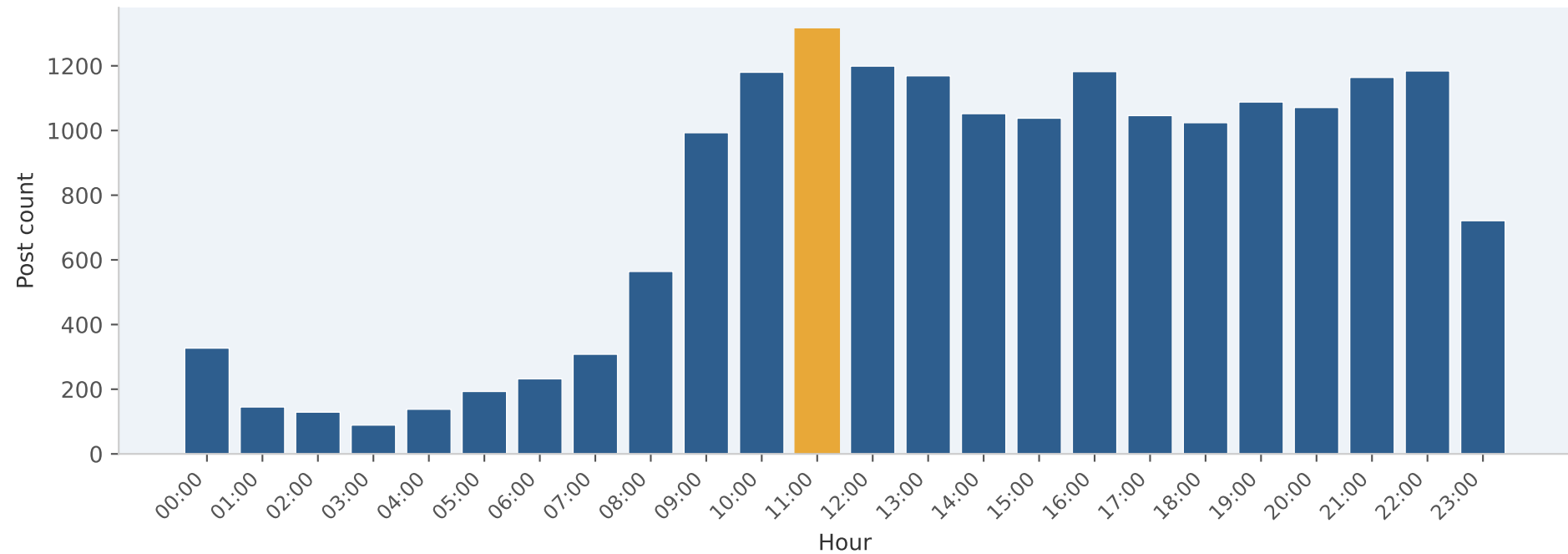
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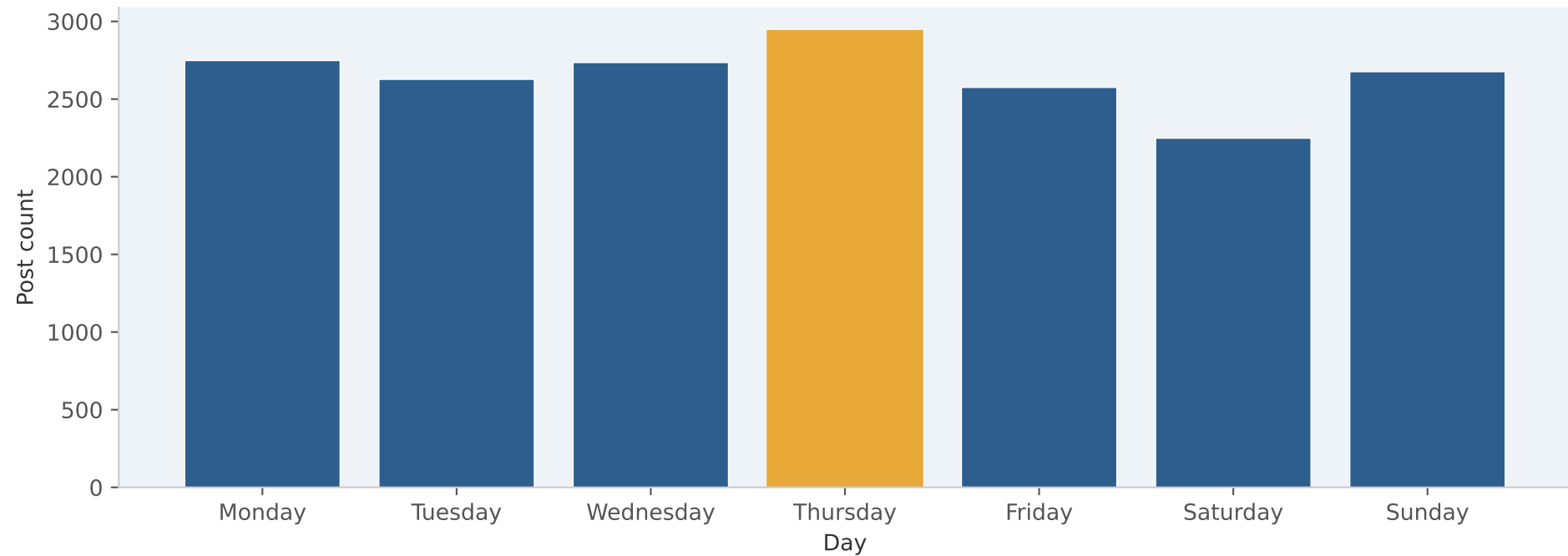
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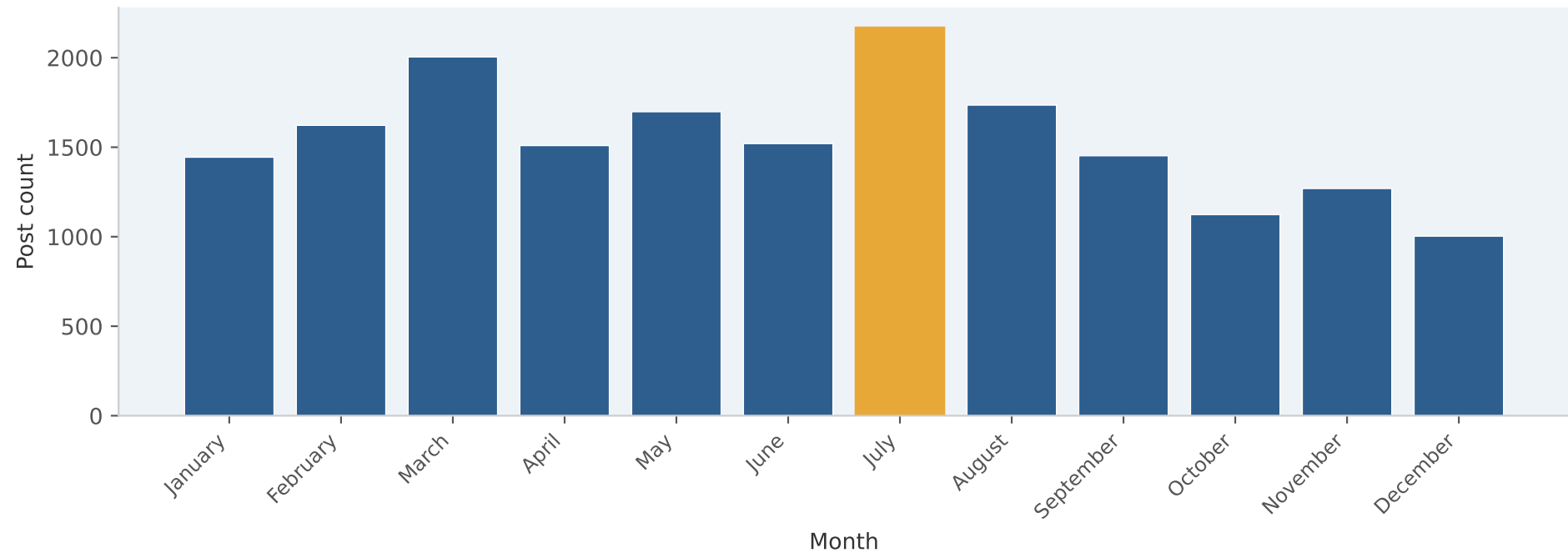
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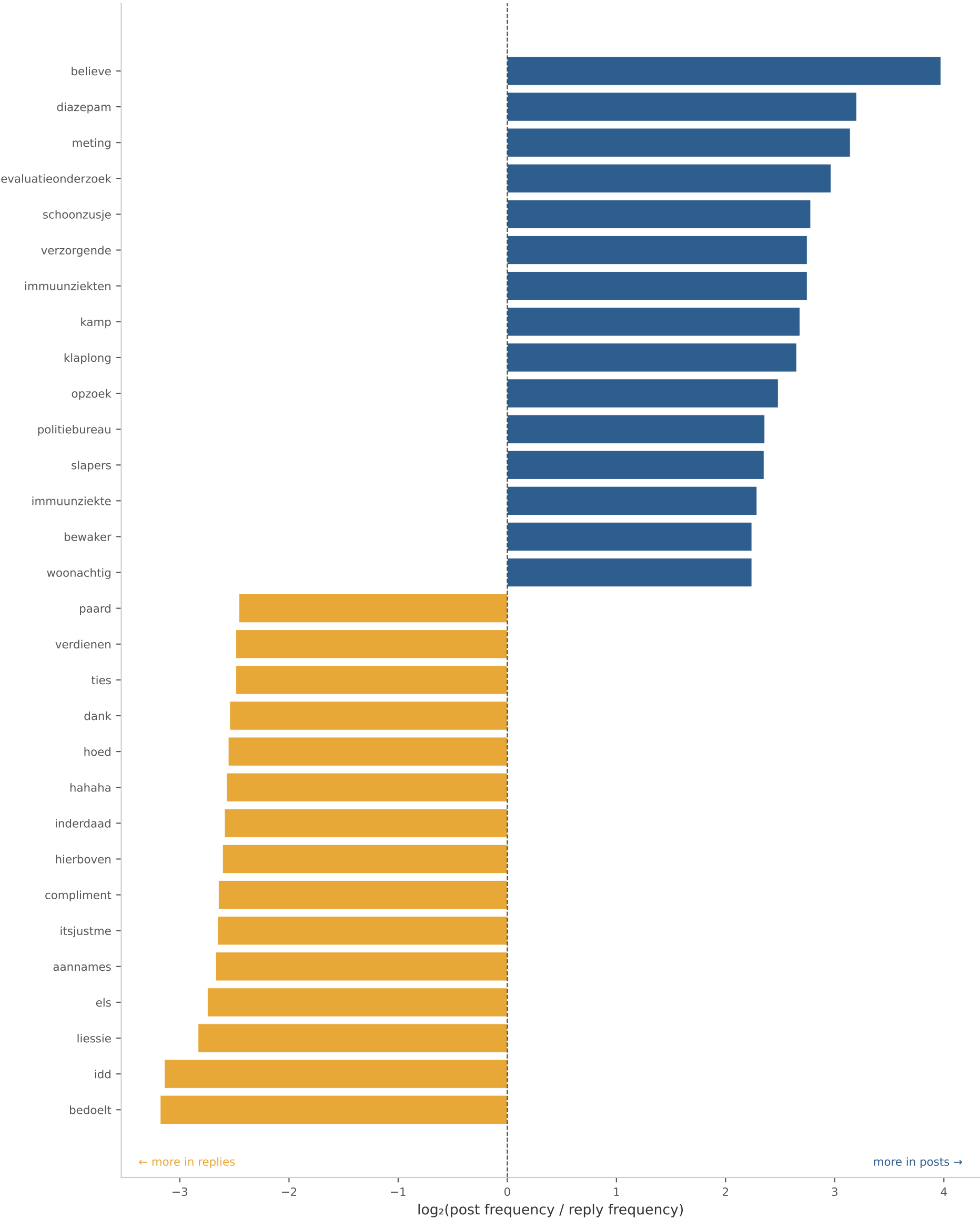


Popular Months



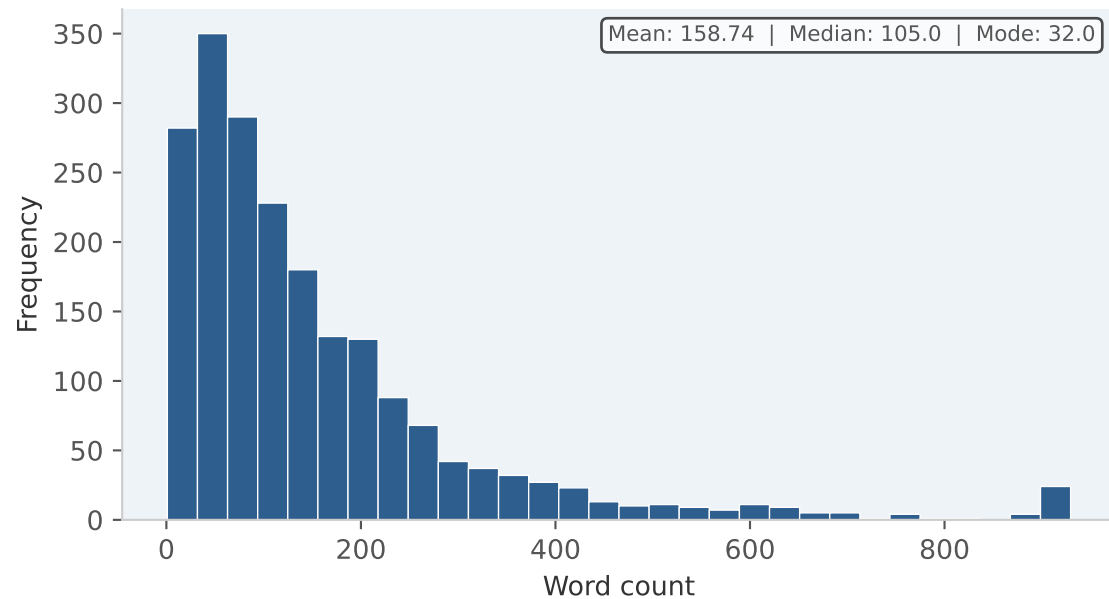
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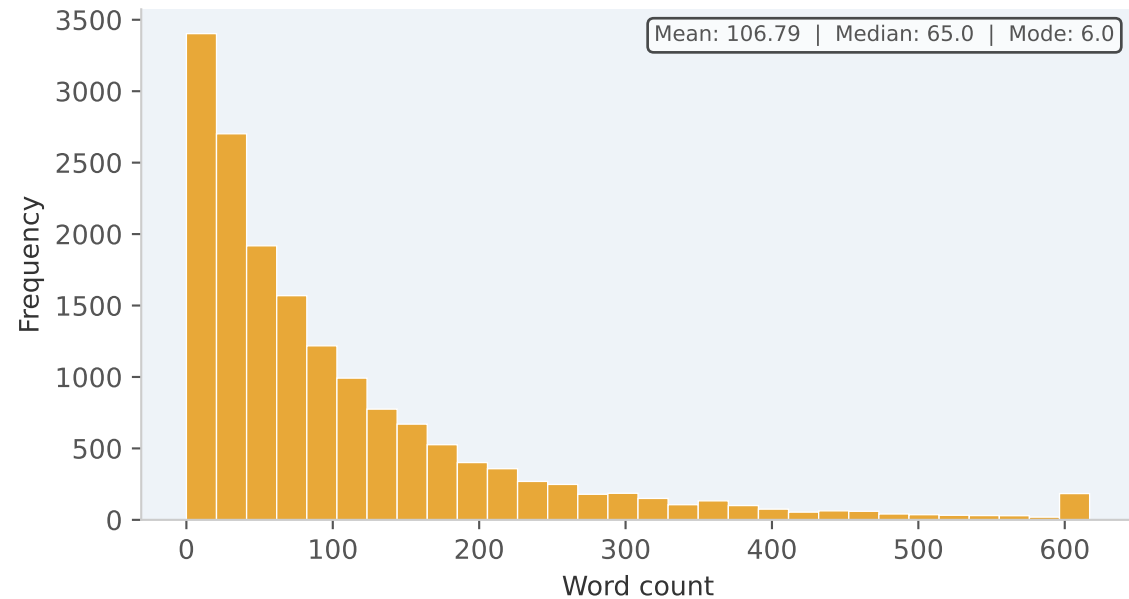


Words per Message: Initial Posts vs Replies

Initial Posts

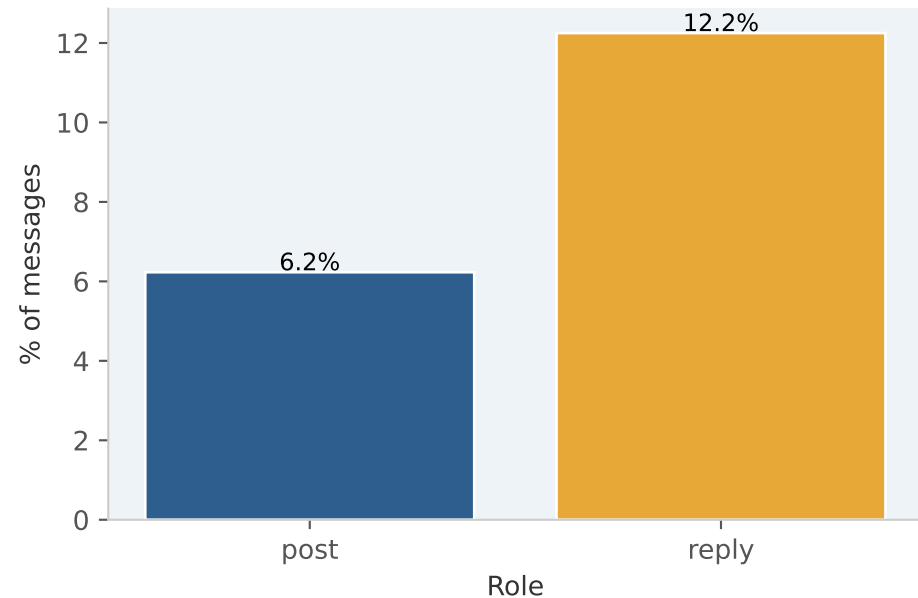


Replies

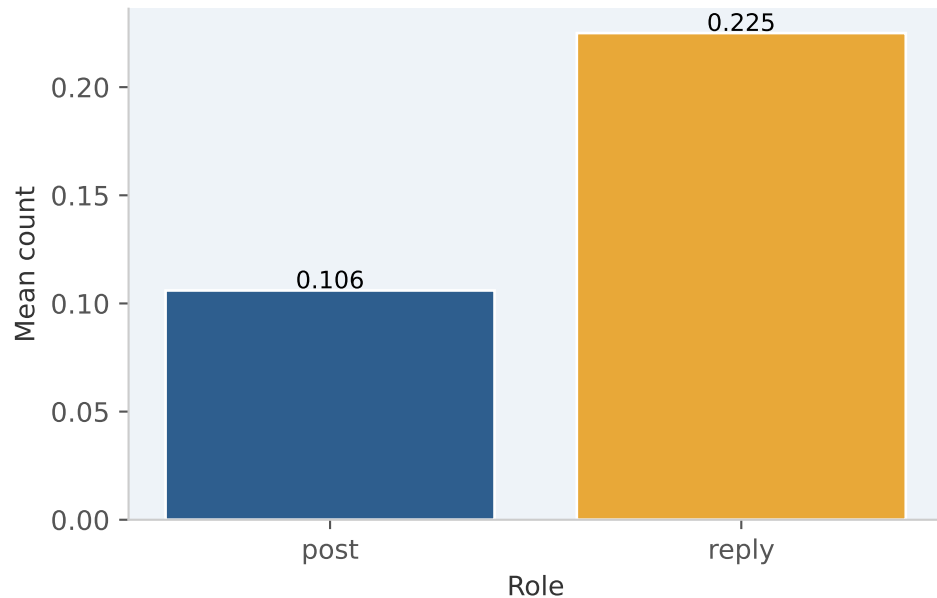


Emoji Use by Message Role

% Messages Containing Emoji



Mean Emoji Count per Message



Top 20 Most Frequent Emojis — Posts vs Replies

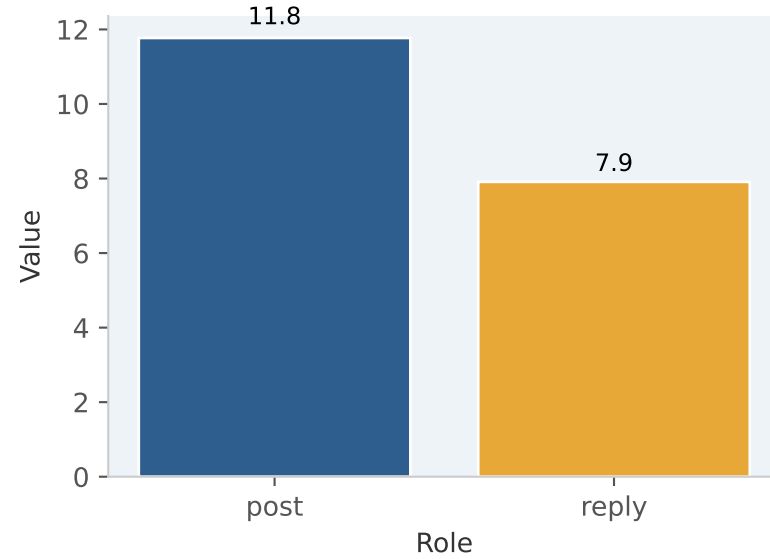
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14	ZZZ	4	1.87	rolling on the floor laughing	60	1.61
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Top 20 Most Frequent Words — Posts vs Replies

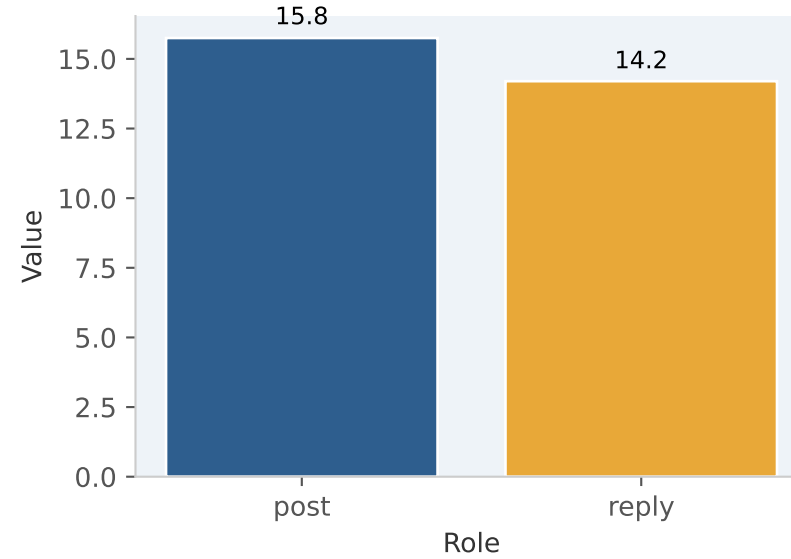
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Sentence Structure — Initial Posts vs Replies

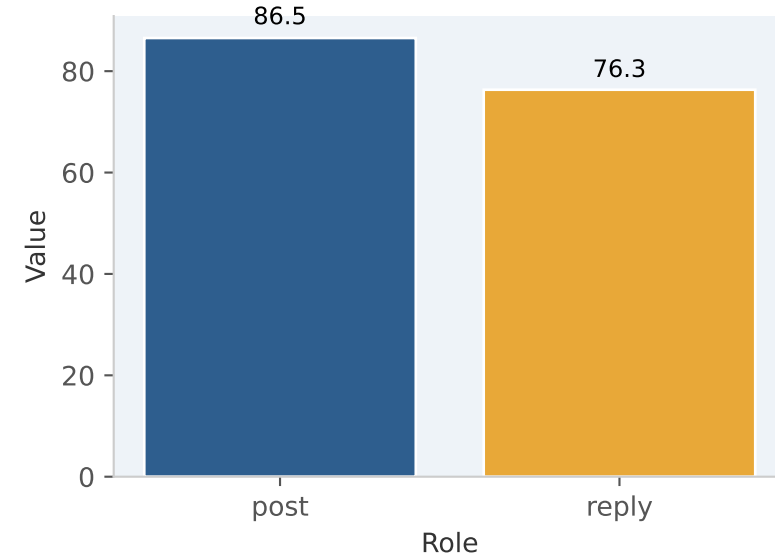
Avg Sentences per Message



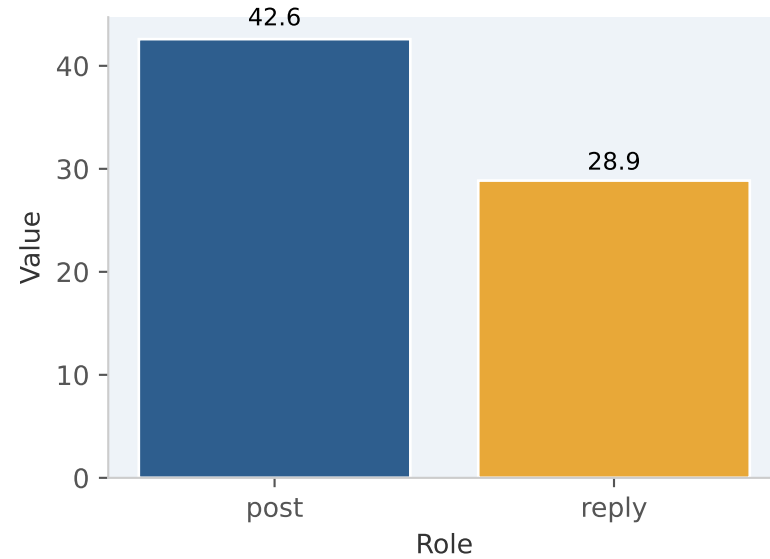
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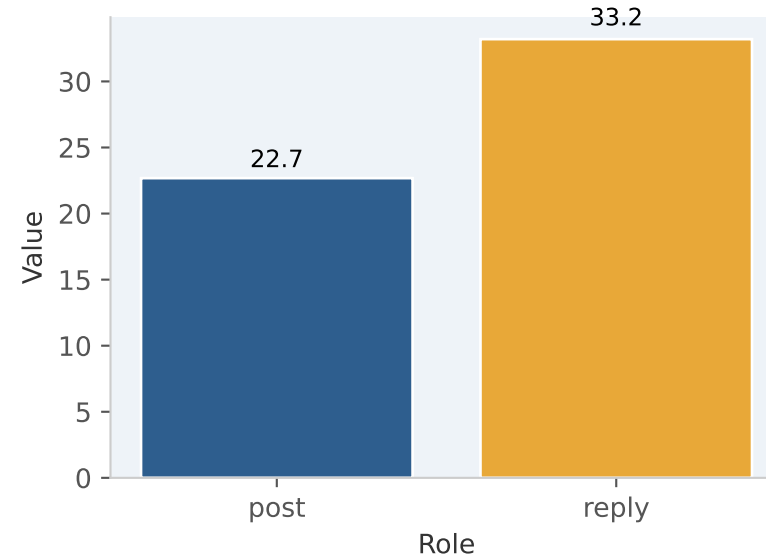
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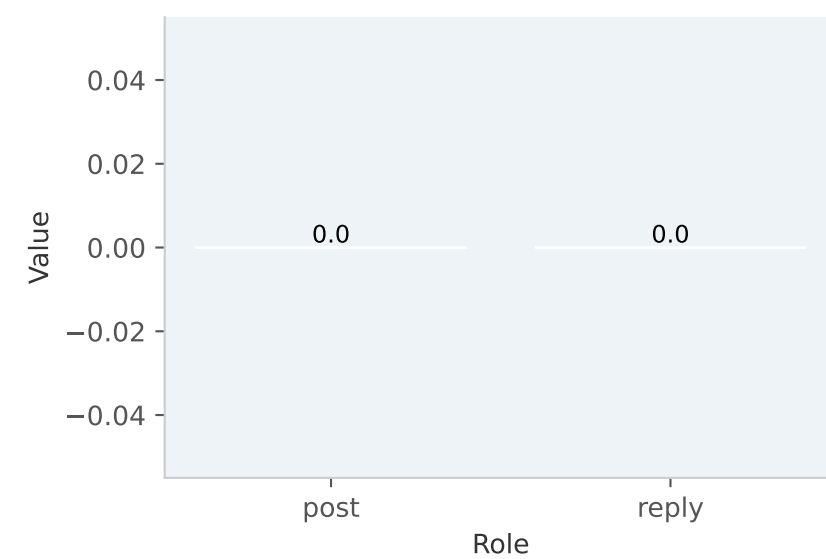
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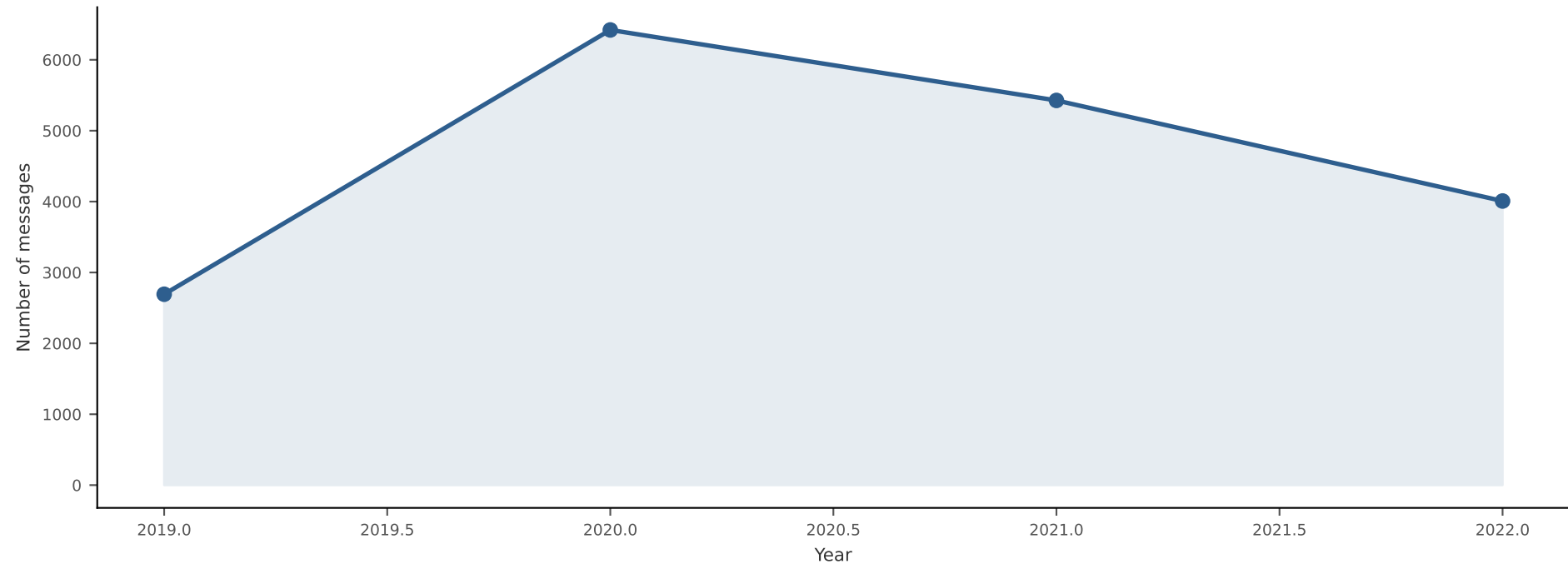
3. Time Series & CDS Prevalence

Depression Connect Forum

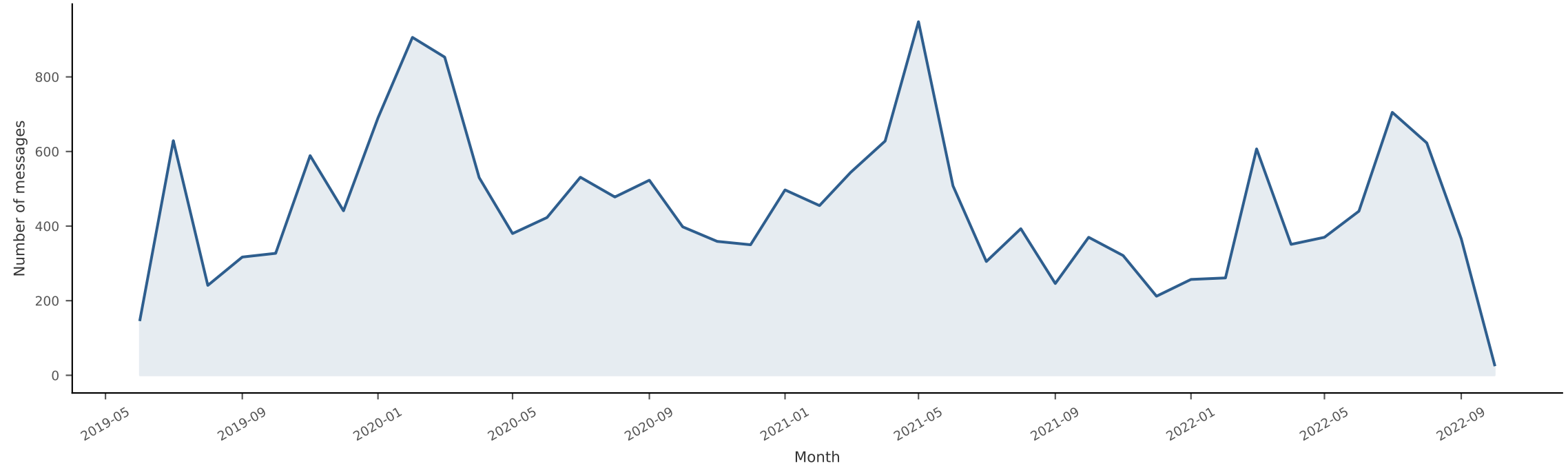
Exploratory Analysis — Time Series & CDS Prevalence

Section 1 — Forum Activity Over Time

Messages per Year

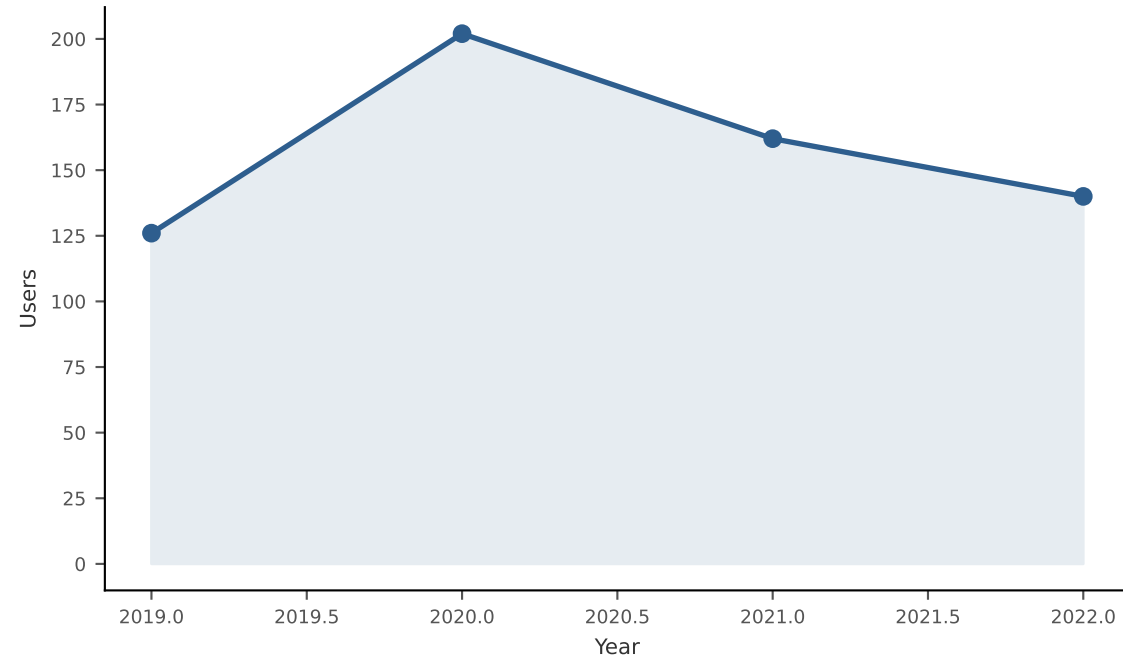


Messages per Month

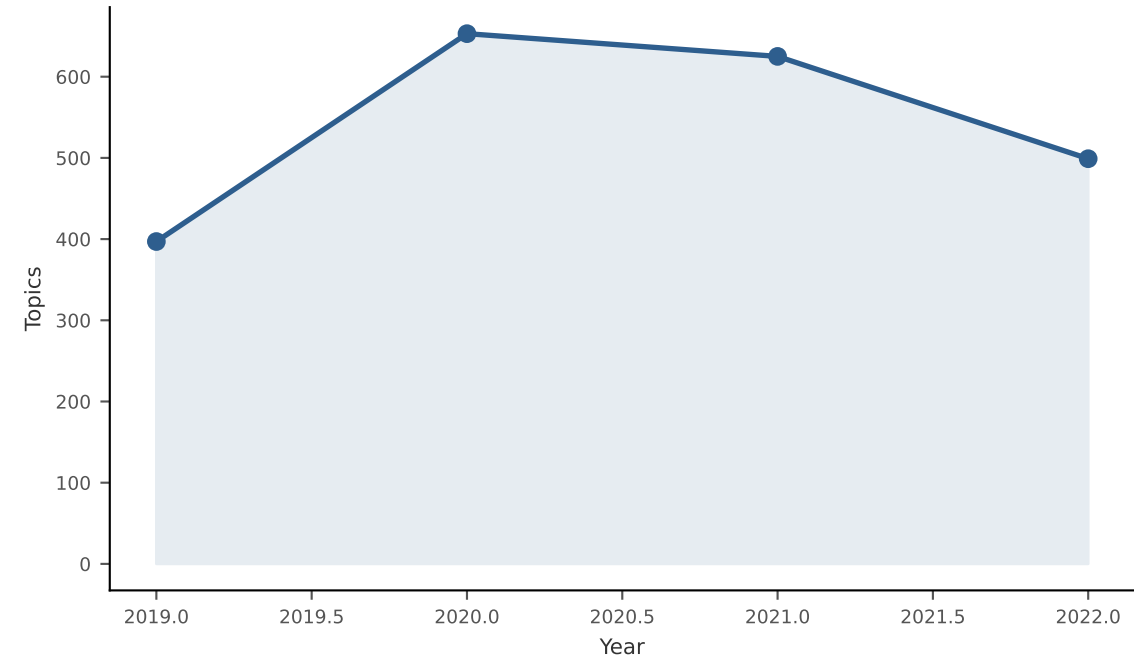


Forum Activity — Year-level Metrics

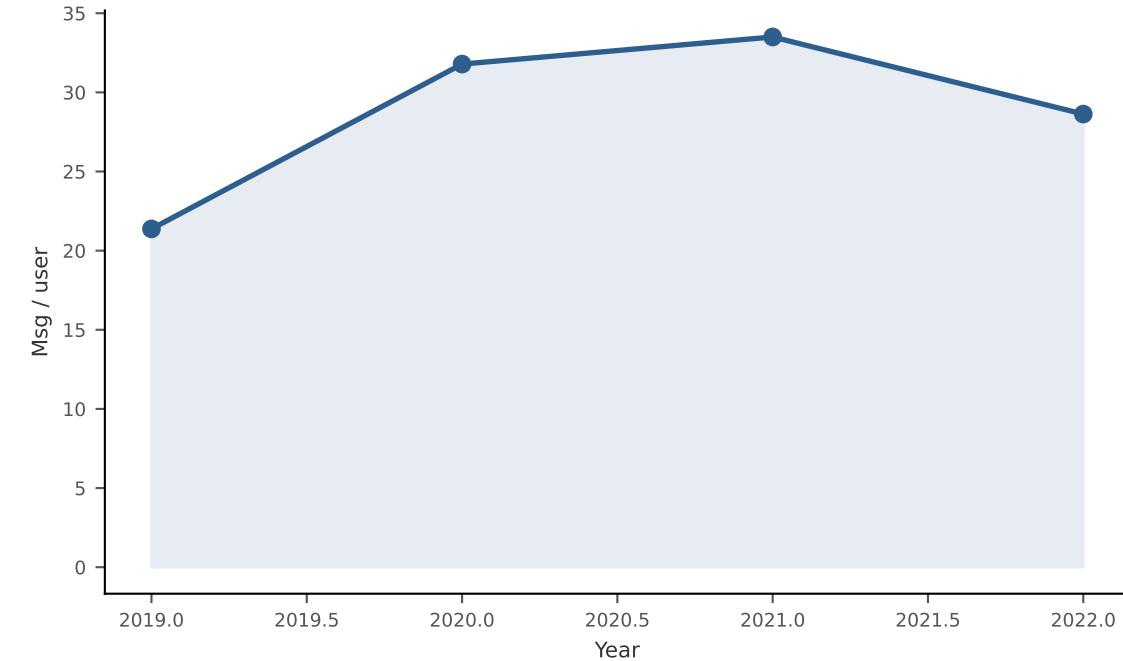
Unique Users per Year



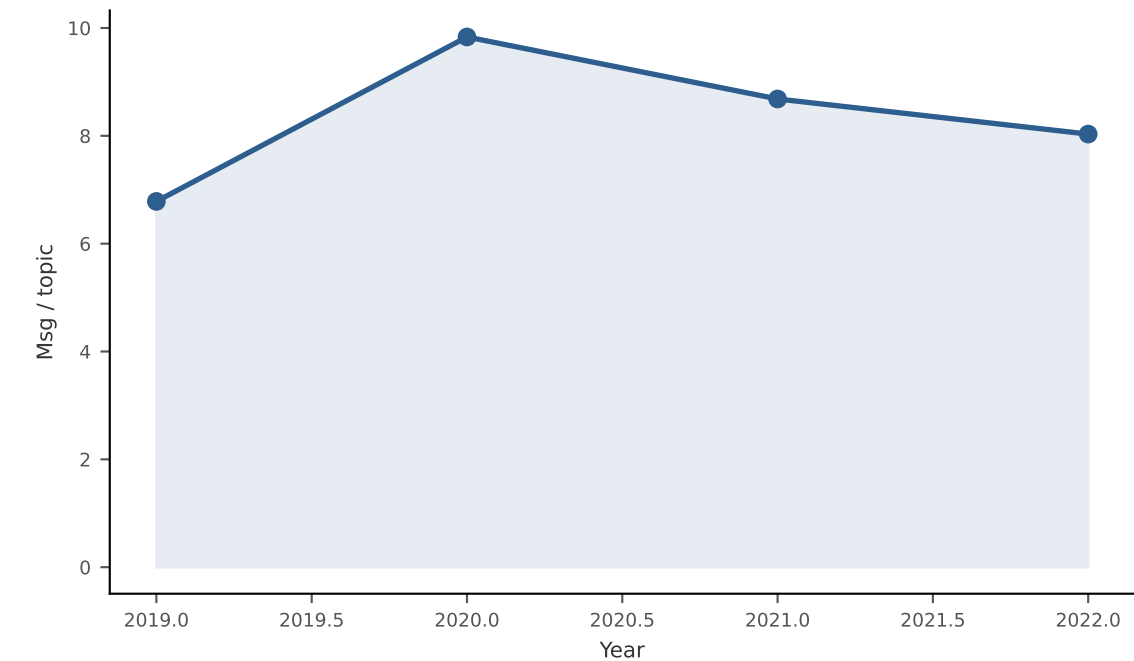
Unique Topics per Year



Messages per User per Year

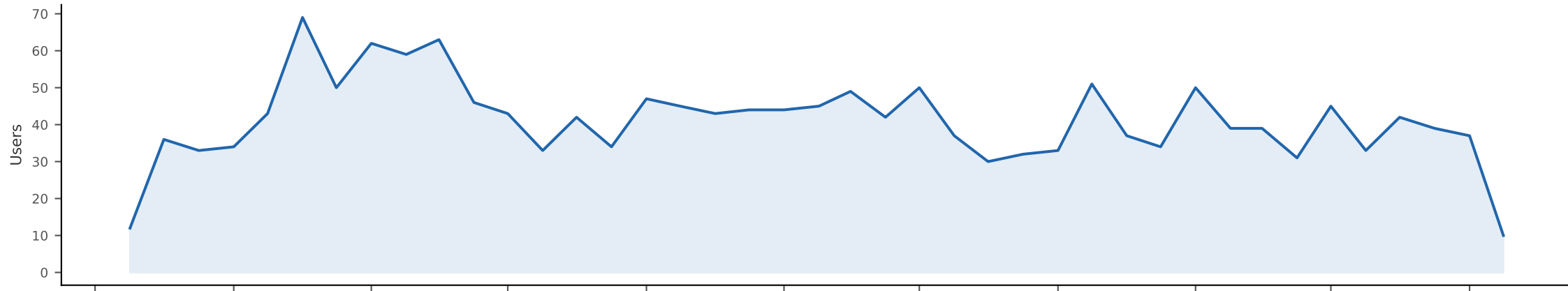


Messages per Topic per Year

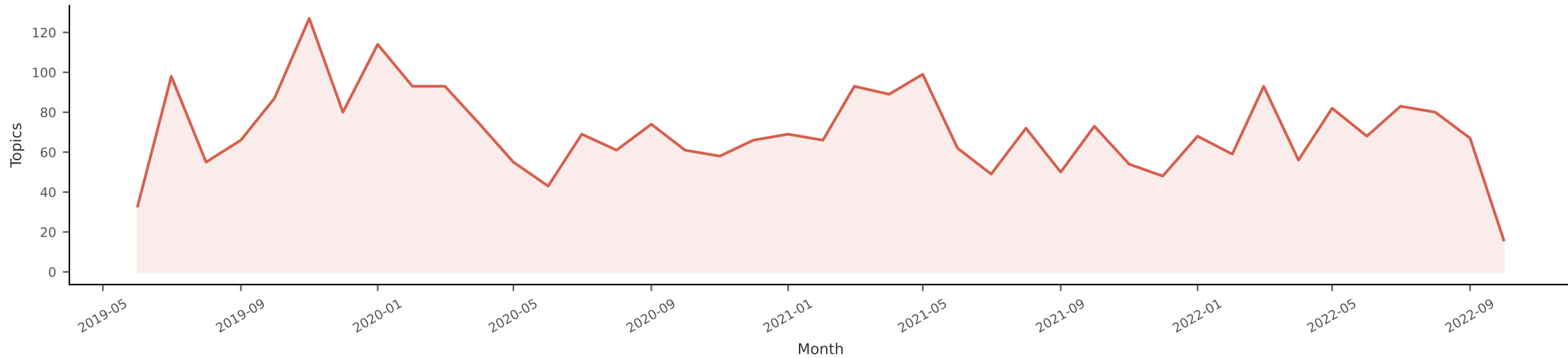


Forum Activity — Month-level Metrics

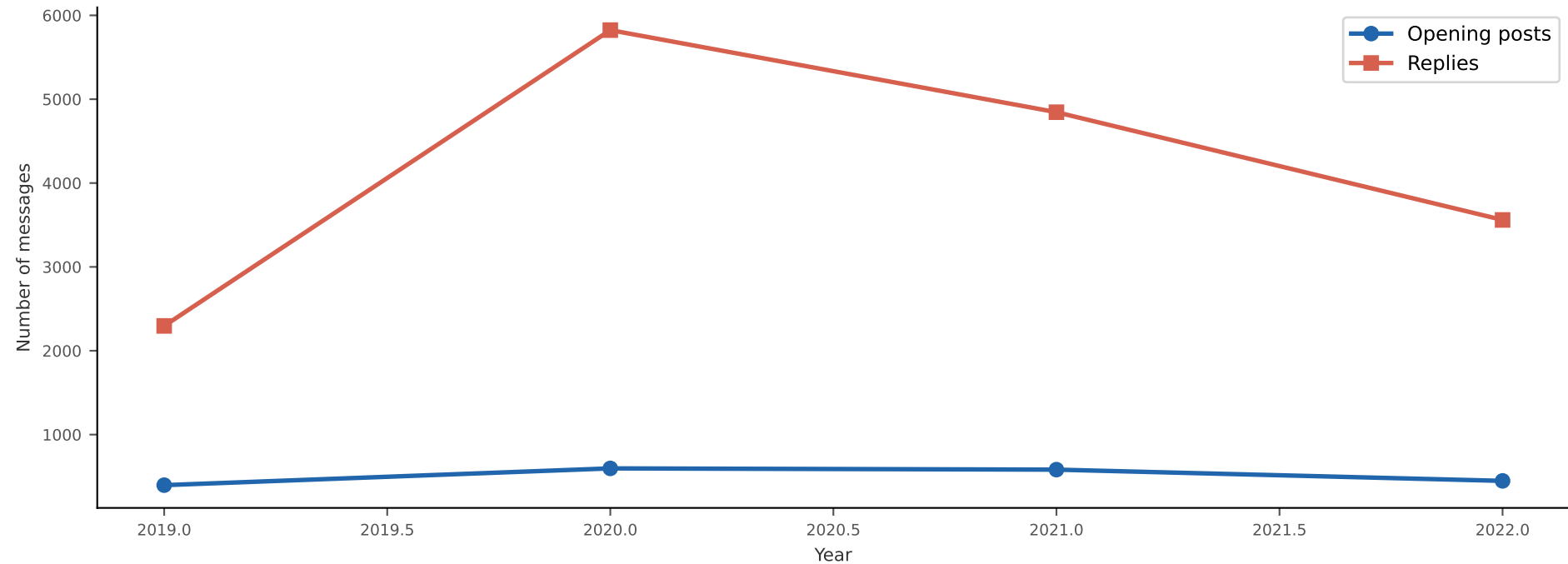
Unique Users per Month



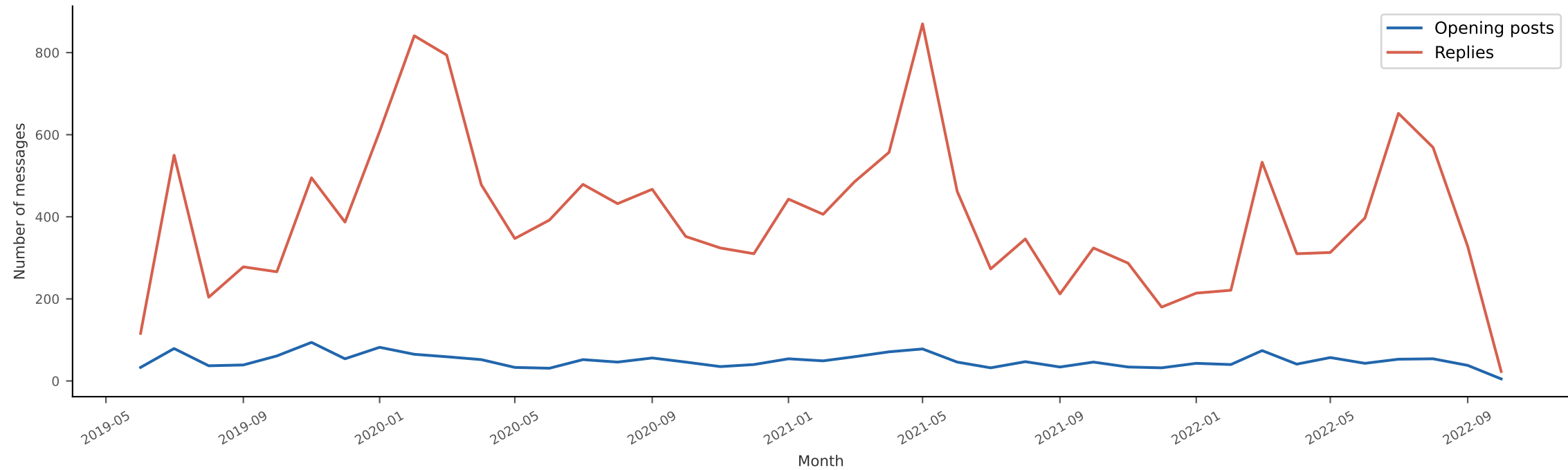
Unique Topics per Month



Message Volume by Role — Year Level



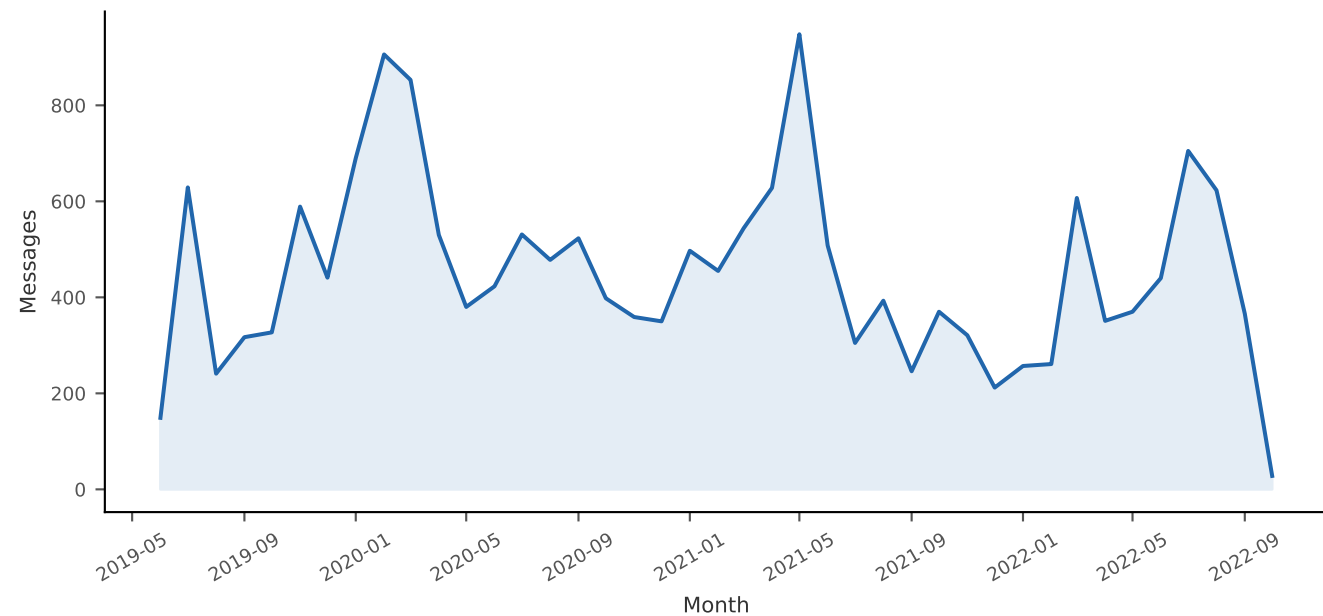
Message Volume by Role — Month Level



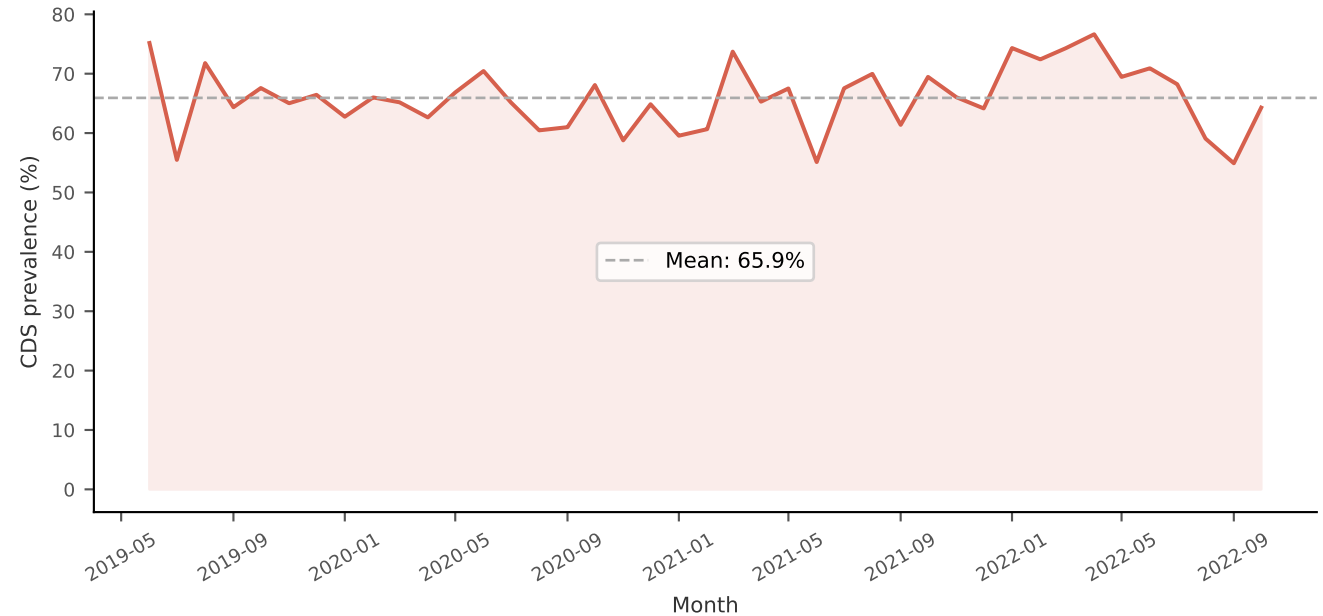
Section 2 — CDS Prevalence Over Time

Message Volume and CDS Prevalence Over Time (Monthly)

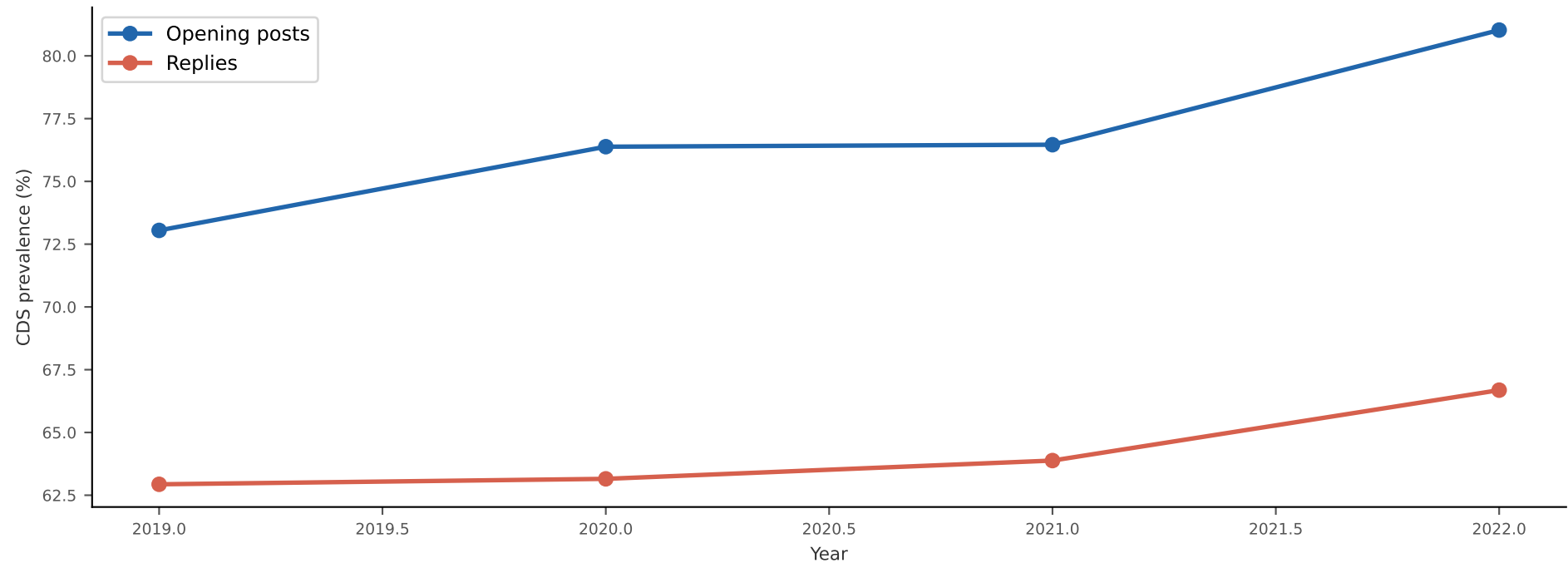
Monthly Message Volume



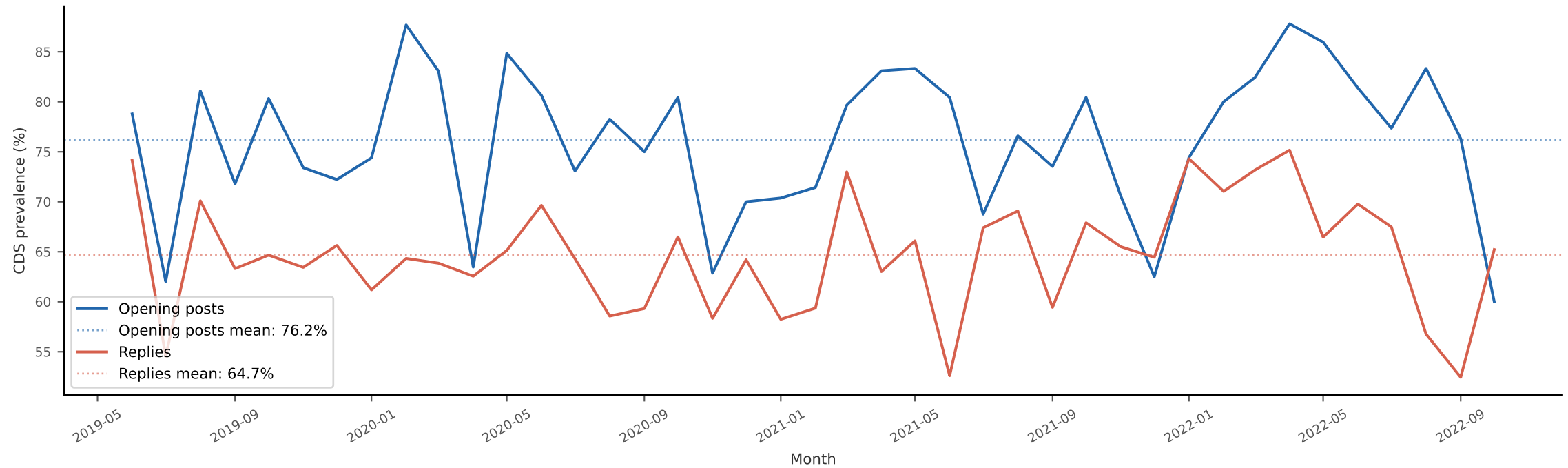
Monthly CDS Prevalence (%)



CDS Prevalence by Role — Year Level

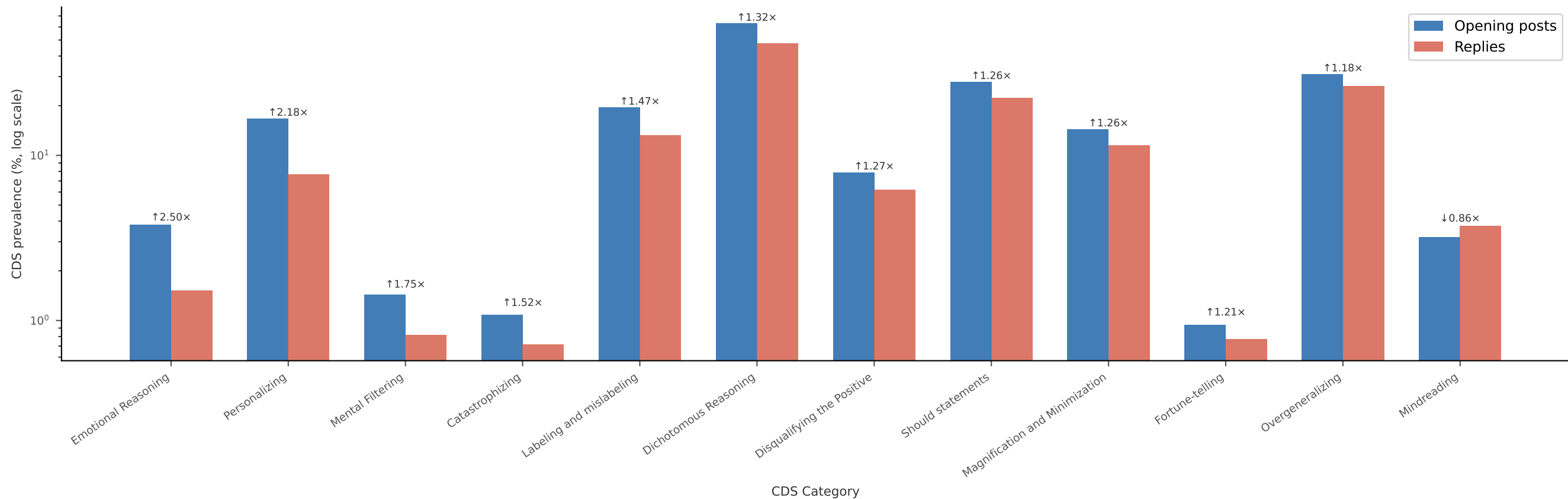


CDS Prevalence by Role — Month Level



Section 3 — CDS Prevalence by Category

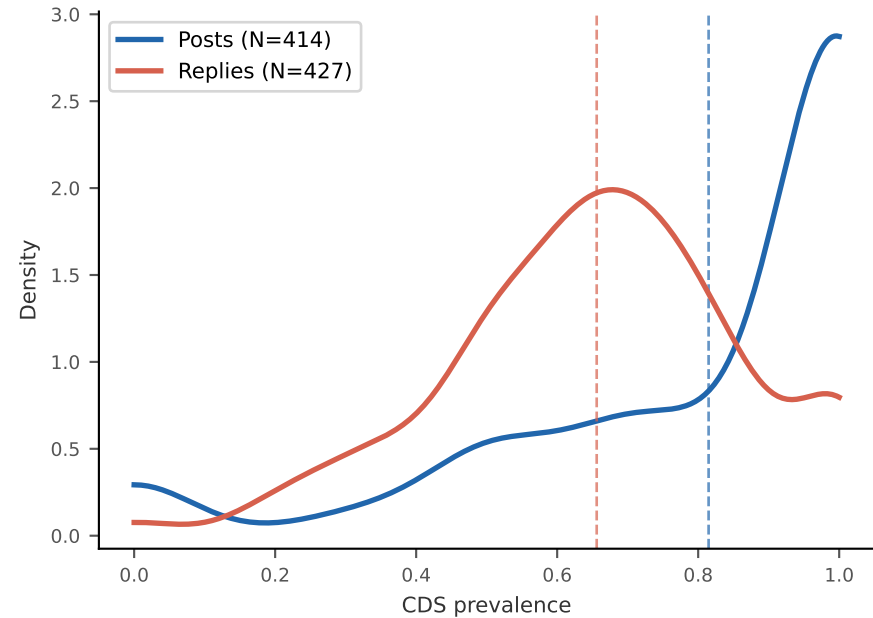
CDS Prevalence by Category — Posts vs Replies
(sorted by prevalence ratio, log scale)



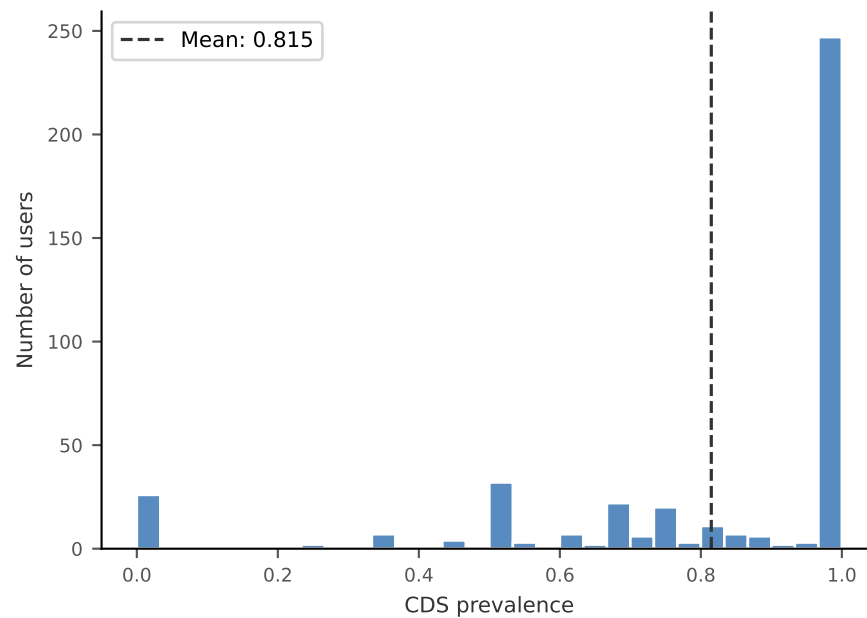
Section 4 — Within-User CDS Distribution

Within-User CDS Prevalence Distribution

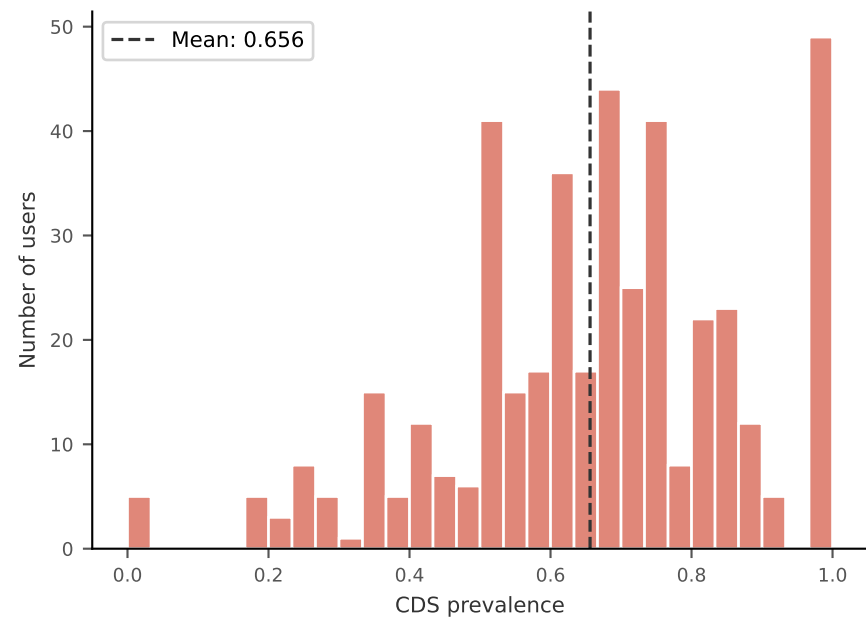
KDE: Within-user CDS Prevalence



Distribution: Posts



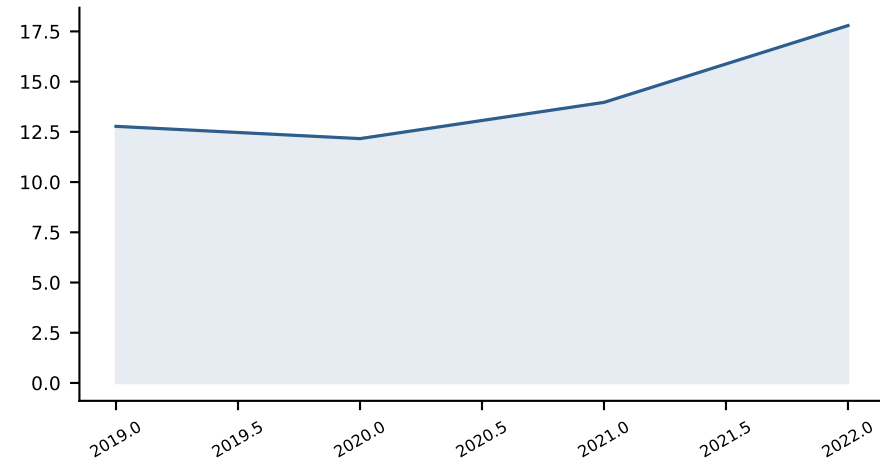
Distribution: Replies



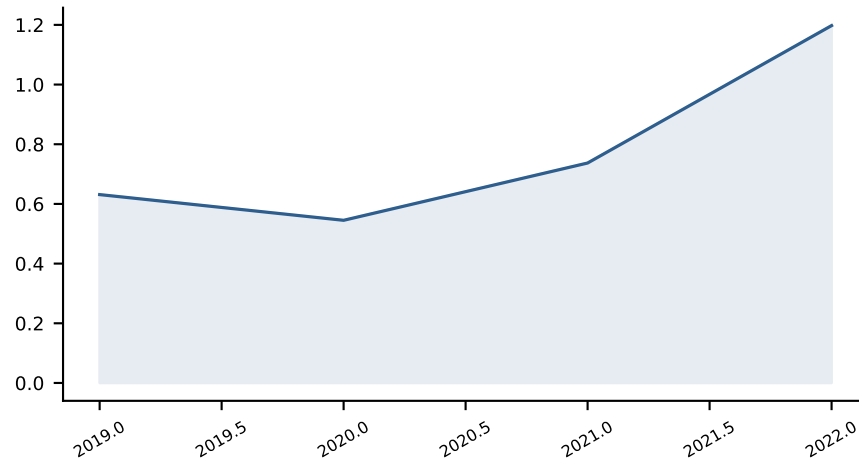
Section 5 — CDS per Category Over Time

CDS Prevalence per Category Over Time (Year Level)

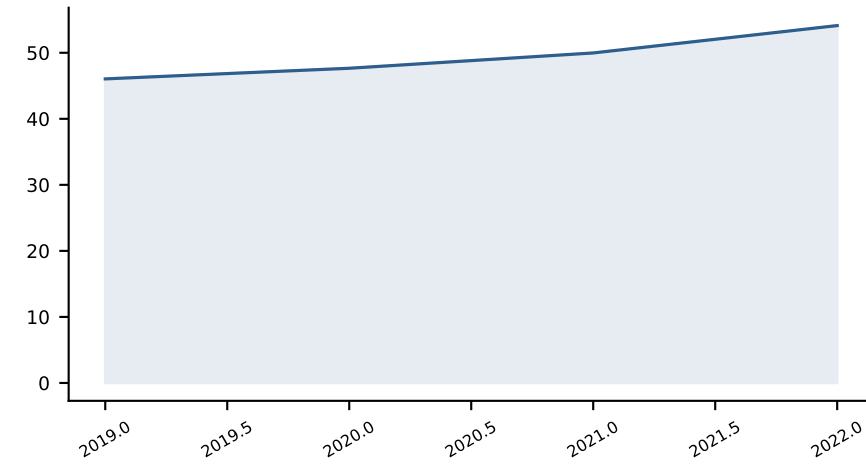
Labeling And Mislabeling



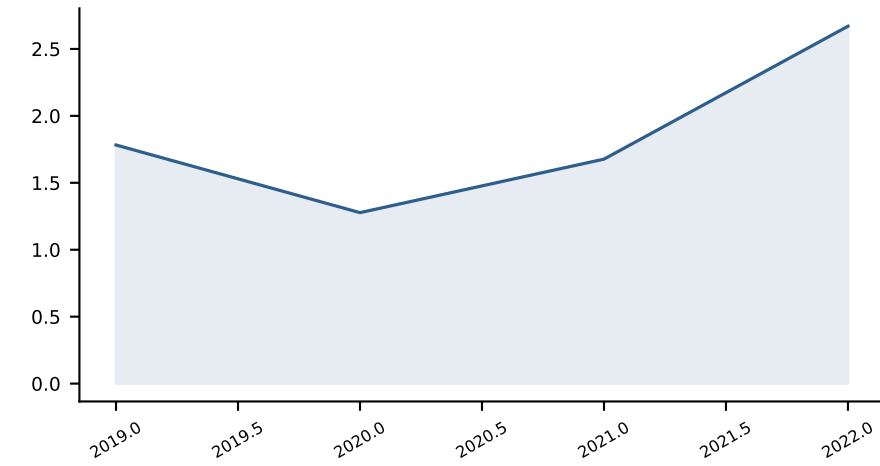
Catastrophizing



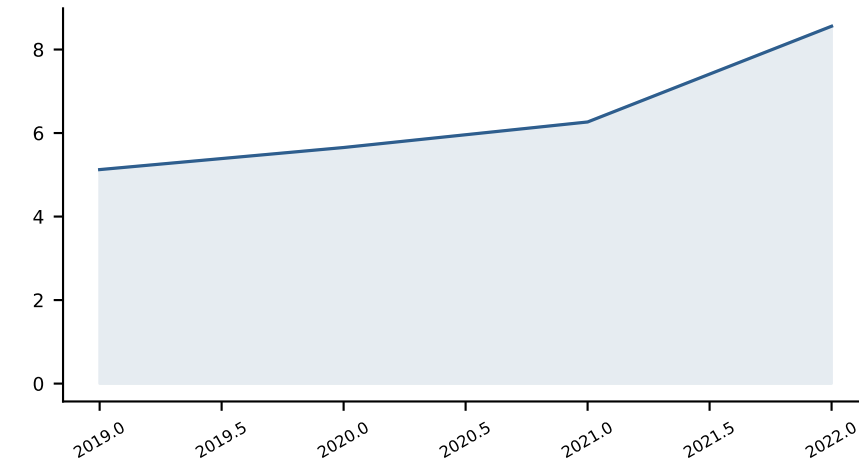
Dichotomous Reasoning



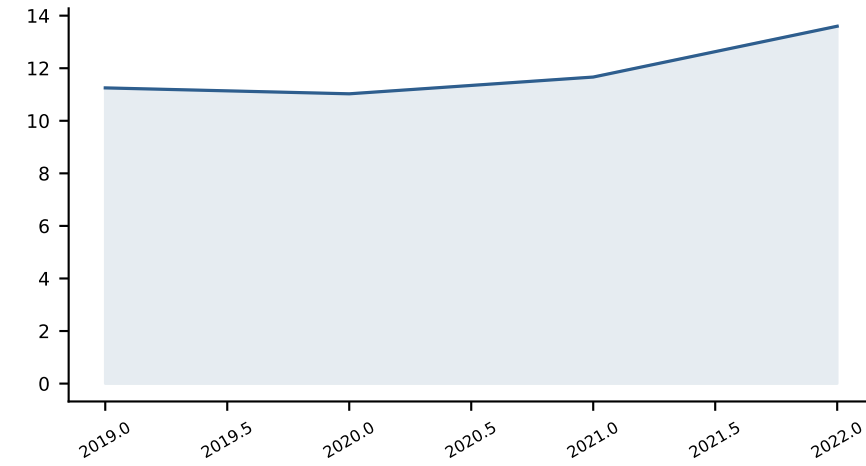
Emotional Reasoning



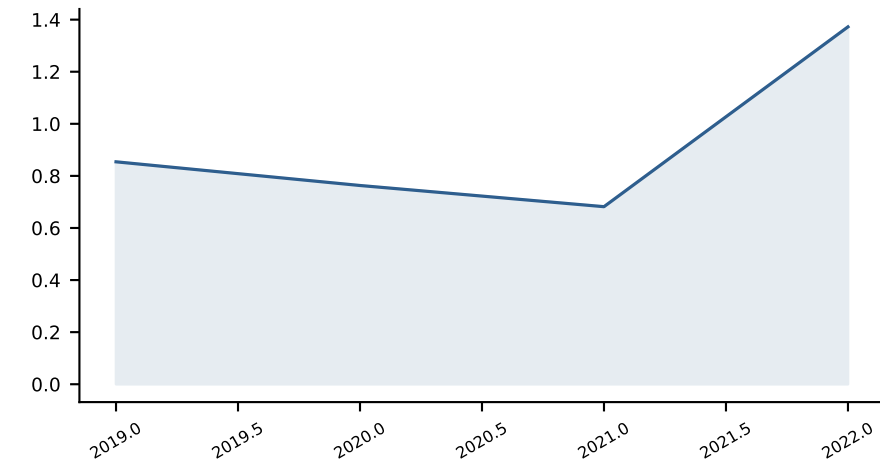
Disqualifying The Positive



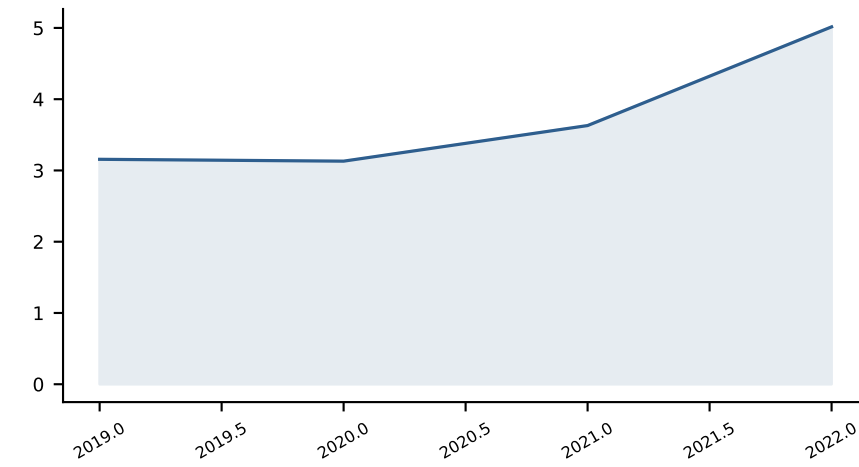
Magnification And Minimization



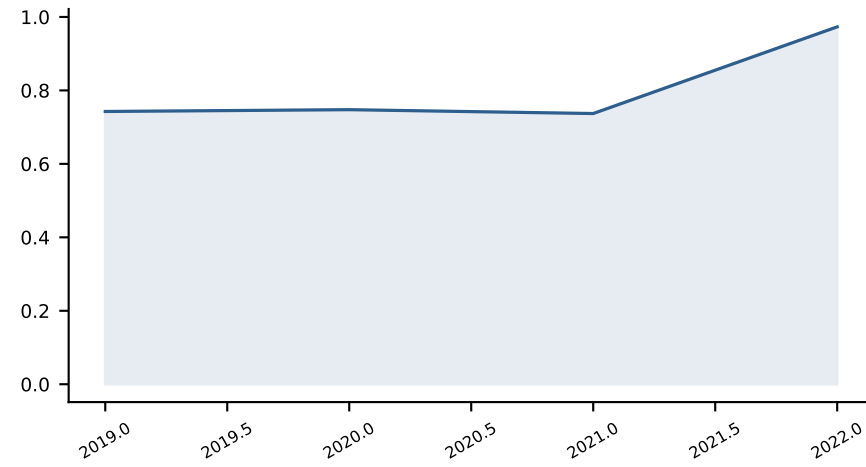
Mental Filtering



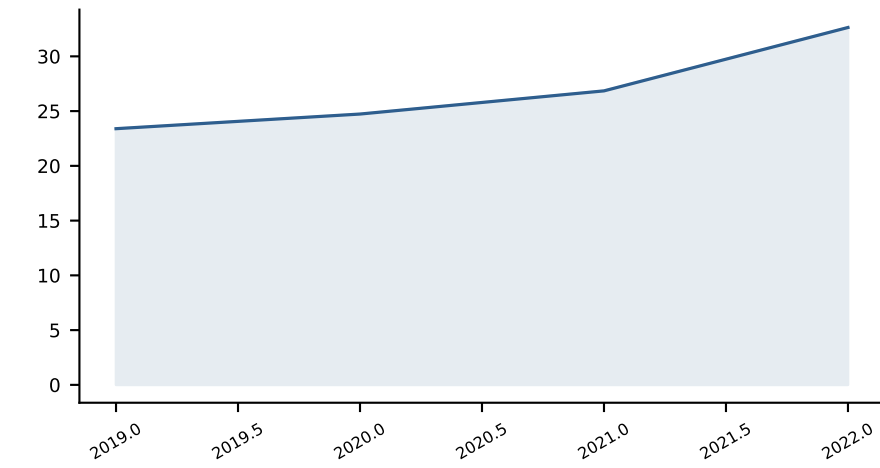
Mindreading



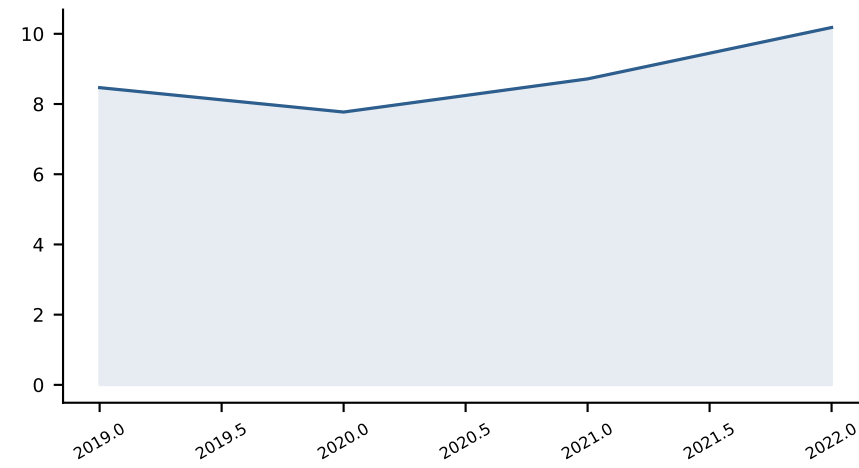
Fortune-Telling



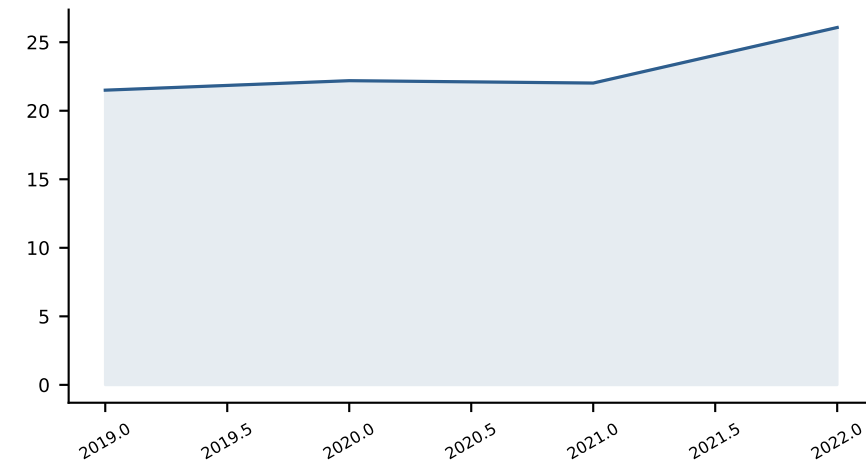
Overgeneralizing



Personalizing

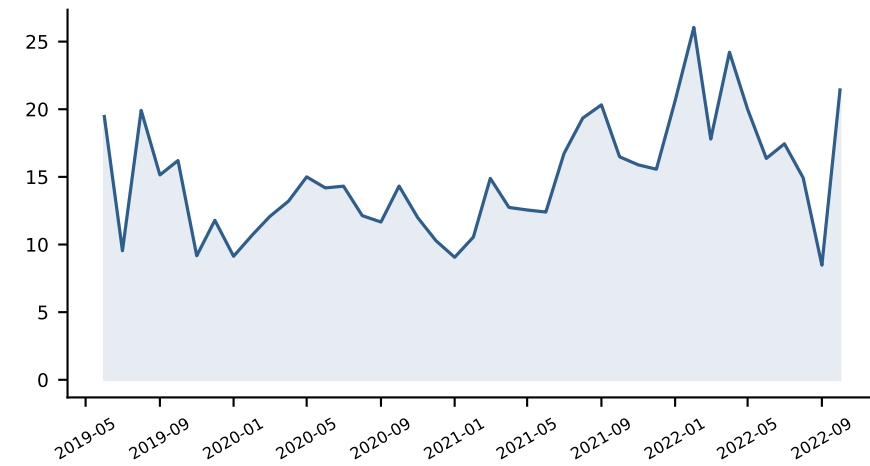


Should Statements

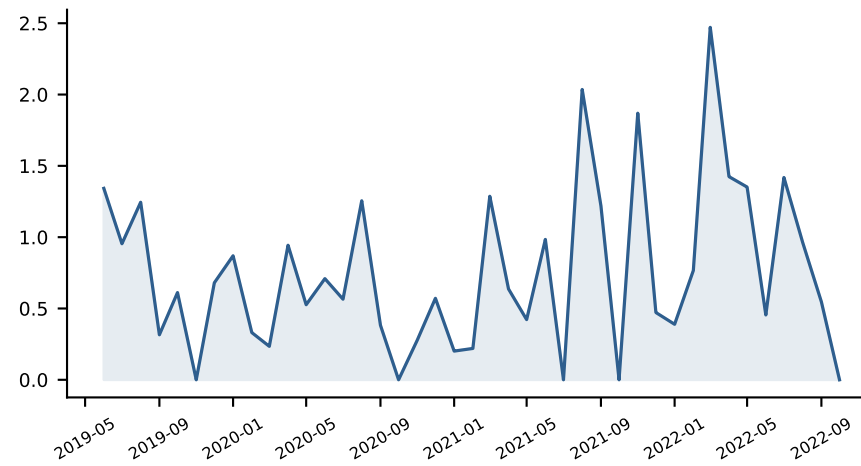


CDS Prevalence per Category Over Time (Month Level)

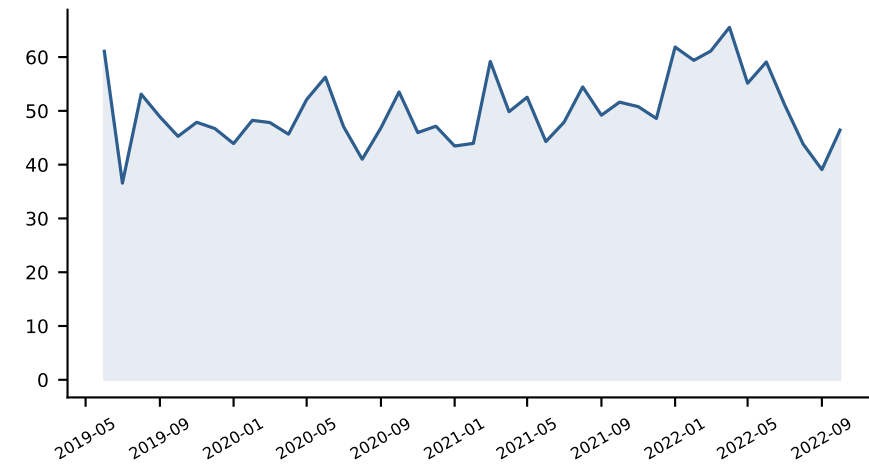
Labeling And Mislabeling



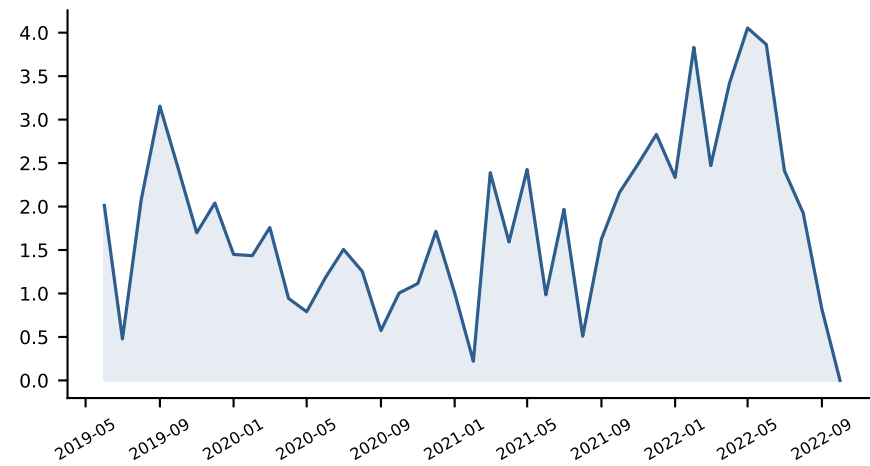
Catastrophizing



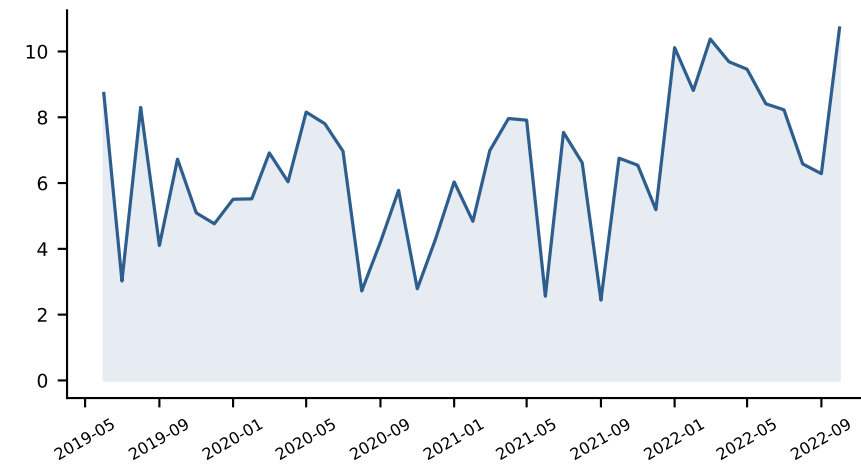
Dichotomous Reasoning



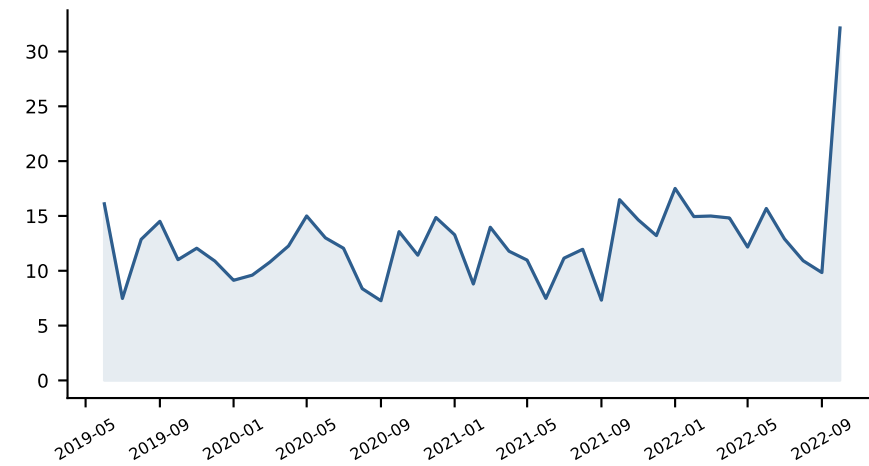
Emotional Reasoning



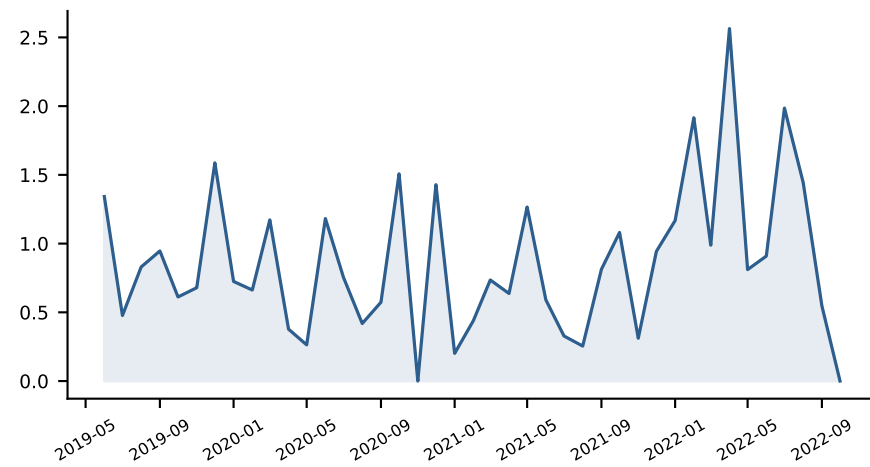
Disqualifying The Positive



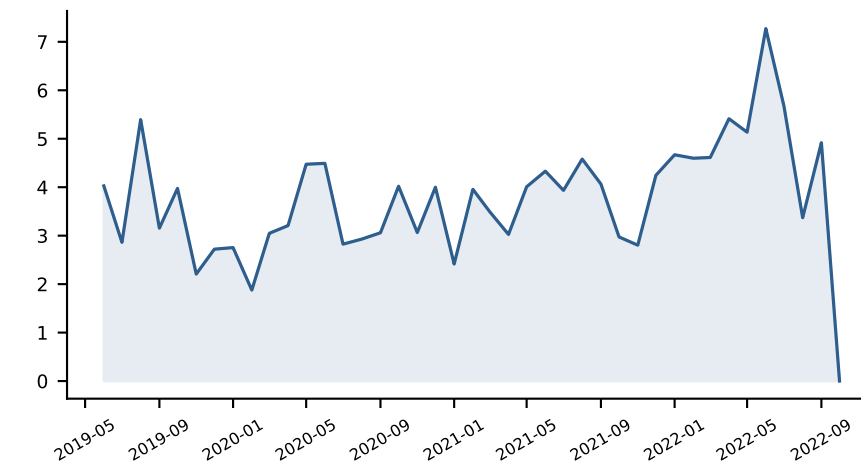
Magnification And Minimization



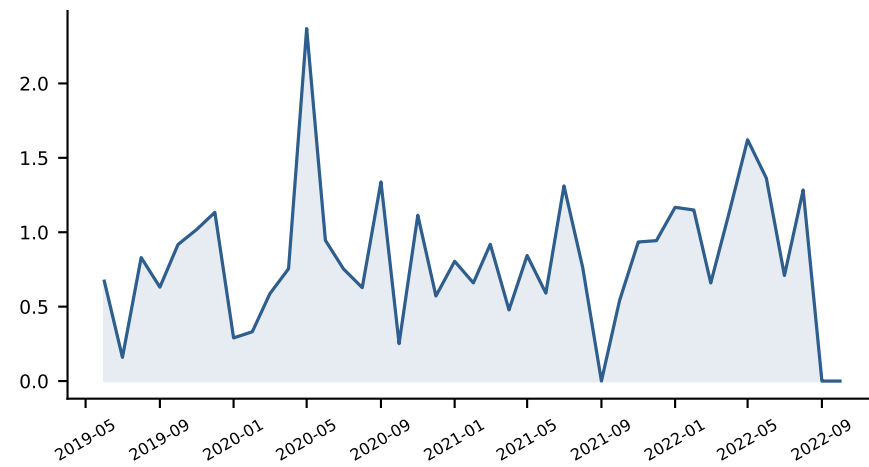
Mental Filtering



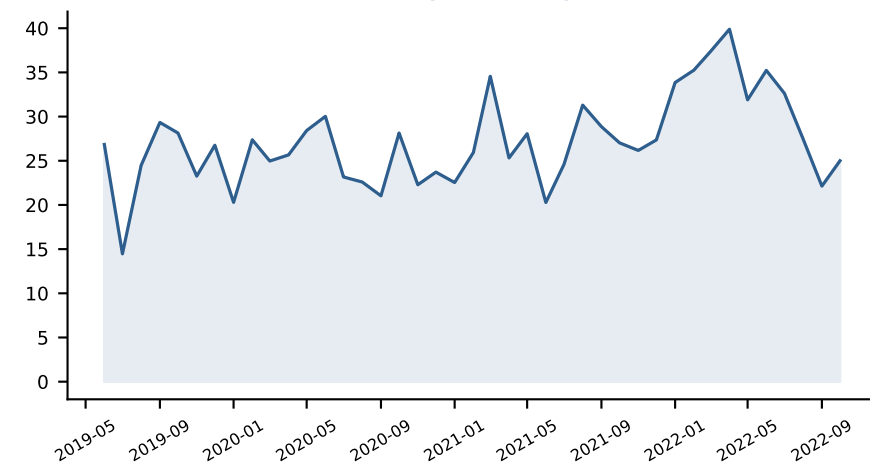
Mindreading



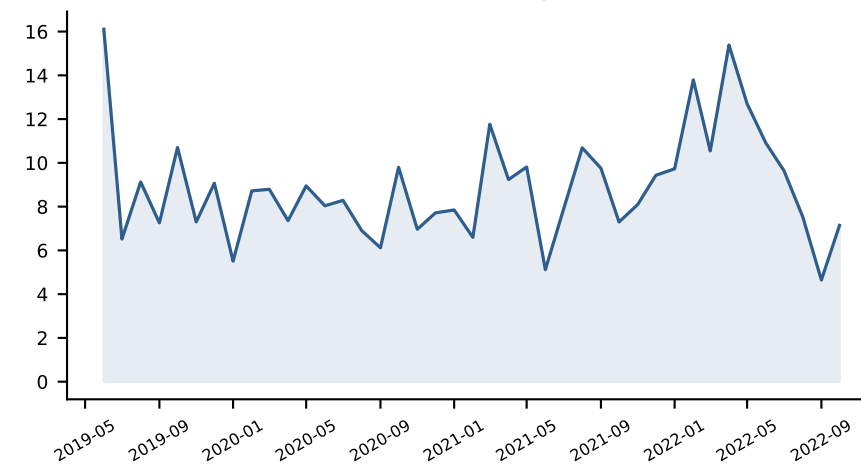
Fortune-Telling



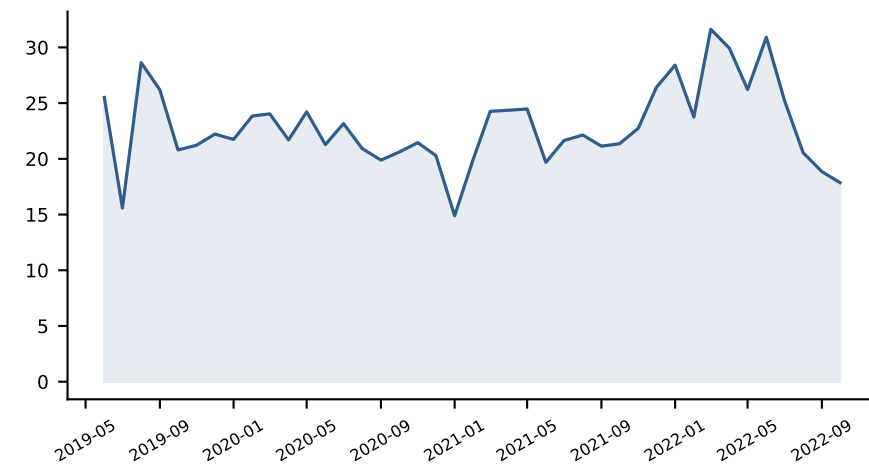
Overgeneralizing



Personalizing



Should Statements



4. Cognitive Distortion (CDS) Prevalence

Depression Connect Forum

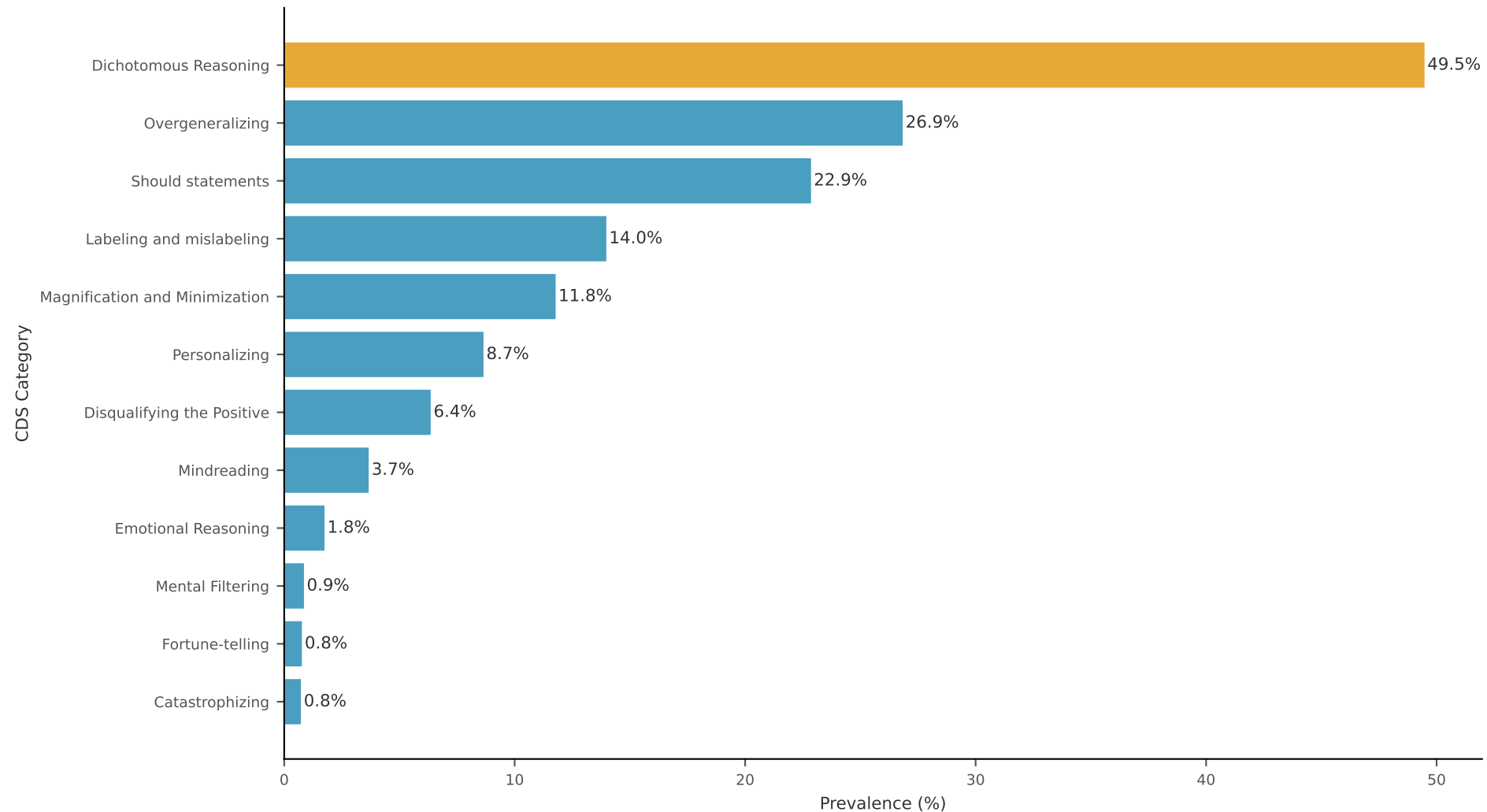
Most Common Cognitive Distortions (CDS Prevalence)

Section 1 — Category Ranking

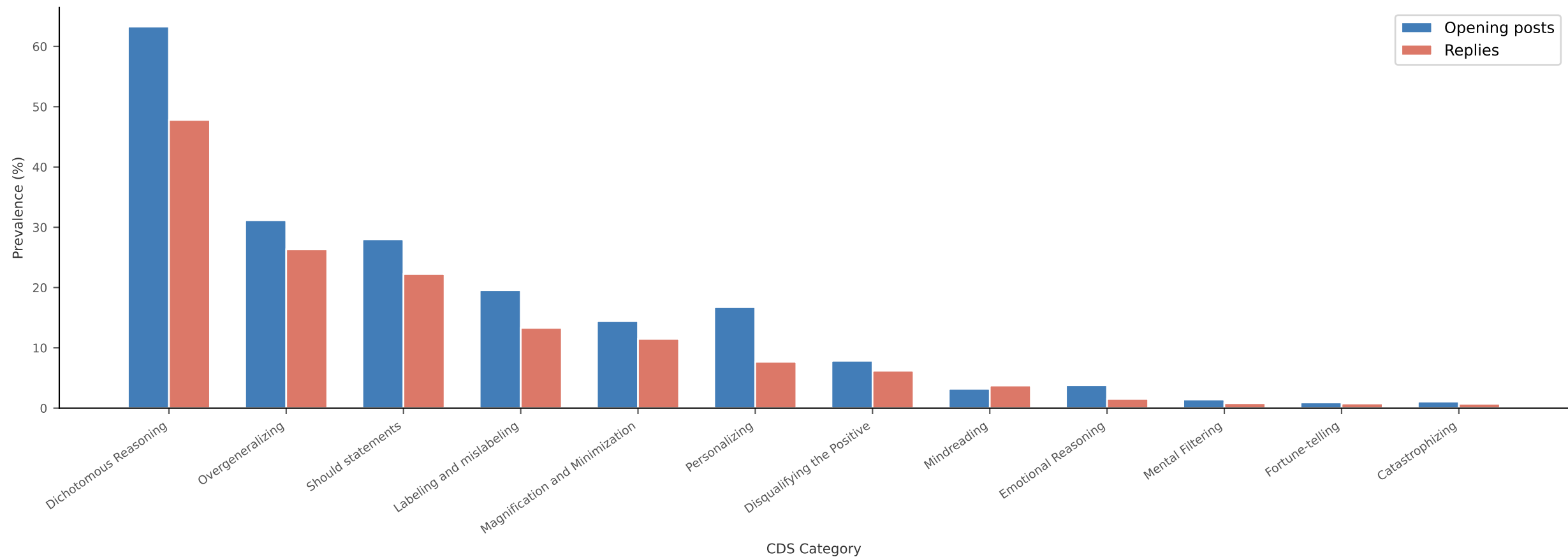
CDS Category Ranking — Mann-Whitney U (per-user rates) with BH correction
(p-BH = Benjamini-Hochberg corrected; yellow rows significant after correction)

Category	Overall %	Posts %	Replies %	Ratio P/R	N users (post)	N users (reply)	p-value (MWU)	p-BH	Rank-biserial r
Dichotomous Reasoning	49.5	63.29	47.81	1.324	414	427	0.0	0.0	-0.4092
Overgeneralizing	26.86	31.18	26.34	1.184	414	427	0.3187	0.3824	-0.0392
Should statements	22.88	28.01	22.25	1.259	414	427	0.7702	0.7702	-0.0114
Labeling and mislabeling	14.0	19.57	13.31	1.47	414	427	0.4357	0.4753	-0.0295
Magnification and Minimization	11.8	14.43	11.48	1.257	414	427	0.0053	0.0091	0.1036
Personalizing	8.67	16.75	7.68	2.181	414	427	0.0896	0.1344	-0.0629
Disqualifying the Positive	6.38	7.86	6.2	1.266	414	427	0.0	0.0	0.1456
Mindreading	3.69	3.21	3.75	0.857	414	427	0.0	0.0	0.1668
Emotional Reasoning	1.77	3.8	1.52	2.505	414	427	0.2557	0.3409	0.0298
Mental Filtering	0.88	1.43	0.82	1.754	414	427	0.0038	0.0076	0.0597
Fortune-telling	0.79	0.94	0.77	1.212	414	427	0.0	0.0	0.0795
Catastrophizing	0.75	1.09	0.71	1.522	414	427	0.0001	0.0002	0.0747

CDS Category Prevalence Ranking (% of messages containing at least one phrase from this category)



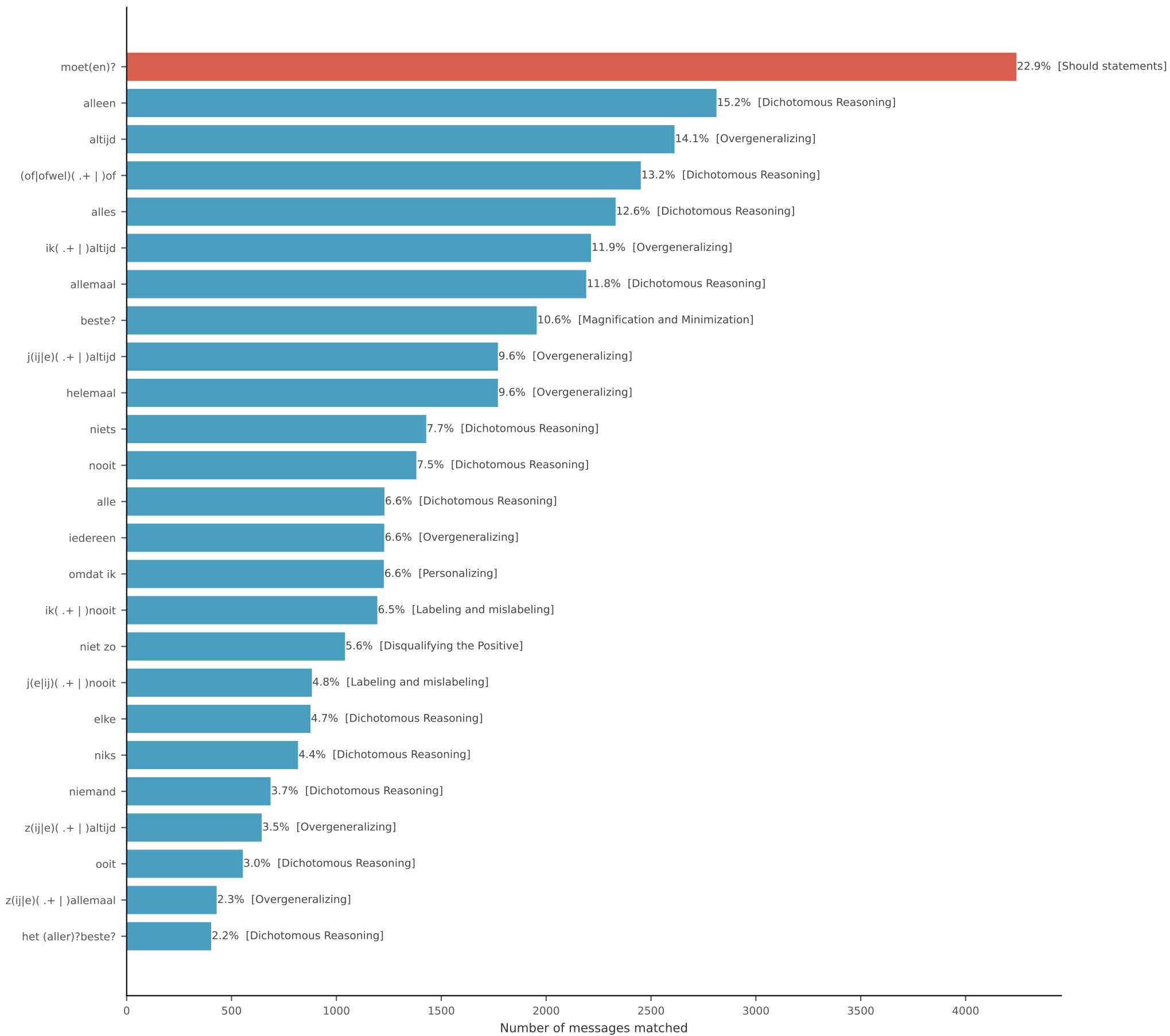
CDS Prevalence by Category — Opening Posts vs Replies



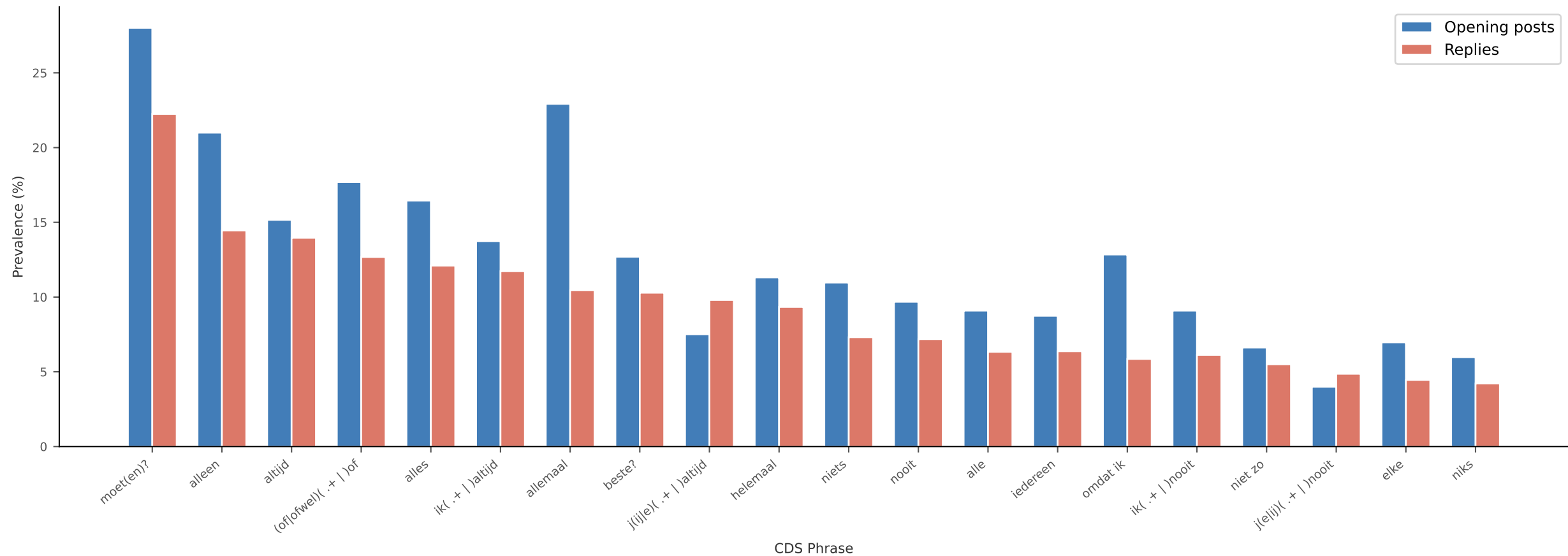
Section 2 — Individual CDS Phrases

Top 25 Most Frequent CDS Phrases (label = prevalence % | category)

CDS Phrase



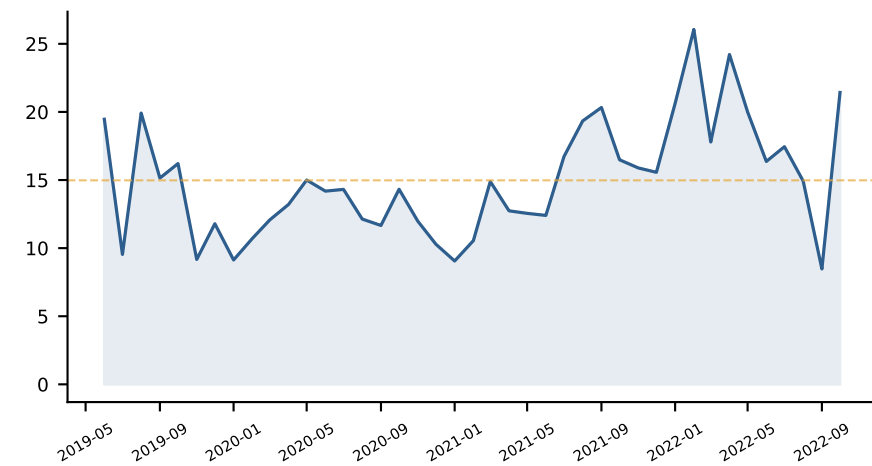
Top 20 CDS Phrases — Opening Posts vs Replies (prevalence %)



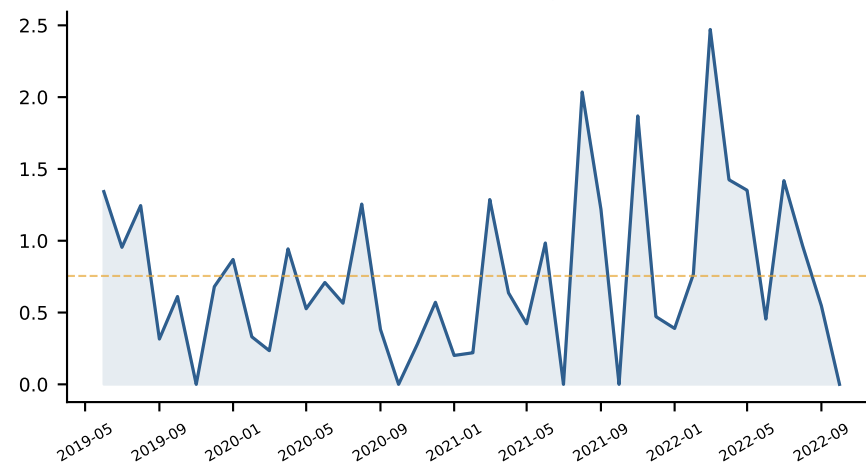
Section 3 — Category Trends Over Time

CDS Category Prevalence Over Time (Monthly, dashed = mean)

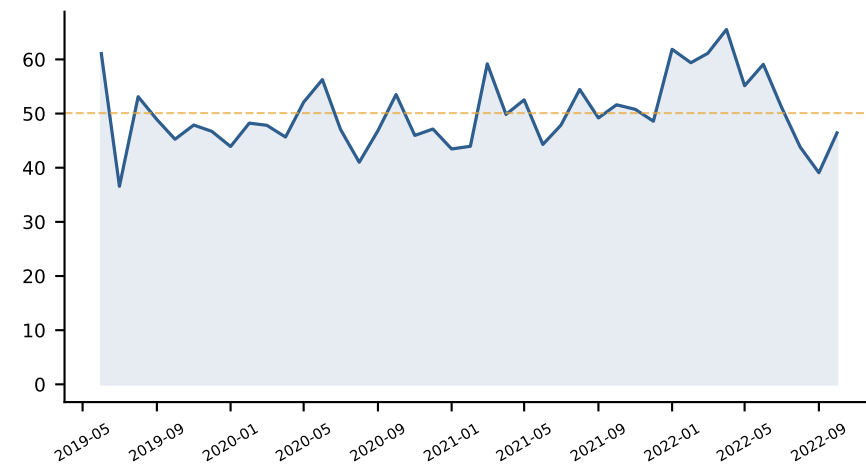
Labeling and mislabeling



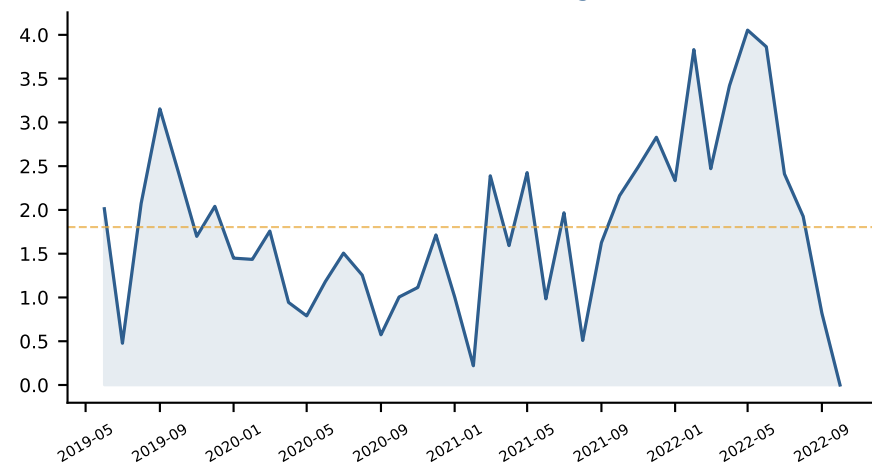
Catastrophizing



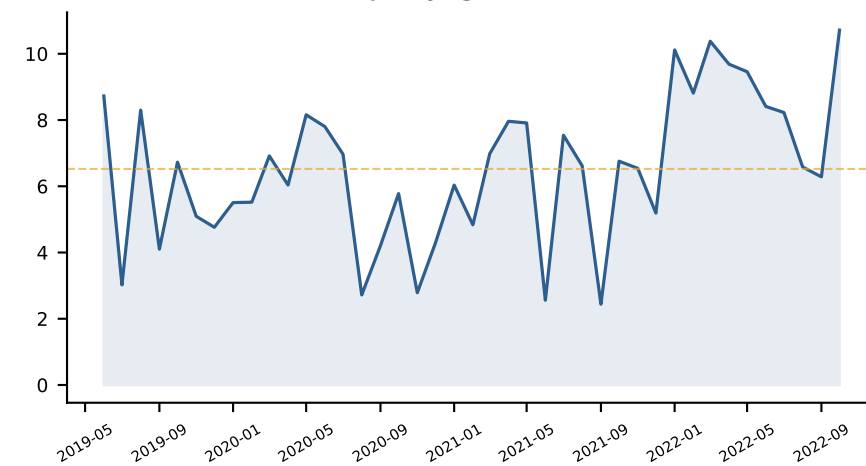
Dichotomous Reasoning



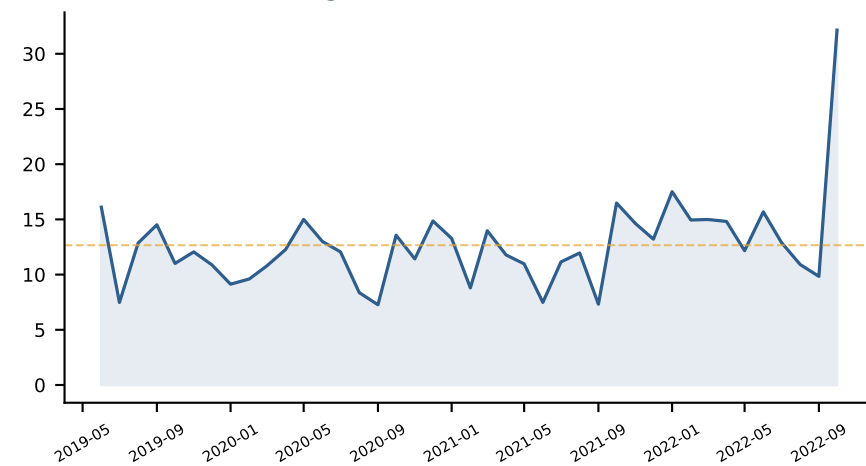
Emotional Reasoning



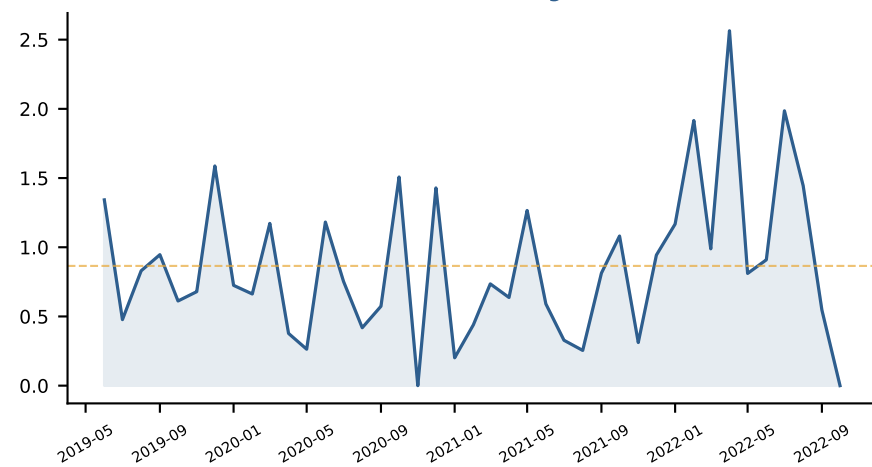
Disqualifying the Positive



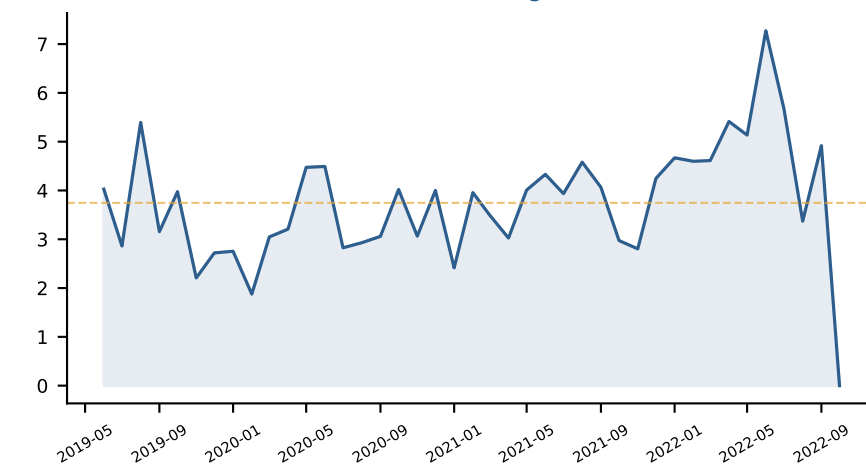
Magnification and Minimization



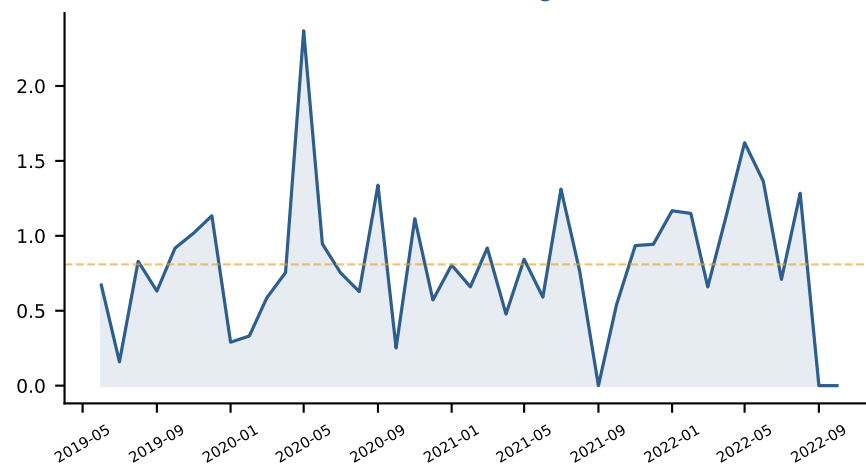
Mental Filtering



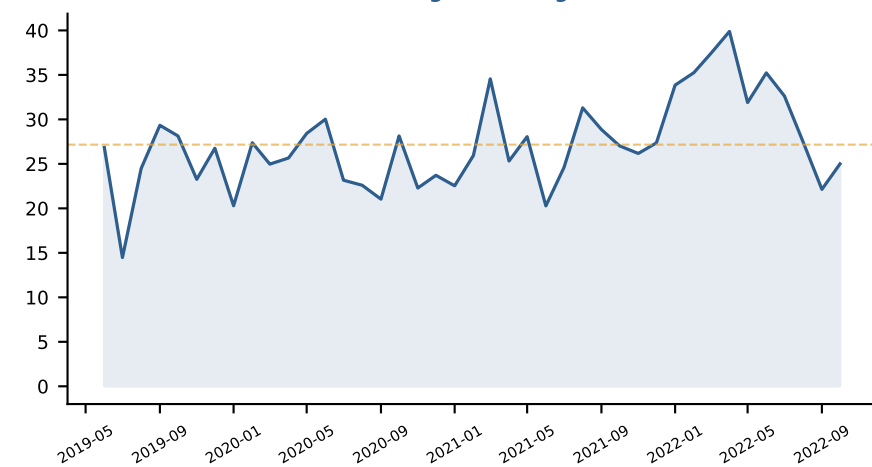
Mindreading



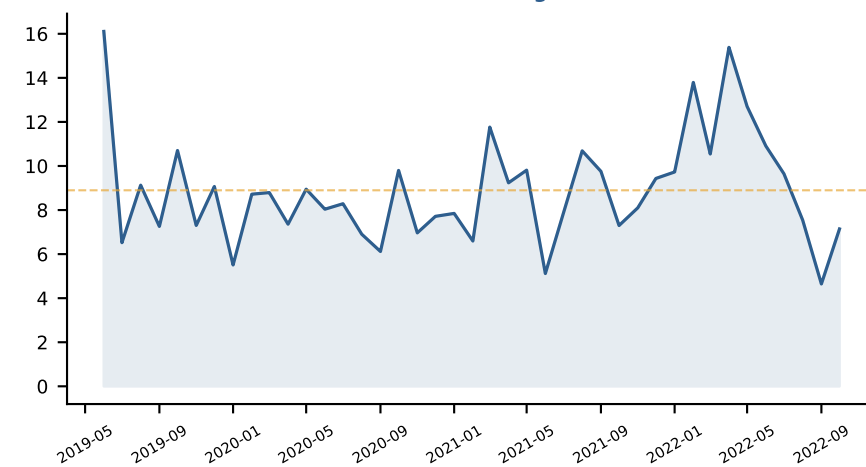
Fortune-telling



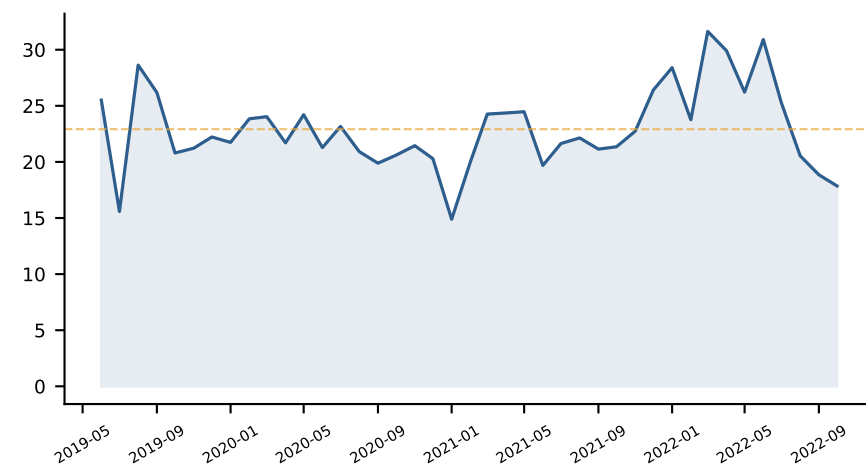
Overgeneralizing



Personalizing



Should statements



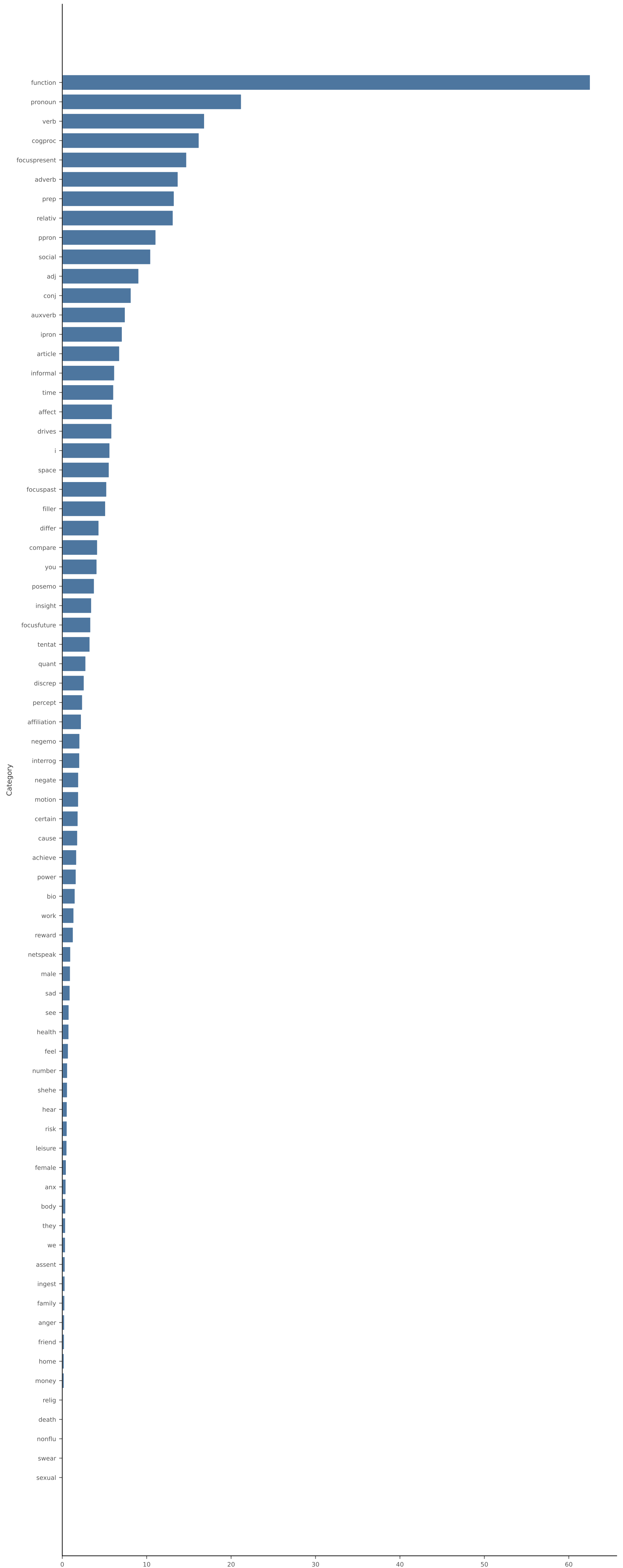
5. LIWC Psycholinguistic Analysis

Depression Connect Forum

LIWC Psycholinguistic Feature Analysis

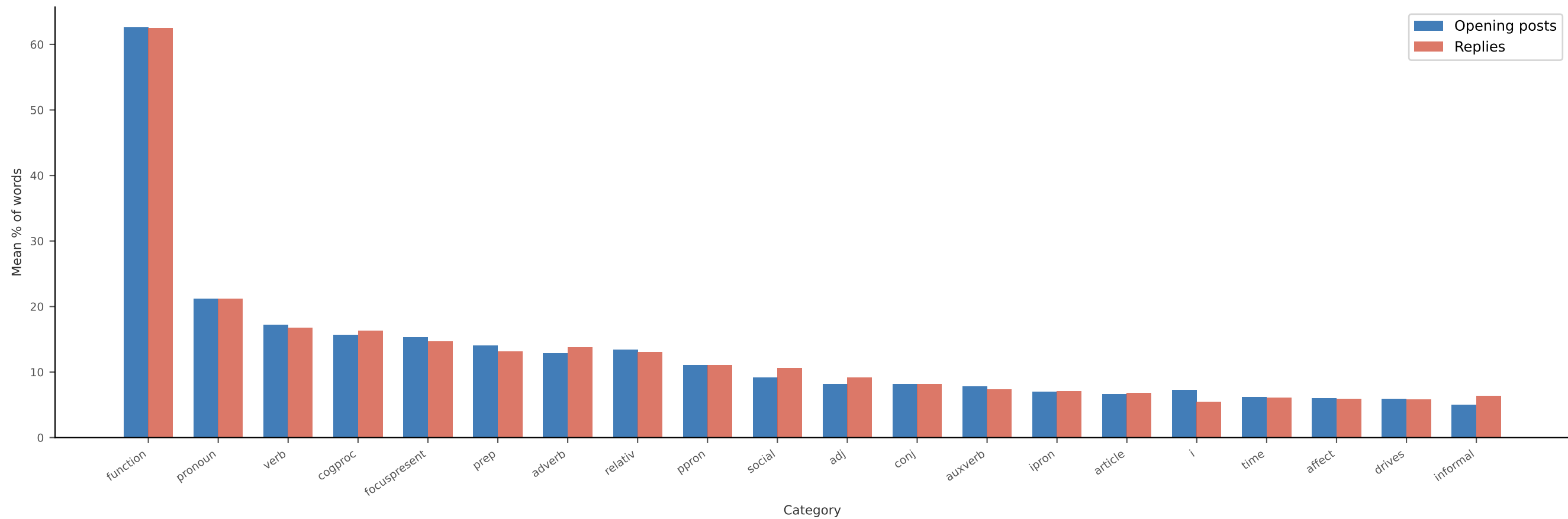
Section 1 — Overall Category Prevalence

Overall LIWC Category Prevalence (mean % of words)



Section 2 — Posts vs Replies

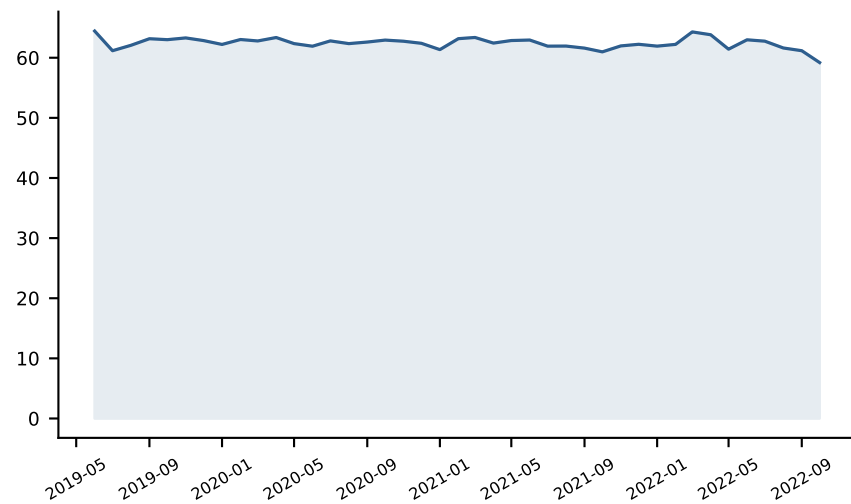
Top 20 LIWC Categories — Posts vs Replies (mean % of words)



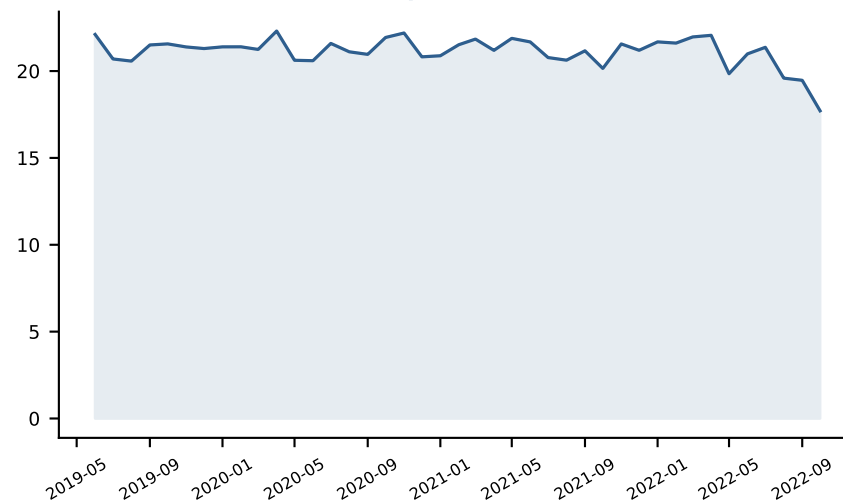
Section 3 — Top Categories Over Time

Top 12 LIWC Categories — Monthly Prevalence Over Time

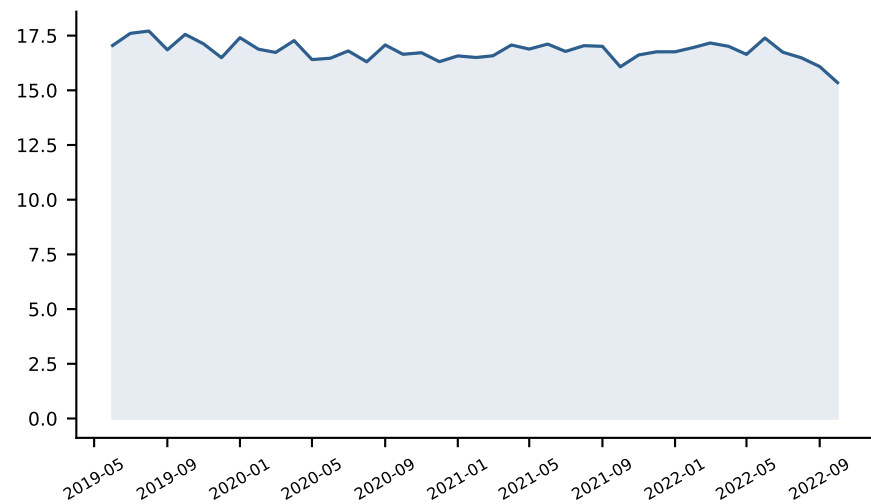
function



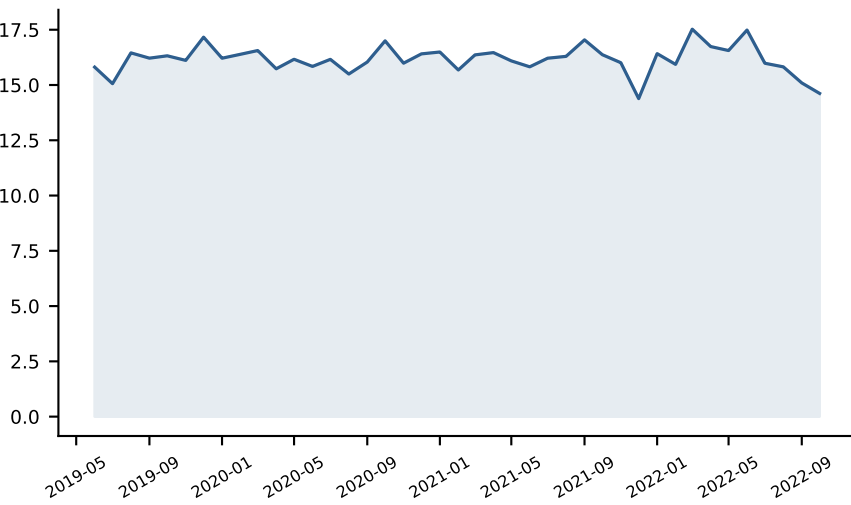
pronoun



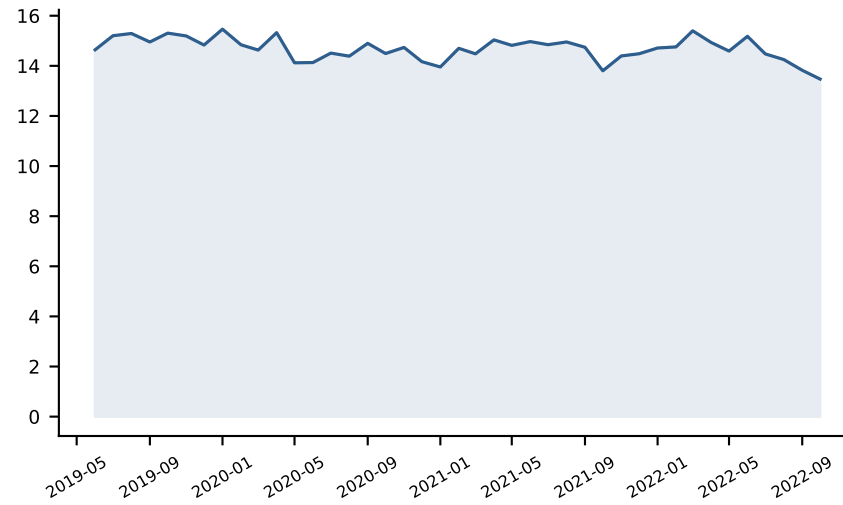
verb



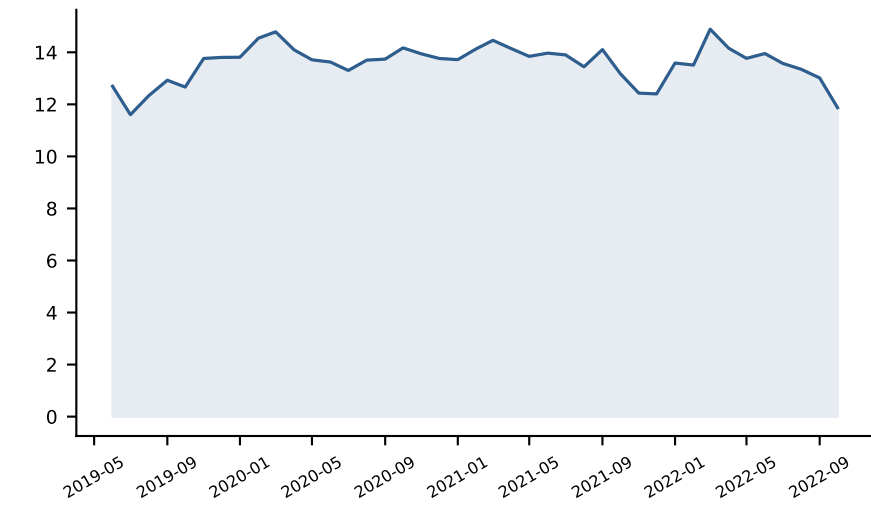
cogproc



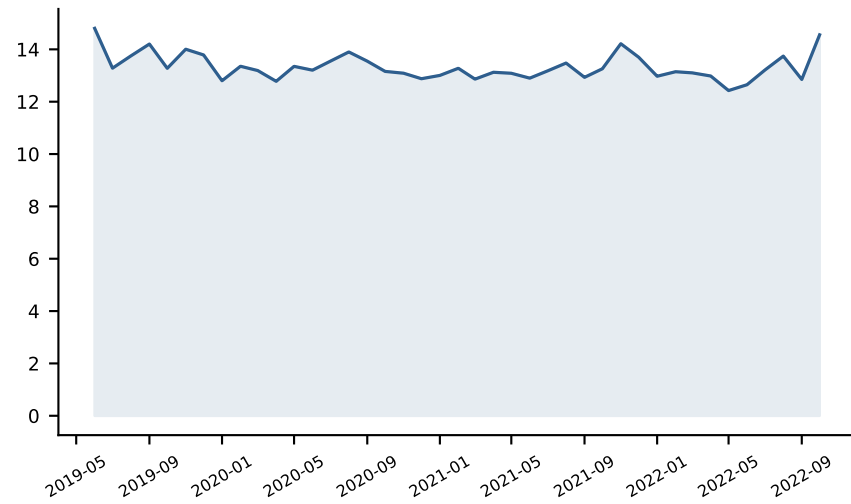
focuspresent



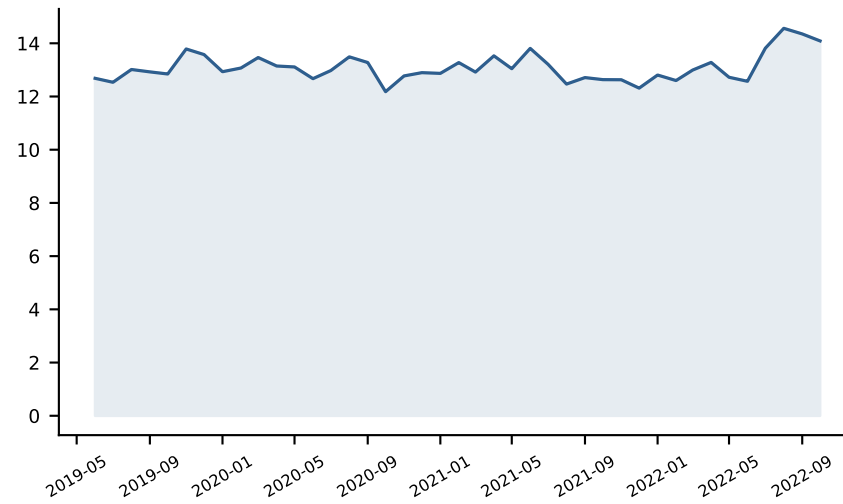
adverb



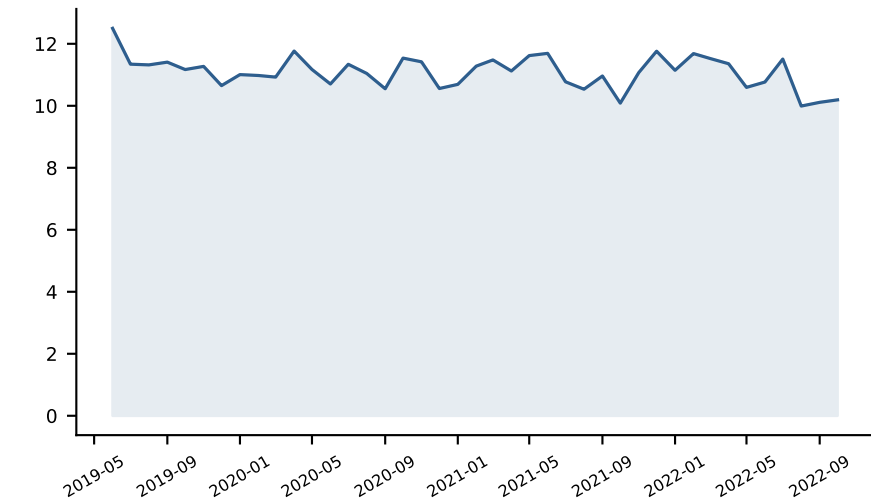
prep



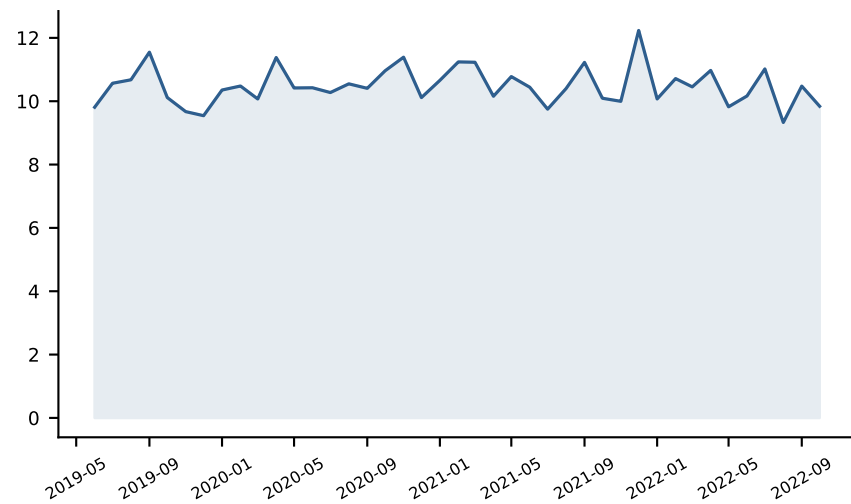
relativ



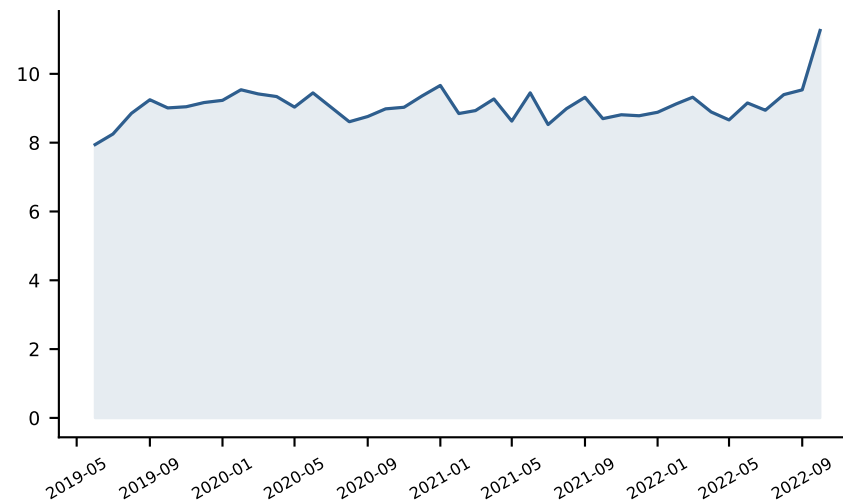
ppron



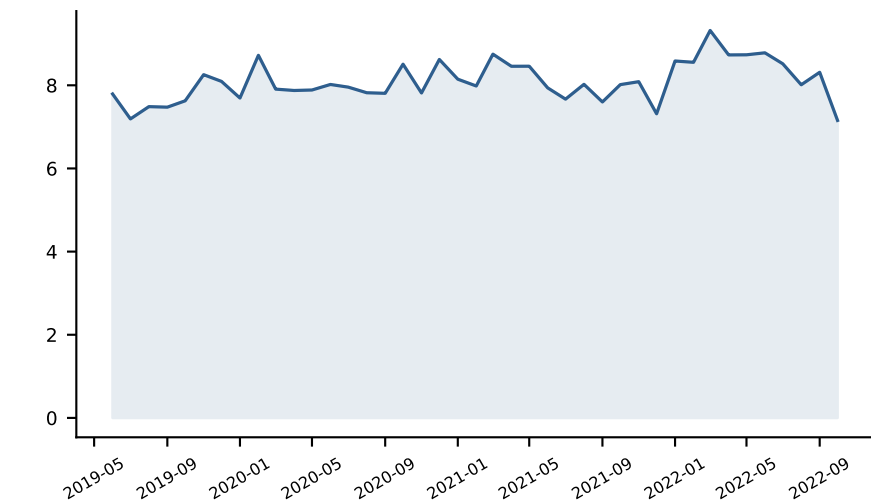
social



adj

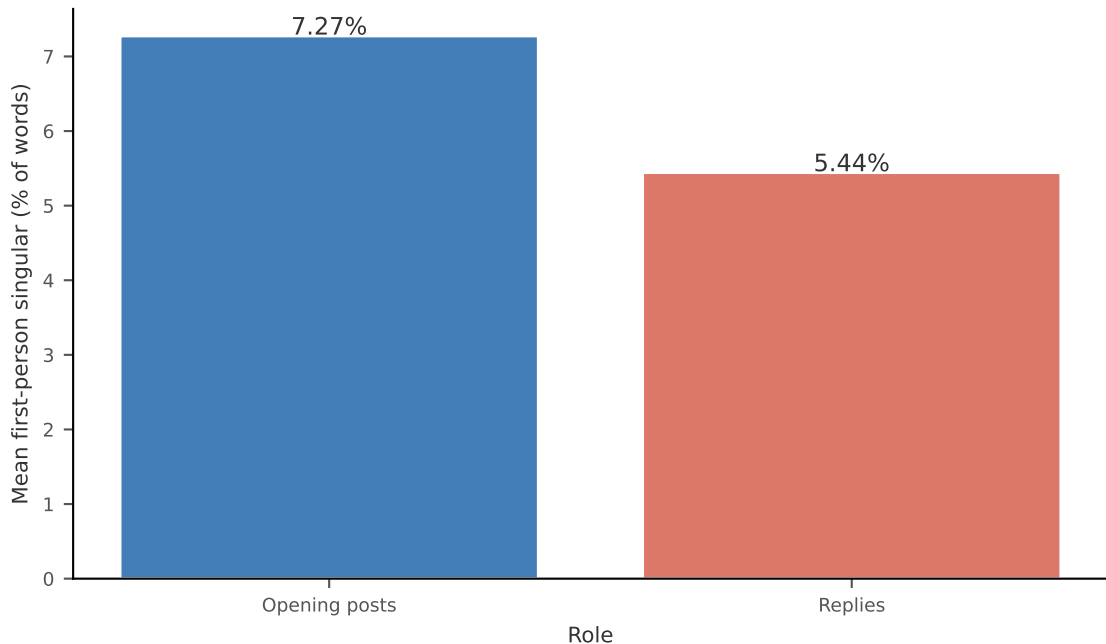


conj



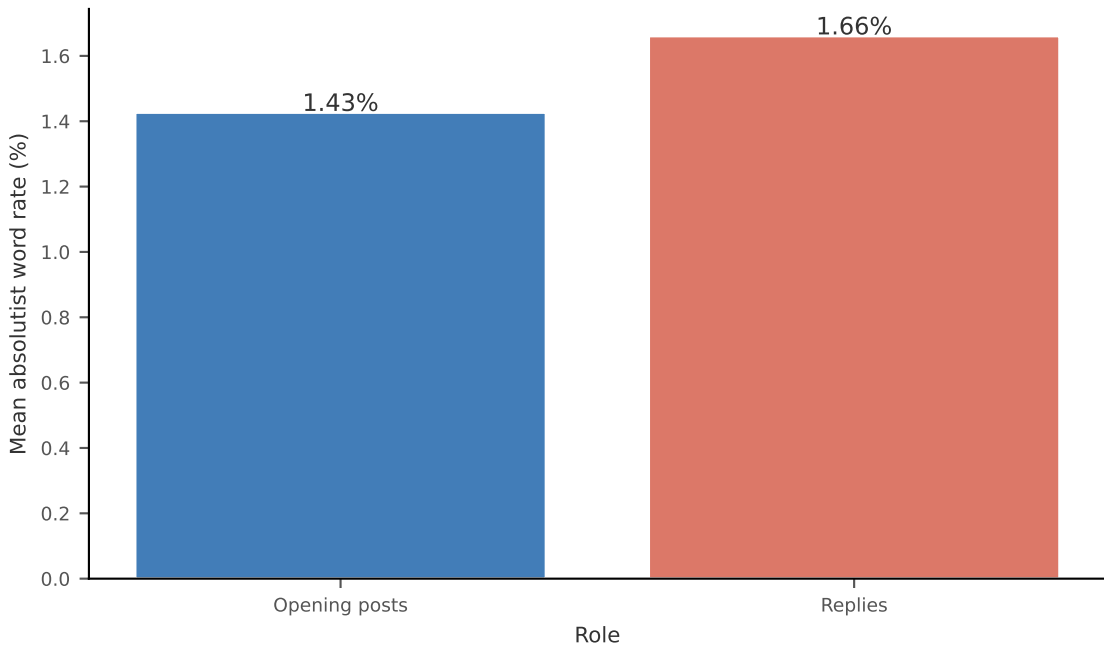
Section 4 — First-Person Singular by Role

First-Person Singular Rate by Role (self-focus marker; Smirnova 2018, Eichstaedt et al. 2018)



Section 5 — Absolutist Words

Absolutist Word Rate by Role (Dutch function-word list; Al-Mosaiwi & Johnstone 2018)



6. Per-User Longitudinal Trends

Depression Connect Forum

Longitudinal Analysis — Top 5 Most Active Posters

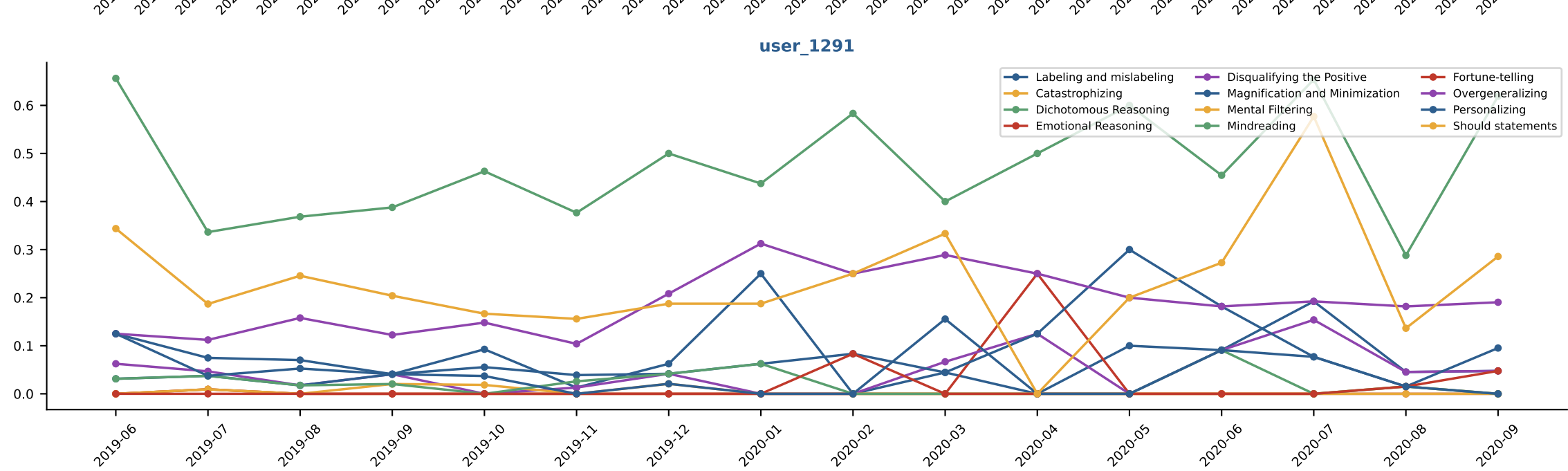
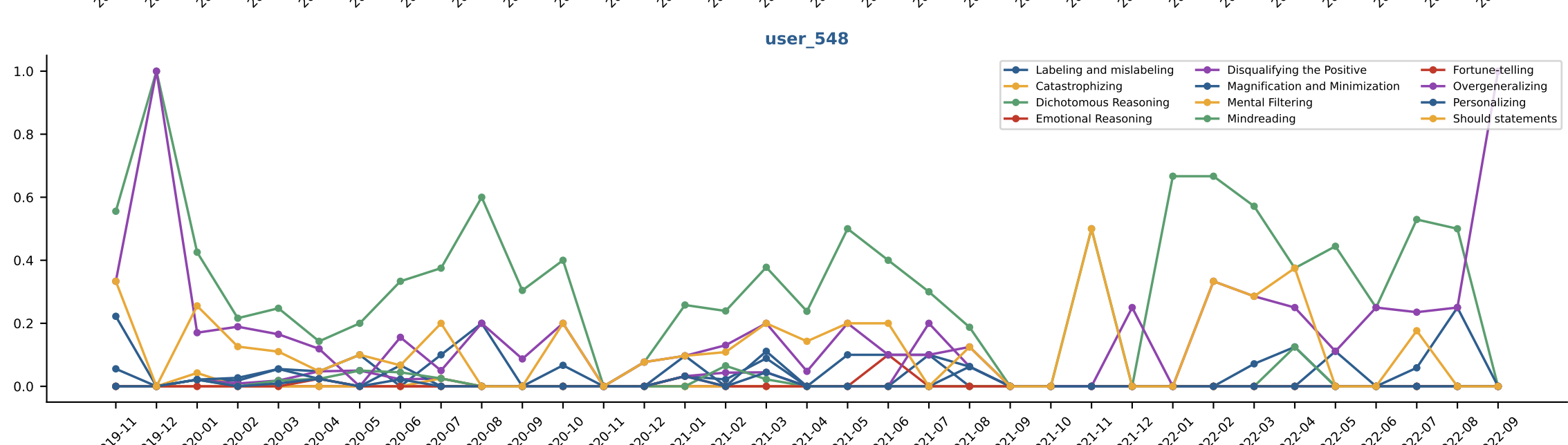
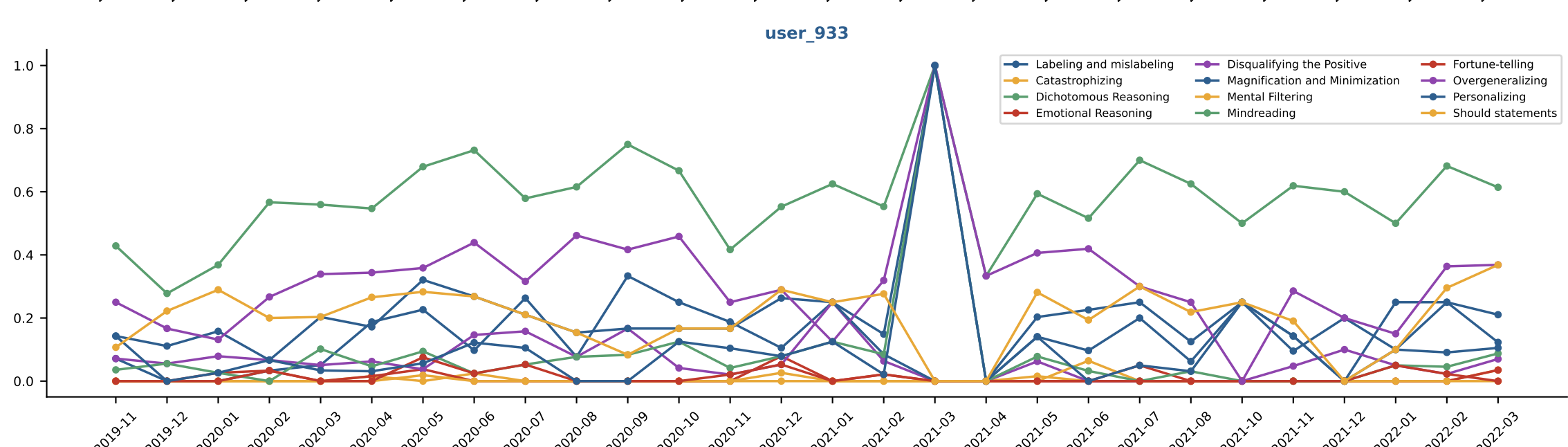
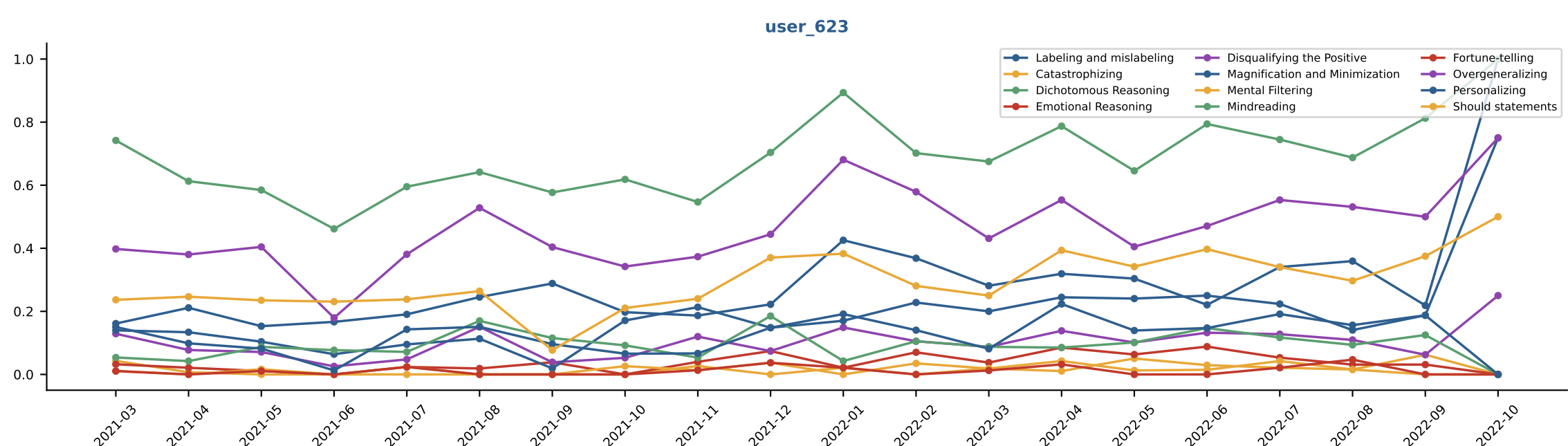
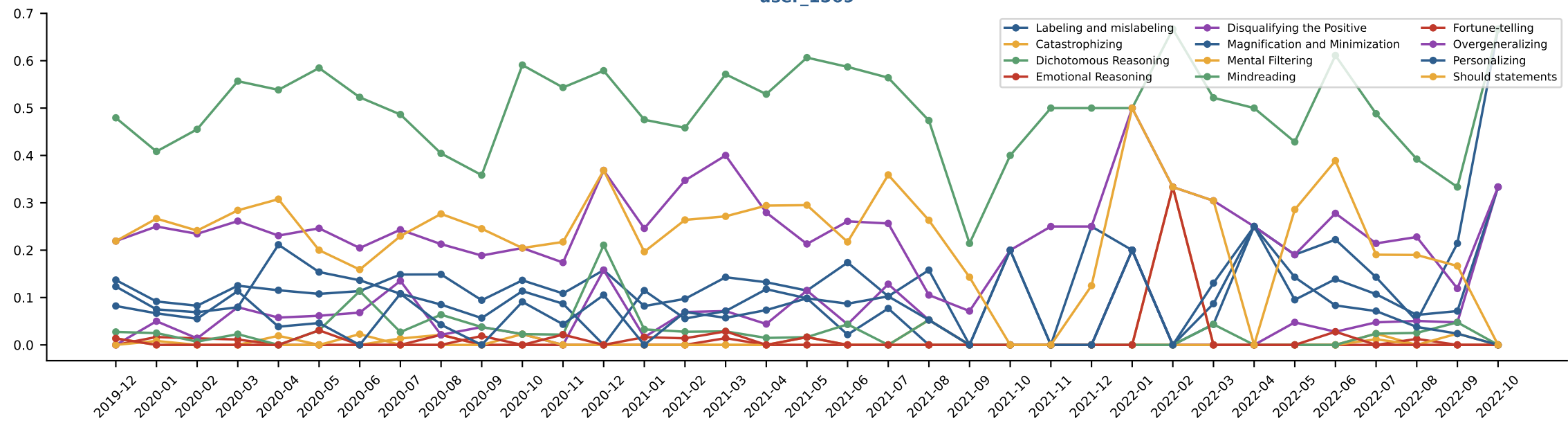
Selected Users — Top by Raw Post Volume

User	Posts	Active days	Posts / active month
user_1569	1646	1030	48.64
user_623	1520	579	79.91
user_933	847	850	30.33
user_548	760	1049	22.05
user_1291	639	454	42.84

Selection is by total post count only. Note this over-weights high-intensity users and can miss long-active, moderate-volume users (see --select sustained).

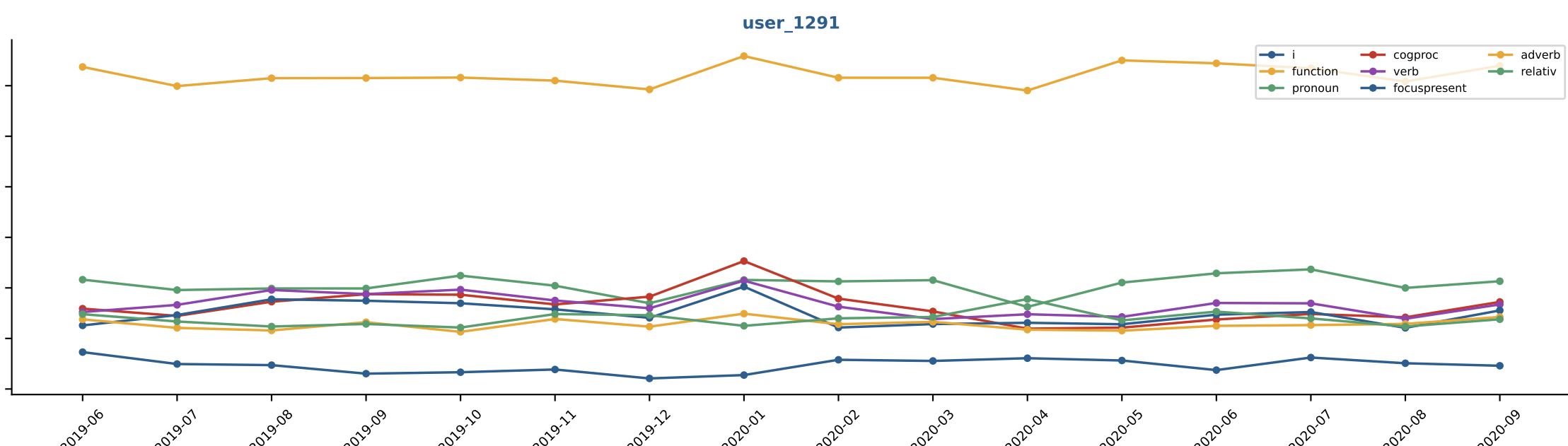
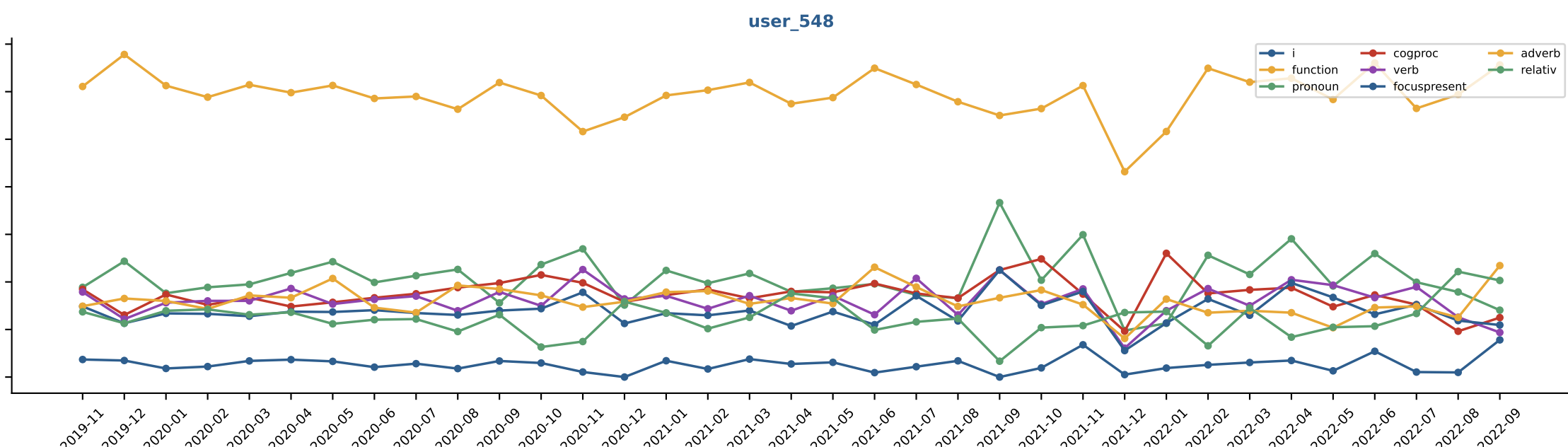
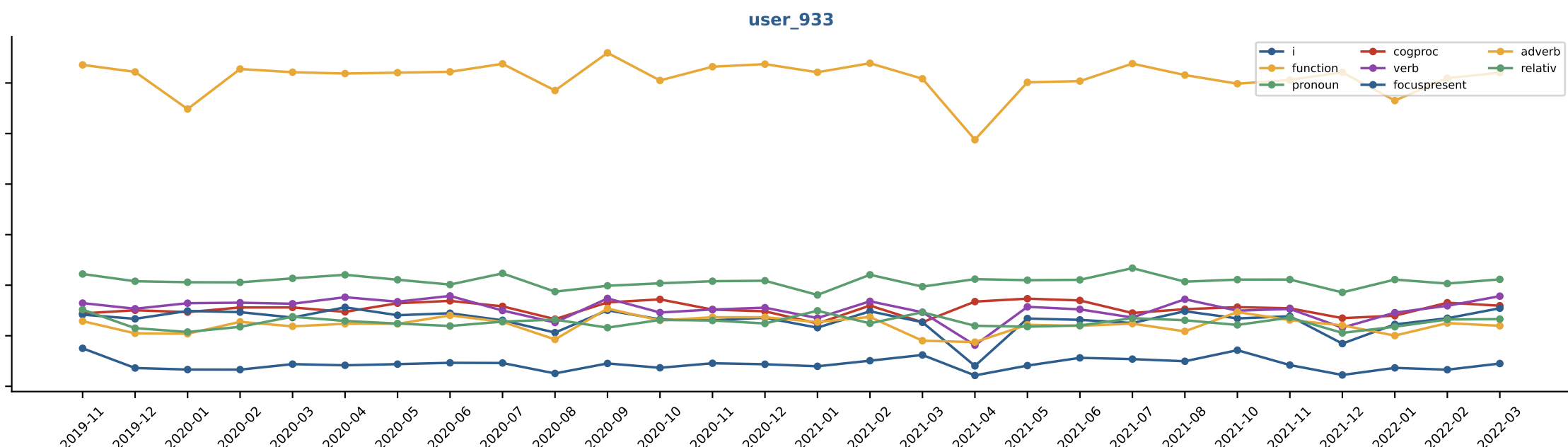
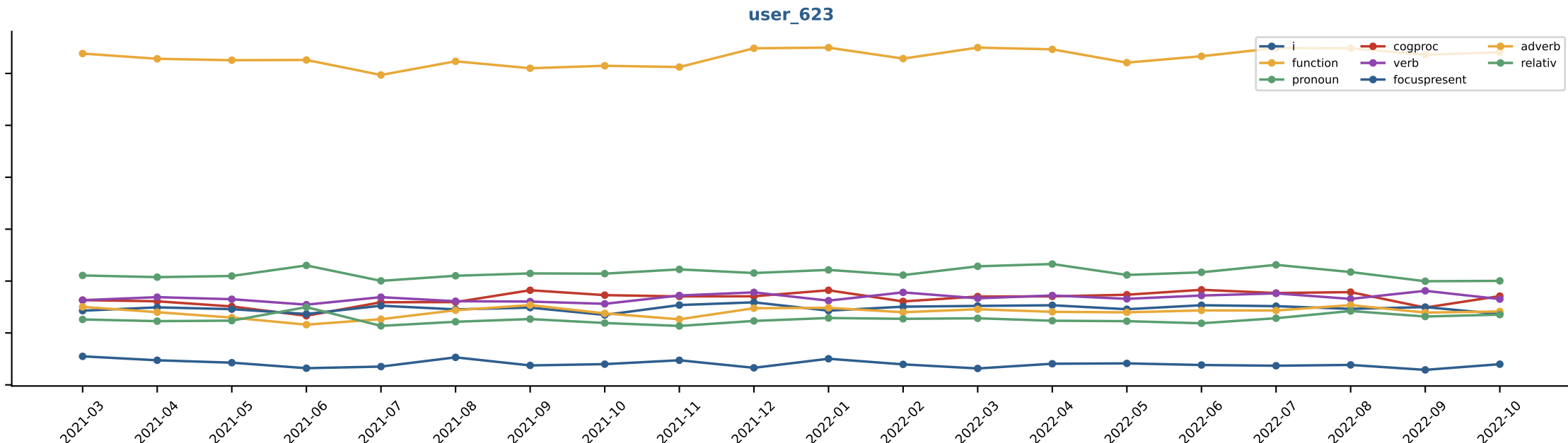
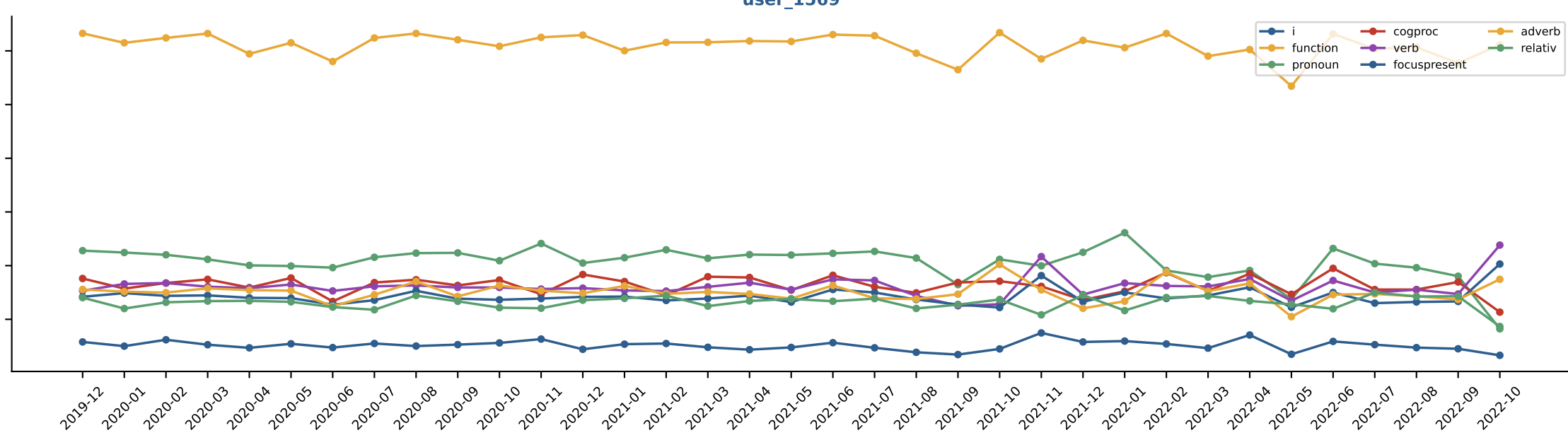
CDS Category Scores Per Month

CDS Categories Over Time (mean per month)



LIWC Category Scores Per Month

Top LIWC Categories Over Time (mean % of words per month)



7. LIWC-22 Validation

LIWC-22 Validation Report

Custom scorer vs LIWC-22 CLI (Dutch LIWC-2015 dictionary)

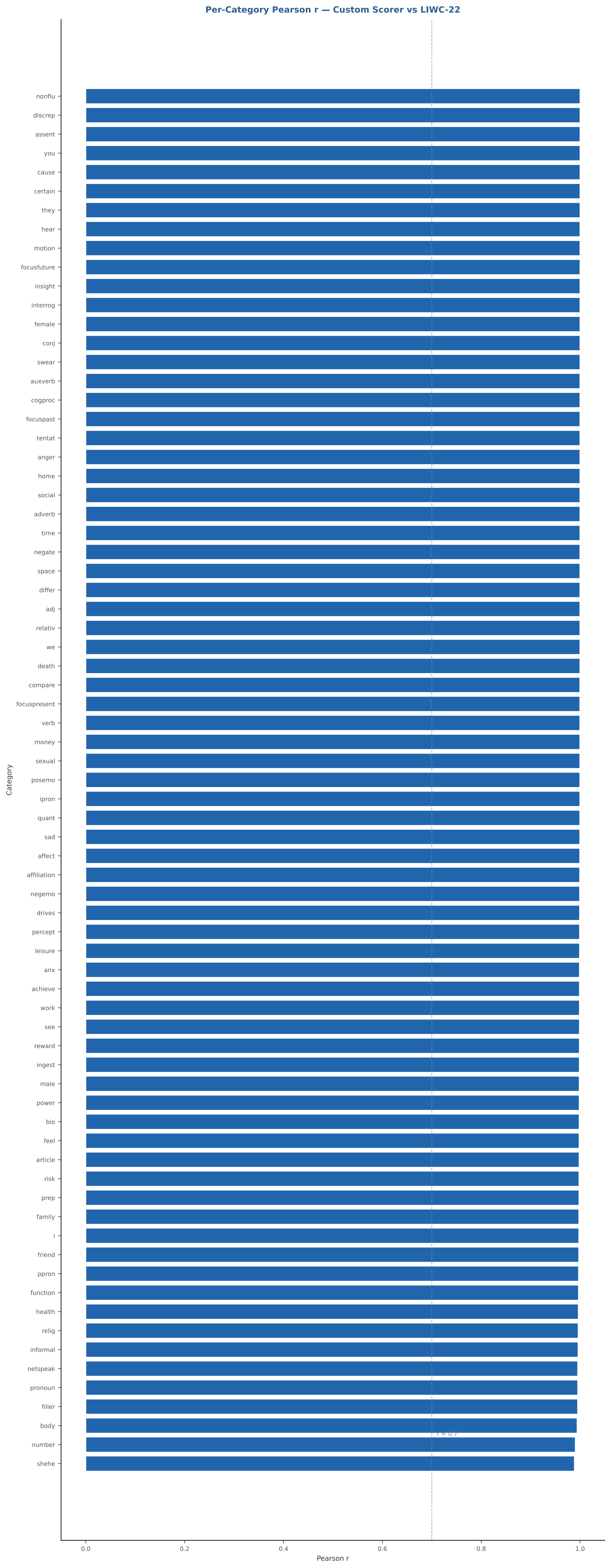
Dataset: Old Data Only

Section 1 — Scoring Agreement

Per-Category Scoring Agreement (sorted by MAE descending)
MAE = mean absolute difference in percentage points; yellow = MAE > 2 pp

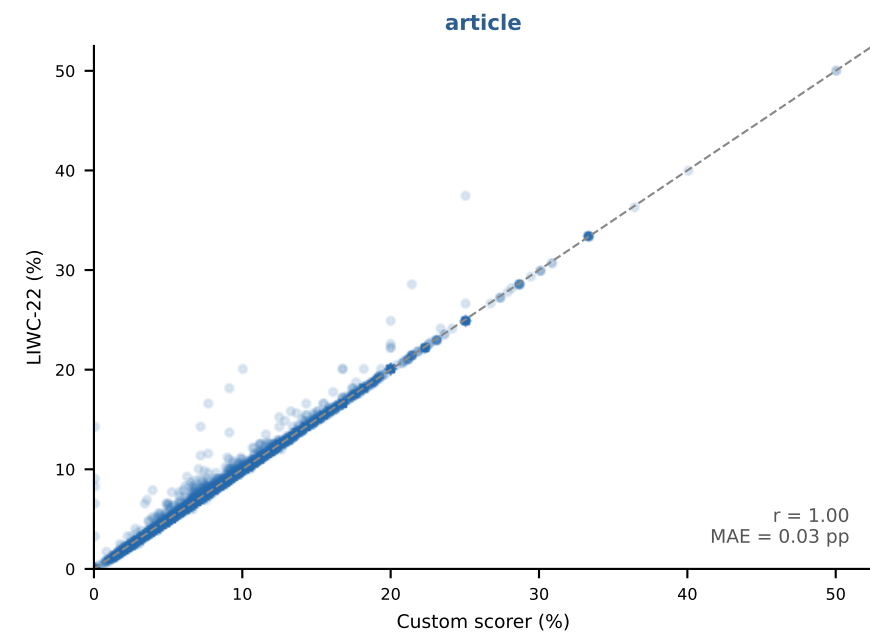
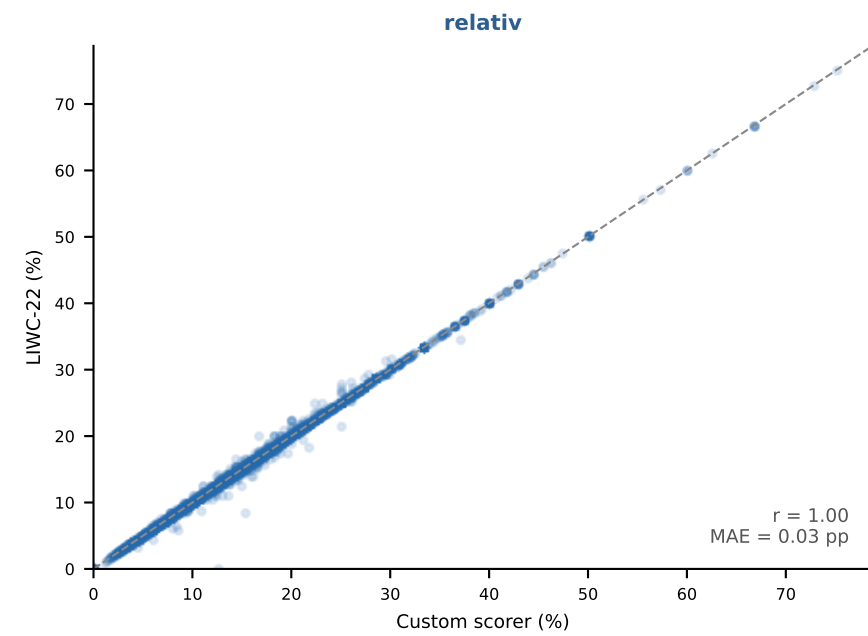
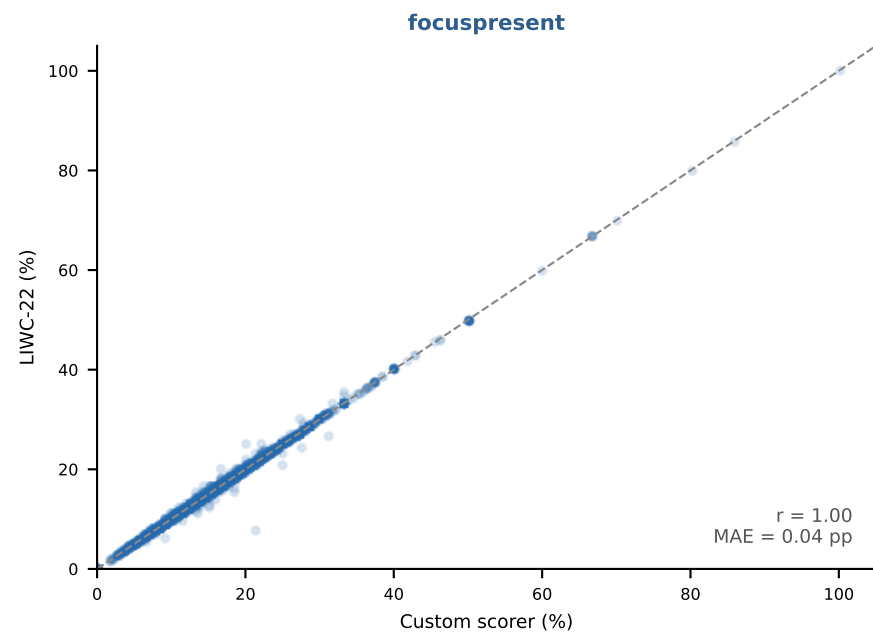
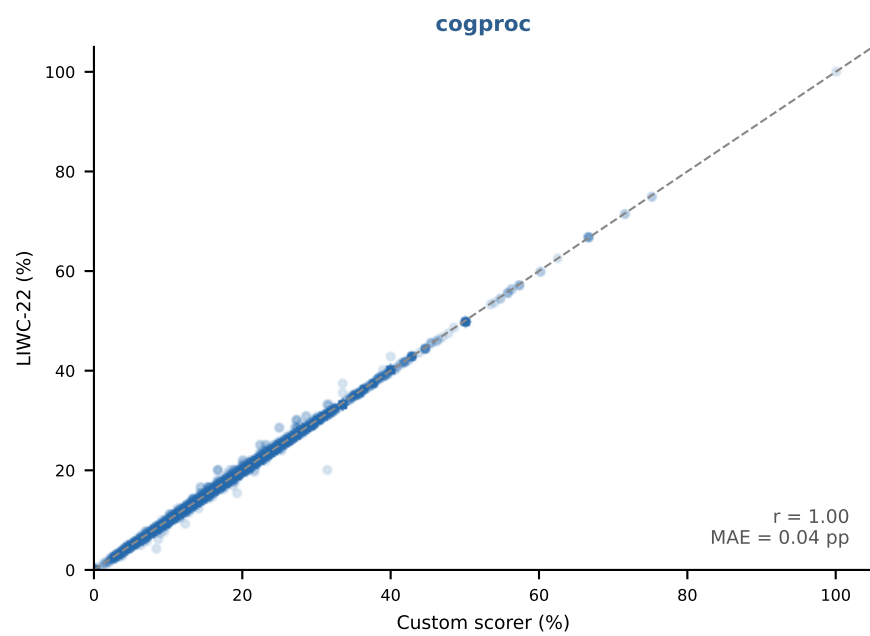
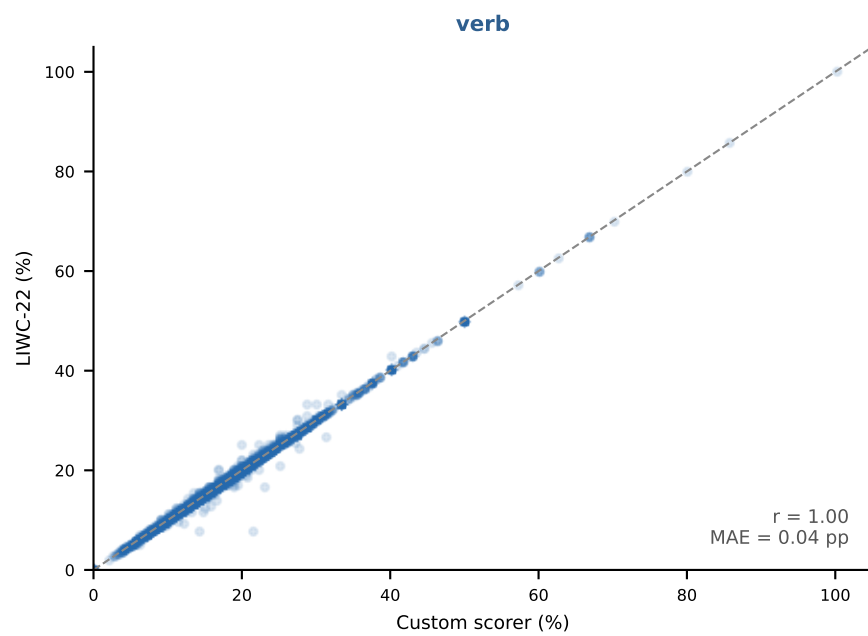
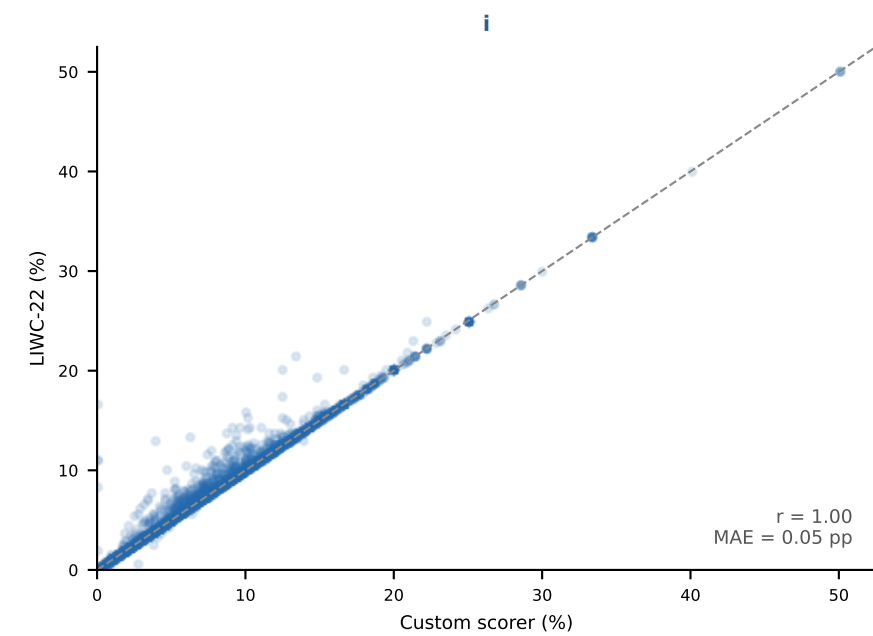
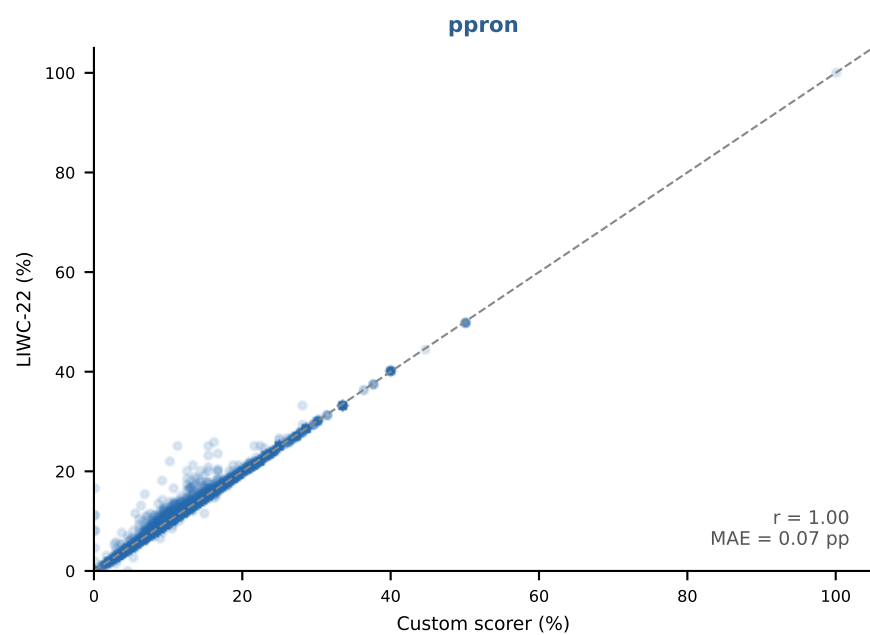
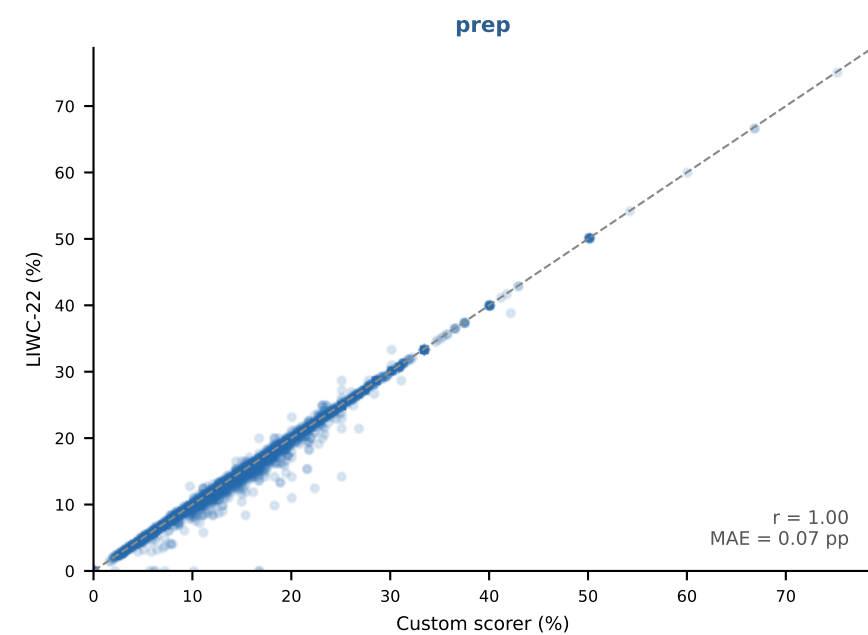
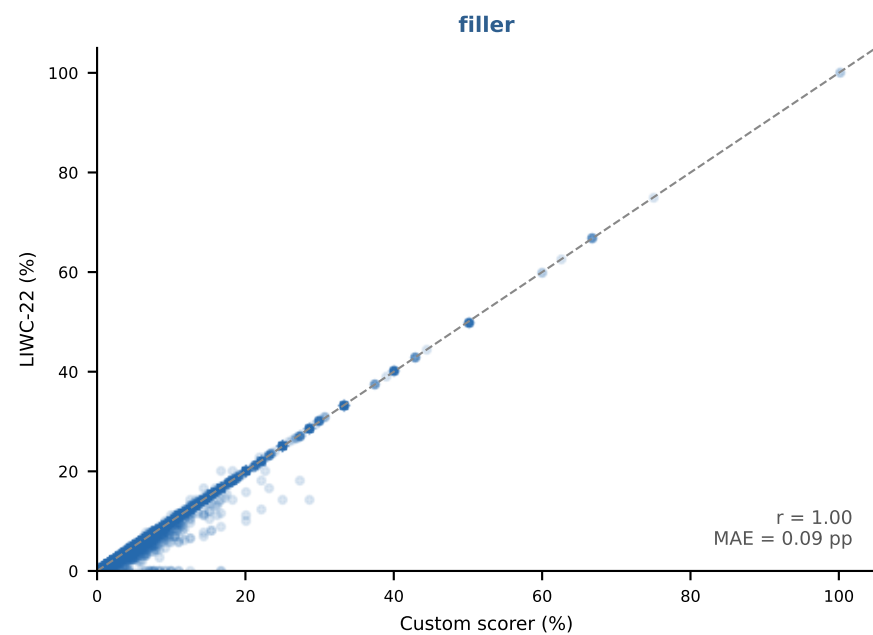
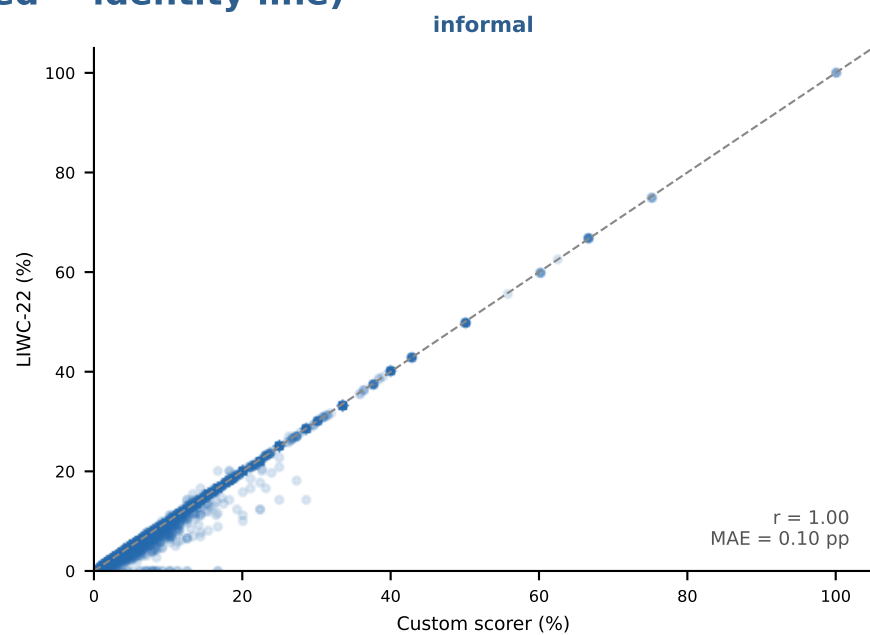
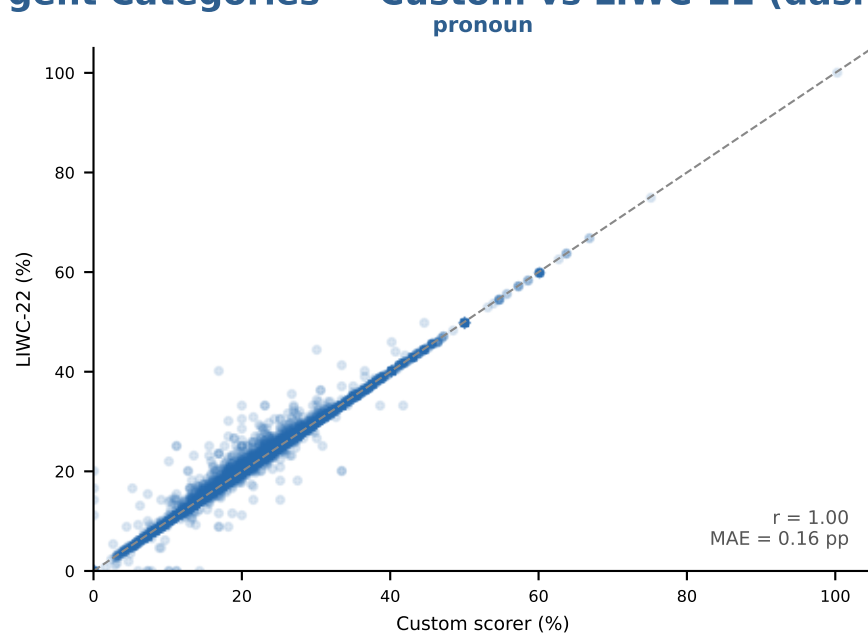
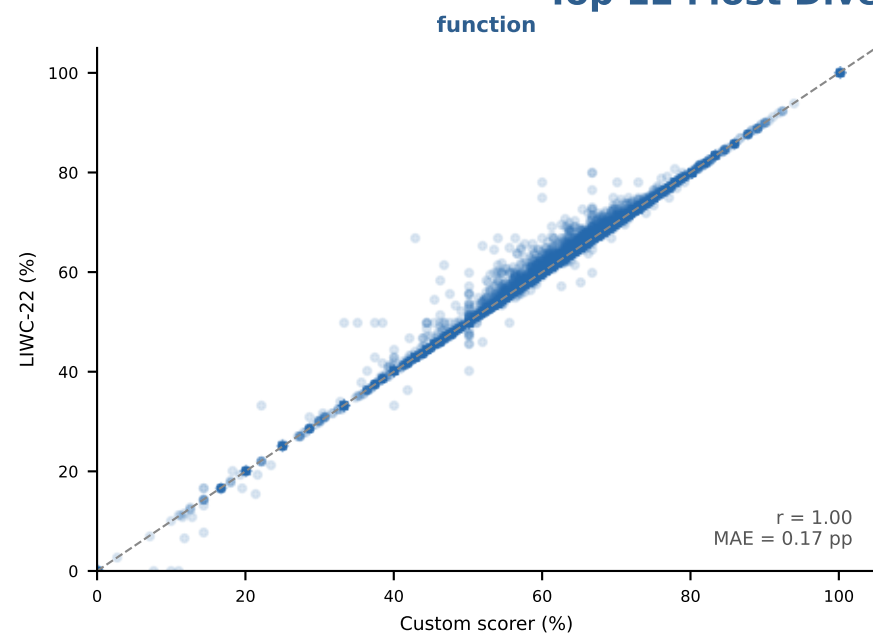
Category	Pearson r	p-value	MAE (pp)	Mean (custom)	Mean (LIWC-22)
function	0.9967	0.0	0.175	62.5499	62.6981
pronoun	0.9952	0.0	0.1579	21.2221	21.3306
informal	0.9959	0.0	0.0956	6.1985	6.118
filler	0.9951	0.0	0.0873	5.1306	5.0565
prep	0.9977	0.0	0.0716	13.263	13.2325
ppron	0.9969	0.0	0.0672	11.0952	11.1567
i	0.9974	0.0	0.0458	5.6415	5.6847
verb	0.9995	0.0	0.041	16.8509	16.8788
cogproc	0.9997	0.0	0.0382	16.2116	16.2401
focuspresent	0.9995	0.0	0.035	14.7364	14.7614
relativ	0.9996	0.0	0.0348	13.1361	13.1565
article	0.998	0.0	0.0328	6.7912	6.8219
adverb	0.9997	0.0	0.0328	13.7237	13.7502
social	0.9997	0.0	0.0286	10.4735	10.4846
drives	0.999	0.0	0.0277	5.8634	5.8593
adj	0.9996	0.0	0.0245	9.0688	9.0822
affect	0.9993	0.0	0.0231	5.9309	5.9309
conj	0.9998	0.0	0.0205	8.158	8.1733
auxverb	0.9997	0.0	0.0195	7.4605	7.4751
ipron	0.9995	0.0	0.0177	7.1067	7.1206
time	0.9997	0.0	0.0177	6.0876	6.097
space	0.9996	0.0	0.0166	5.5569	5.5664
netspeak	0.9952	0.0	0.0164	0.9939	0.9817
affiliation	0.9992	0.0	0.0153	2.2607	2.2553
focuspast	0.9997	0.0	0.0151	5.2641	5.2724
work	0.9987	0.0	0.0138	1.3776	1.3706
differ	0.9996	0.0	0.0137	4.345	4.3511
negemo	0.9991	0.0	0.0131	2.0815	2.0771
shehe	0.9885	0.0	0.0125	0.6129	0.6242
percept	0.9989	0.0	0.0121	2.3986	2.3972
posemo	0.9995	0.0	0.0116	3.8	3.8046
compare	0.9995	0.0	0.0115	4.1779	4.1861
tentat	0.9997	0.0	0.0107	3.2832	3.2914
insight	0.9998	0.0	0.0101	3.4707	3.4757
male	0.9983	0.0	0.0099	0.9629	0.9578
bio	0.9981	0.0	0.0098	1.5215	1.5184
number	0.9903	0.0	0.0097	0.6251	0.6189
quant	0.9994	0.0	0.0089	2.7921	2.7964
you	0.9999	0.0	0.0089	4.111	4.1172
focusfuture	0.9998	0.0	0.0086	3.3704	3.3761
power	0.9982	0.0	0.0078	1.6448	1.6452
achieve	0.9987	0.0	0.0073	1.6997	1.7009
see	0.9987	0.0	0.0071	0.8034	0.8002
discrep	0.9999	0.0	0.0068	2.5951	2.5997
negate	0.9996	0.0	0.0064	1.935	1.9372
interrog	0.9998	0.0	0.0059	2.0615	2.0657
leisure	0.9989	0.0	0.0058	0.5533	0.5508
motion	0.9998	0.0	0.0057	1.9271	1.9294
health	0.9964	0.0	0.0057	0.7871	0.7848
reward	0.9986	0.0	0.0054	1.3024	1.3029
cause	0.9999	0.0	0.0054	1.8189	1.8218
certain	0.9999	0.0	0.0053	1.8697	1.873
risk	0.9977	0.0	0.0046	0.5766	0.5749
sad	0.9994	0.0	0.004	0.9238	0.9244
body	0.994	0.0	0.0038	0.4109	0.4094
anx	0.9987	0.0	0.0035	0.4438	0.4424
feel	0.998	0.0	0.003	0.7216	0.722
family	0.9975	0.0	0.0026	0.2986	0.2975
friend	0.9973	0.0	0.0026	0.2519	0.2512
ingest	0.9985	0.0	0.0022	0.3246	0.3239
hear	0.9998	0.0	0.0022	0.5838	0.5846
female	0.9998	0.0	0.002	0.4766	0.4773
we	0.9996	0.0	0.0017	0.3749	0.3752
they	0.9999	0.0	0.0013	0.3877	0.3884
anger	0.9997	0.0	0.0012	0.2724	0.2727
money	0.9995	0.0	0.0012	0.2274	0.2275
home	0.9997	0.0	0.0012	0.2282	0.2282
relig	0.9961	0.0	0.0012	0.0803	0.0796
assent	0.9999	0.0	0.0011	0.3372	0.3376
death	0.9996	0.0	0.0005	0.0786	0.0785
swear	0.9998	0.0	0.0003	0.0429	0.0428
sexual	0.9995	0.0	0.0002	0.0213	0.0212
nonflu	0.9999	0.0	0.0002	0.0547	0.0547

Per-Category Pearson r — Custom Scorer vs LIWC-22



Section 2 — Divergence Analysis

Top 12 Most Divergent Categories — Custom vs LIWC-22 (dashed = identity line)



Section 3 — LIWC-22 Summary Variables

LIWC-22 Summary Variables — Not Available with External Dictionary

The four LIWC-22 summary variables are not available in this run:

Analytic · Clout · Authentic · Tone

These scores are computed by LIWC-22 from a proprietary regression model trained on the built-in English dictionary. They do not generalise to external dictionary files, so LIWC-22 omits them when a .dicx file is supplied (as here, with the Dutch LIWC-2015 dictionary).

To obtain these variables you would need to run LIWC-22 with its built-in English dictionary (-d LIWC22), which is not appropriate for Dutch text.

See docs/statistical_decisions.md §5 for context.

Section 4 — Category Coverage Differences

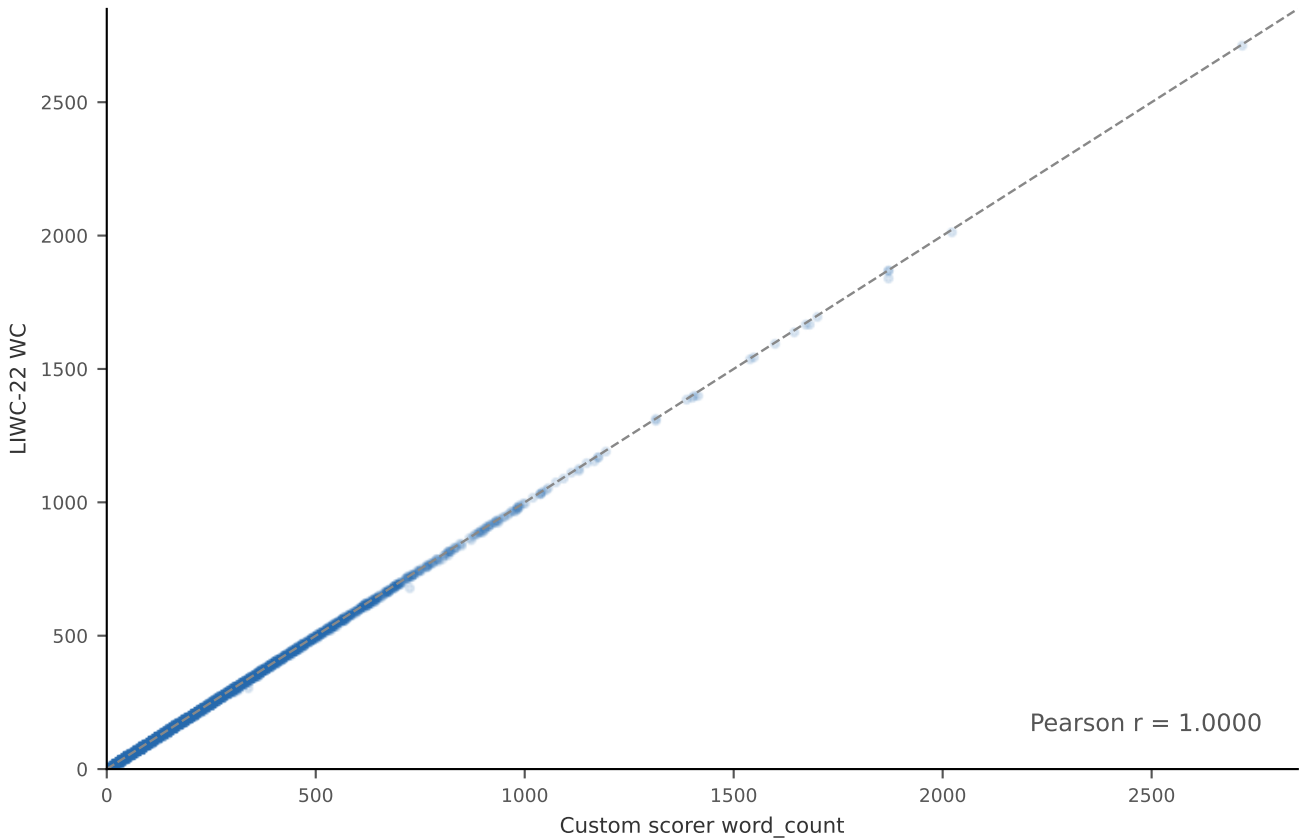
Category Coverage Differences

(custom-only includes fps_dutch fallback; LIWC-22-only includes structural metrics like WC, Emoji, AllPunc)

Custom scorer only	LIWC-22 only
—	—

Section 5 — Diagnostics

Tokenisation Sanity Check — Word Count Agreement (custom scorer re.findall vs LIWC-22 WC; dashed = identity)



8. Pandemic-Period Comparison

Pandemic-Period Comparison

Psycholinguistic markers pre / during / post pandemic
after Yahya & Abdul Rahim (2023), with per-user rank-based statistics

Dataset: Old Data Only

Section 1 — Diagnostics

Period Definitions

```
pre      : PostDate < 2020-03-11
during   : 2020-03-11 ≤ PostDate < 2022-03-23
post     : PostDate ≥ 2022-03-23
```

Cutoff = WHO pandemic declaration (2020-03-11).

End = end of most Dutch COVID restrictions (2022-03-23).

Dataset Variant × Pandemic Period — messages and unique users
⚠ **Period effects are confounded with dataset variant (old export ≈ pre+during, new export ≈ post); yellow = near-empty cell**

variant	period	messages	users
old	pre	4505	169
old	during	11102	291
old	post	2943	96
new_only	pre	94	15
new_only	during	318	23
new_only	post	25092	520
combined	pre	4865	177
combined	during	11960	307
combined	post	6592	247

Users by Number of Periods Appeared In
Users in >1 period violate between-group independence — see the single-period sensitivity analysis in Section 4

periods_appeared_in	n_users	pct_users
1	318	74.5
2	89	20.8
3	20	4.7

Section 2 — Dictionary Introspection

Yahya & Abdul Rahim (2023) §2.3 Categories vs Available Dictionaries
Only available categories are analyzed; '—' = genuinely absent from that source (omission, not a bug)

yahya_group	liwc2015	liwc22	dic_crosscheck
pronouns	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you
function_words	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant
other_grammar	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb
tense	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent
emotion	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad
informal	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear
summary	—	—	—
absolutist	absolutist	—	—

Section 3 — Main Results (all users)

Kruskal-Wallis Omnibus per Feature (per-user rates)
effect_size = epsilon-squared; p_bh = BH over pooled omnibus + post-hoc family; yellow = significant

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
i_liwc22	169	291	96	omnibus	9.7306	0.00771	0.0175		0.184479	False
sad_liwc2015	169	291	96	omnibus	9.2763	0.009676	0.0167		0.184479	False
sad_liwc22	169	291	96	omnibus	9.1888	0.010108	0.0166		0.184479	False
i_liwc2015	169	291	96	omnibus	10.0542	0.006558	0.0181		0.184479	False
ppron_liwc2015	169	291	96	omnibus	7.9128	0.019132	0.0143		0.215846	False
netspeak_liwc22	169	291	96	omnibus	7.627	0.022071	0.0137		0.215846	False
netspeak_liwc2015	169	291	96	omnibus	7.4884	0.023654	0.0135		0.215846	False
ppron_liwc22	169	291	96	omnibus	7.6033	0.022334	0.0137		0.215846	False
prep_liwc22	169	291	96	omnibus	6.5633	0.037565	0.0118		0.228523	False
pronoun_liwc2015	169	291	96	omnibus	6.9183	0.031456	0.0125		0.228523	False
pronoun_liwc22	169	291	96	omnibus	6.5734	0.037377	0.0118		0.228523	False
prep_liwc2015	169	291	96	omnibus	6.6142	0.036622	0.0119		0.228523	False
informal_liwc22	169	291	96	omnibus	5.4247	0.066381	0.0098		0.372755	False
informal_liwc2015	169	291	96	omnibus	4.9998	0.082093	0.009		0.428058	False
quant_liwc2015	169	291	96	omnibus	2.8017	0.246382	0.005		0.539319	False
shehe_liwc2015	169	291	96	omnibus	2.7987	0.246763	0.005		0.539319	False
anx_liwc22	169	291	96	omnibus	2.9797	0.225402	0.0054		0.539319	False
quant_liwc22	169	291	96	omnibus	2.8156	0.244684	0.0051		0.539319	False
function_liwc22	169	291	96	omnibus	3.7842	0.150754	0.0068		0.539319	False
interrog_liwc22	169	291	96	omnibus	3.7826	0.150878	0.0068		0.539319	False
shehe_liwc22	169	291	96	omnibus	2.6841	0.261313	0.0048		0.539319	False
ipron_liwc22	169	291	96	omnibus	3.0943	0.212857	0.0056		0.539319	False
compare_liwc22	169	291	96	omnibus	2.7725	0.250015	0.005		0.539319	False
swear_liwc2015	169	291	96	omnibus	3.8183	0.148209	0.0069		0.539319	False
negate_liwc22	169	291	96	omnibus	3.0507	0.217546	0.0055		0.539319	False
filler_liwc22	169	291	96	omnibus	3.1052	0.211693	0.0056		0.539319	False
conj_liwc2015	169	291	96	omnibus	3.5193	0.172101	0.0063		0.539319	False
anx_liwc2015	169	291	96	omnibus	3.258	0.196129	0.0059		0.539319	False
swear_liwc22	169	291	96	omnibus	3.8162	0.148361	0.0069		0.539319	False
compare_liwc2015	169	291	96	omnibus	2.7462	0.253322	0.0049		0.539319	False
filler_liwc2015	169	291	96	omnibus	2.6488	0.265966	0.0048		0.539319	False
conj_liwc22	169	291	96	omnibus	3.4181	0.18104	0.0062		0.539319	False
interrog_liwc2015	169	291	96	omnibus	3.7834	0.150814	0.0068		0.539319	False
ipron_liwc2015	169	291	96	omnibus	3.0614	0.216386	0.0055		0.539319	False
function_liwc2015	169	291	96	omnibus	3.9407	0.139408	0.0071		0.539319	False
negate_liwc2015	169	291	96	omnibus	3.0605	0.216481	0.0055		0.539319	False
focuspresent_liwc22	169	291	96	omnibus	2.3885	0.302927	0.0043		0.574196	False
assent_liwc22	169	291	96	omnibus	2.3749	0.305006	0.0043		0.574196	False
assent_liwc2015	169	291	96	omnibus	2.3634	0.306762	0.0043		0.574196	False
focuspresent_liwc2015	169	291	96	omnibus	2.2269	0.328432	0.004		0.599389	False
anger_liwc22	169	291	96	omnibus	1.9617	0.374993	0.0035		0.622147	False
focuspast_liwc2015	169	291	96	omnibus	2.0522	0.358398	0.0037		0.622147	False
focuspast_liwc22	169	291	96	omnibus	2.0303	0.362346	0.0037		0.622147	False
anger_liwc2015	169	291	96	omnibus	1.9657	0.374237	0.0035		0.622147	False
auxverb_liwc22	169	291	96	omnibus	1.7739	0.411902	0.0032		0.632661	False
they_liwc2015	169	291	96	omnibus	1.7682	0.413077	0.0032		0.632661	False
they_liwc22	169	291	96	omnibus	1.7825	0.410149	0.0032		0.632661	False
auxverb_liwc2015	169	291	96	omnibus	1.7129	0.424663	0.0031		0.632661	False
number_liwc22	169	291	96	omnibus	1.7524	0.416359	0.0032		0.632661	False
focusfuture_liwc2015	169	291	96	omnibus	1.5024	0.47179	0.0027		0.675307	False
focusfuture_liwc22	169	291	96	omnibus	1.5296	0.465421	0.0028		0.675307	False
absolutist_liwc2015	169	291	96	omnibus	1.137	0.56638	0.002		0.736076	False
adverb_liwc2015	169	291	96	omnibus	1.1077	0.574744	0.002		0.736076	False
article_liwc22	169	291	96	omnibus	1.2256	0.541837	0.0022		0.736076	False
article_liwc2015	169	291	96	omnibus	1.1873	0.552302	0.0021		0.736076	False
number_liwc2015	169	291	96	omnibus	1.2578	0.533186	0.0023		0.736076	False
adverb_liwc22	169	291	96	omnibus	1.1419	0.564987	0.0021		0.736076	False
verb_liwc22	169	291	96	omnibus	0.3649	0.833246	0.0007		0.919775	False
affect_liwc2015	169	291	96	omnibus	0.2684	0.874423	0.0005		0.919775	False
negemo_liwc2015	169	291	96	omnibus	0.3289	0.848374	0.0006		0.919775	False
nonflu_liwc2015	169	291	96	omnibus	0.4427	0.801456	0.0008		0.919775	False
posemo_liwc2015	169	291	96	omnibus	0.539	0.763744	0.001		0.919775	False
verb_liwc2015	169	291	96	omnibus	0.2991	0.861096	0.0005		0.919775	False
adj_liwc2015	169	291	96	omnibus	0.3506	0.83919	0.0006		0.919775	False
negemo_liwc22	169	291	96	omnibus	0.2681	0.874566	0.0005		0.919775	False
posemo_liwc22	169	291	96	omnibus	0.5524	0.758655	0.001		0.919775	False
affect_liwc22	169	291	96	omnibus	0.2526	0.881348	0.0005		0.919775	False
nonflu_liwc22	169	291	96	omnibus	0.3327	0.846735	0.0006		0.919775	False
you_liwc22	169	291	96	omnibus	0.2378	0.887914	0.0004		0.919775	False
adj_liwc22	169	291	96	omnibus	0.3453	0.841418	0.0006		0.919775	False
you_liwc2015	169	291	96	omnibus	0.2228	0.894575	0.0004		0.919775	False
we_liwc22	169	291	96	omnibus	0.072	0.96464	0.0001		0.970085	False
we_liwc2015	169	291	96	omnibus	0.0607	0.970085	0.0001		0.970085	False

No features survived omnibus BH correction — no post-hoc tests run

Section 4 — Sensitivity: Single-Period Users Only

Same Design, Restricted to Users Appearing in Exactly One Period
Removes the between-group dependence of multi-period users; compare significance patterns with Section 3

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
netspeak_liwc2015	91.0	182.0	45.0	omnibus	14.547	0.000694	0.0459		0.018267	True
netspeak_liwc22	91.0	182.0	45.0	omnibus	13.863	0.000977	0.0437		0.019286	True
swear_liwc22	91.0	182.0	45.0	omnibus	9.1426	0.010344	0.0288		0.116744	False
informal_liwc22	91.0	182.0	45.0	omnibus	9.2583	0.009763	0.0292		0.116744	False
swear_liwc2015	91.0	182.0	45.0	omnibus	9.1494	0.01031	0.0289		0.116744	False
informal_liwc2015	91.0	182.0	45.0	omnibus	8.5249	0.014088	0.0269		0.139119	False
adj_liwc2015	91.0	182.0	45.0	omnibus	4.8756	0.087352	0.0154		0.462613	False
adj_liwc22	91.0	182.0	45.0	omnibus	4.7972	0.090845	0.0151		0.462613	False
ipron_liwc22	91.0	182.0	45.0	omnibus	5.5501	0.062346	0.0175		0.462613	False
pronoun_liwc2015	91.0	182.0	45.0	omnibus	4.7354	0.093694	0.0149		0.462613	False
ipron_liwc2015	91.0	182.0	45.0	omnibus	5.3258	0.069746	0.0168		0.462613	False
absolutist_liwc2015	91.0	182.0	45.0	omnibus	5.3788	0.067921	0.017		0.462613	False
filler_liwc22	91.0	182.0	45.0	omnibus	4.5851	0.101007	0.0145		0.468054	False
pronoun_liwc22	91.0	182.0	45.0	omnibus	4.4765	0.106645	0.0141		0.468054	False
focusfuture_liwc22	91.0	182.0	45.0	omnibus	3.54	0.170329	0.0112		0.482354	False
i_liwc2015	91.0	182.0	45.0	omnibus	3.7926	0.150121	0.012		0.482354	False
filler_liwc2015	91.0	182.0	45.0	omnibus	3.9493	0.138813	0.0125		0.482354	False
interrog_liwc2015	91.0	182.0	45.0	omnibus	3.4094	0.181828	0.0108		0.482354	False
sad_liwc22	91.0	182.0	45.0	omnibus	3.204	0.20149	0.0101		0.482354	False
i_liwc22	91.0	182.0	45.0	omnibus	3.2506	0.196849	0.0103		0.482354	False
number_liwc22	91.0	182.0	45.0	omnibus	4.2691	0.118299	0.0135		0.482354	False
interrog_liwc22	91.0	182.0	45.0	omnibus	3.4382	0.179228	0.0108		0.482354	False
number_liwc2015	91.0	182.0	45.0	omnibus	3.9936	0.135767	0.0126		0.482354	False
focusfuture_liwc2015	91.0	182.0	45.0	omnibus	3.4815	0.175393	0.011		0.482354	False
article_liwc22	91.0	182.0	45.0	omnibus	3.3392	0.188321	0.0105		0.482354	False
article_liwc2015	91.0	182.0	45.0	omnibus	3.2542	0.196501	0.0103		0.482354	False
sad_liwc2015	91.0	182.0	45.0	omnibus	3.2375	0.198145	0.0102		0.482354	False
we_liwc2015	91.0	182.0	45.0	omnibus	2.8017	0.246393	0.0088		0.486626	False
we_liwc22	91.0	182.0	45.0	omnibus	2.9183	0.232435	0.0092		0.486626	False
assent_liwc2015	91.0	182.0	45.0	omnibus	2.9444	0.229419	0.0093		0.486626	False
ppron_liwc2015	91.0	182.0	45.0	omnibus	2.828	0.243167	0.0089		0.486626	False
assent_liwc22	91.0	182.0	45.0	omnibus	2.9605	0.227579	0.0093		0.486626	False
anx_liwc22	91.0	182.0	45.0	omnibus	3.0622	0.216298	0.0097		0.486626	False
anx_liwc2015	91.0	182.0	45.0	omnibus	2.8333	0.242521	0.0089		0.486626	False
ppron_liwc22	91.0	182.0	45.0	omnibus	2.158	0.339935	0.0068		0.583674	False
focuspast_liwc22	91.0	182.0	45.0	omnibus	2.1154	0.347249	0.0067		0.583674	False
prep_liwc22	91.0	182.0	45.0	omnibus	2.1521	0.340935	0.0068		0.583674	False
focuspast_liwc2015	91.0	182.0	45.0	omnibus	2.228	0.328246	0.007		0.583674	False
nonflu_liwc22	91.0	182.0	45.0	omnibus	2.3179	0.313808	0.0073		0.583674	False
prep_liwc2015	91.0	182.0	45.0	omnibus	2.2921	0.317895	0.0072		0.583674	False
nonflu_liwc2015	91.0	182.0	45.0	omnibus	2.319	0.313639	0.0073		0.583674	False
function_liwc22	91.0	182.0	45.0	omnibus	1.6268	0.443344	0.0051		0.72967	False
function_liwc2015	91.0	182.0	45.0	omnibus	1.487	0.475441	0.0047		0.766527	False
adverb_liwc22	91.0	182.0	45.0	omnibus	1.2132	0.545197	0.0038		0.812653	False
negate_liwc2015	91.0	182.0	45.0	omnibus	1.253	0.534456	0.004		0.812653	False
negate_liwc22	91.0	182.0	45.0	omnibus	1.2251	0.541955	0.0039		0.812653	False
adverb_liwc2015	91.0	182.0	45.0	omnibus	1.2741	0.528858	0.004		0.812653	False
posemo_liwc22	91.0	182.0	45.0	omnibus	0.7431	0.68965	0.0023		0.889588	False
posemo_liwc2015	91.0	182.0	45.0	omnibus	0.727	0.695237	0.0023		0.889588	False
conj_liwc22	91.0	182.0	45.0	omnibus	0.7734	0.67928	0.0024		0.889588	False
conj_liwc2015	91.0	182.0	45.0	omnibus	0.7296	0.694319	0.0023		0.889588	False
quant_liwc2015	91.0	182.0	45.0	omnibus	0.6866	0.709419	0.0022		0.889588	False
anger_liwc22	91.0	182.0	45.0	omnibus	0.8653	0.648791	0.0027		0.889588	False
shehe_liwc22	91.0	182.0	45.0	omnibus	0.8462	0.655022	0.0027		0.889588	False
shehe_liwc2015	91.0	182.0	45.0	omnibus	0.8247	0.662076	0.0026		0.889588	False
anger_liwc2015	91.0	182.0	45.0	omnibus	0.8826	0.643204	0.0028		0.889588	False
quant_liwc22	91.0	182.0	45.0	omnibus	0.7132	0.700047	0.0022		0.889588	False
focuspresent_liwc22	91.0	182.0	45.0	omnibus	0.4046	0.816849	0.0013		0.942002	False
compare_liwc2015	91.0	182.0	45.0	omnibus	0.421	0.810172	0.0013		0.942002	False
compare_liwc22	91.0	182.0	45.0	omnibus	0.4592	0.794866	0.0014		0.942002	False
auxverb_liwc22	91.0	182.0	45.0	omnibus	0.3878	0.823722	0.0012		0.942002	False
auxverb_liwc2015	91.0	182.0	45.0	omnibus	0.3509	0.839096	0.0011		0.942002	False
focuspresent_liwc2015	91.0	182.0	45.0	omnibus	0.333	0.846609	0.0011		0.942002	False
affect_liwc22	91.0	182.0	45.0	omnibus	0.414	0.813021	0.0013		0.942002	False
affect_liwc2015	91.0	182.0	45.0	omnibus	0.4316	0.805897	0.0014		0.942002	False
they_liwc2015	91.0	182.0	45.0	omnibus	0.1776	0.915027	0.0006		0.967971	False
negemo_liwc22	91.0	182.0	45.0	omnibus	0.1562	0.924857	0.0005		0.967971	False
verb_liwc2015	91.0	182.0	45.0	omnibus	0.1164	0.943465	0.0004		0.967971	False
verb_liwc22	91.0	182.0	45.0	omnibus	0.1332	0.935579	0.0004		0.967971	False
they_liwc22	91.0	182.0	45.0	omnibus	0.1954	0.906943	0.0006		0.967971	False
negemo_liwc2015	91.0	182.0	45.0	omnibus	0.1881	0.910217	0.0006		0.967971	False
you_liwc2015	91.0	182.0	45.0	omnibus	0.0052	0.997406	0.0		0.997906	False
you_liwc22	91.0	182.0	45.0	omnibus	0.0042	0.997906	0.0		0.997906	False